



**COMPONENT MAINTENANCE MANUAL
WITH ILLUSTRATED PARTS LIST**

**DC STARTER-GENERATOR
23079 SERIES**

**MODELS 23079-000, -000-1, -000-2, -002, -002-1, -003, -004,
-005, -006, -007, -009, -010**

Original Issue: July 1, 1976
Revision 9: February 28, 2003

"THESE COMMODITIES, SOFTWARE, OR TECHNOLOGY WERE EXPORTED FROM THE UNITED STATES IN ACCORDANCE WITH THE EXPORT ADMINISTRATION REGULATION. DIVERSION CONTRARY TO U.S LAW IS PROHIBITED."

© Copyright 2003 Goodrich Corporation (formerly TRW Aeronautical Systems, Lucas Aerospace, Lucas Aerospace and also Lear Siegler) - Copyright in this manual and associated documents and drawings belongs to Goodrich Corporation. All rights reserved. No reproduction of all or part of this manual shall be made without prior written consent of Goodrich Corporation. This document contains information that may be confidential. Disclosure to others of information contained herein requires the written consent of Goodrich Corporation. This manual produced at the Aurora, OH plant.

TITLE PAGE Feb 28/03



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

EXCLUSIVE DISTRIBUTORS

Genuine Goodrich Corporation component parts and technical publications are to be purchased from the following Exclusive Distributors:

<u>REGION</u>	<u>ADDRESS</u>	<u>COMMUNICATIONS</u>
AMERICAS:	Goodrich Corporation 30 Van Nostrand Avenue Englewood, NJ 07631-4396 USA	PHONE (1) 201 567 6400 AOG (1) 800 735 7899 (1) 201 567 6411 FAX (1) 201 894 1965 Technical Publications Library
EUROPE/ MIDEAST/ AFRICA:	Goodrich Control Systems Limited Stratford Road Solihull B90 4LA England	PHONE (44) 121 451 5788 AOG (44) 121 451 5904 FAX (44) 121 451 6133 AOG aog.helpdesk@goodrich.com
ASIA/ PACIFIC:	Goodrich Control Systems Pte. Ltd. 35 & 37 Loyang Way Republic of Singapore 508733	PHONE (65) 6545 9975 AOG (65) 6545 6253 FAX (65) 6543 0419
	Goodrich Control Systems, Pty. Ltd. 84-92 Epsom Road Zetland, Sydney NSW 2017 Australia	PHONE (61) 2 9315 3700 AOG (61) 2 9937 2553 FAX (61) 2 9313 7676
	Goodrich TAECO Aeronautical Systems (Xiamen) Co. Ltd. GAOQI International Airport Xiamen 361006 Fujian Peoples Republic of China	PHONE (86) 592 573 0089 AOG (86) 592 573 1098 FAX (86) 592 573 0090 AOG burk.xu@goodrich.com

Parts and assemblies not authorized for repair in the field must be sent to the Goodrich Corporation, Repair and Overhaul (R&O) at the location below.

<u>REGION</u>	<u>ADDRESS</u>	<u>COMMUNICATIONS</u>
ALL:	Goodrich Corporation 777 Lena Drive Aurora, OH 44202 USA	PHONE (1) 330 995 1000 FAX (1) 330 995 1117

CAUTION: USE OF PARTS, MATERIALS, OR PROCESSES NOT AUTHORIZED BY GOODRICH CORPORATION FOR THE MAINTENANCE OR SERVICE OF THESE UNITS WILL HAVE AN AFFECT ON THEIR CONTINUED AIRWORTHINESS AND INVALIDATES THEIR CERTIFICATION.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

HIGHLIGHTS

TO: Holders of Component Maintenance Manual with IPL for DC Starter-Generator Models 23079-000, -000-1, -000-2, -002, -002-1, -003, -004, -005, -006, -007, -009, and -010.

Attached to this transmittal letter is Revision No. 9 of the Component Maintenance Manual with IPL (basic issue dated July 1, 1976).

REVISION 9, DATED 28 FEBRUARY 2003

Remove the complete existing manual and replace with this full revision of the manual. Retain the Highlights page(s) in the front of the manual for future reference.

Documents

	Name	Modification Date
Newer Document:	Introduction.fm	March 21, 2002 2:39 pm
Older Document:	Introduction.fm	September 4, 2001 3:57 pm

Text Content Changes

Page Numbers

Composite	Newer	Older	Change	Text
xxv	xxv	xxv	Replaced:	"ND" With: "Non-Destructive Test (..."
xxv	xxv	xxv	Inserted:	"Magnetic particle inspection of the ..."
xxv	xxv	xxv	Deleted:	"ISPEC 2200 ("
xxv	xxv	xxv	Replaced:	"2200) Information Standards for..." With: "No. 100 (Revision..."

Documents

	Name	Modification Date
Newer Document:	00001-Description.fm	March 19, 2002 9:51 am
Older Document:	00001-Description.fm	December 18, 2000 2:12 pm

Text Content Changes

Page Numbers

Composite	Newer	Older	Change	Text
3	3	3	Replaced:	"Weight" With: "Weight without QAD kit"
3	3	3	Deleted:	"23079-010"
3	3	3	Replaced:	"32" With: "31"
3	3	3	Replaced:	"9" With: "2"
3	3	3	Replaced:	"92" With: "15"
3	3	3	Replaced:	"30" With: "45"
3	3	3	Replaced:	"65" With: "72"
3	3	3	Inserted:	"23079-010"
3	3	3	Replaced:	"70" With: "85"
3	3	3	Replaced:	"95" With: "99"
6	6	6	Replaced:	"23079 Series DC Starter-..." With: "Models



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

			23079-006,..."
6	6	6	Replaced: "23079 Series DC Starter-..." With: "Models 23079-000,..."

Documents

	Name	Modification Date
Newer Document:	01001-Testing.fm	March 21, 2002 2:18 pm
Older Document:	01001-Testing.fm	December 18, 2000 2:12 pm

Text Content Changes

Page Numbers

Composite	Newer	Older	Change	Text
1006	1006	1006	Inserted:	"Maximum Speed for Regulation Test..."
1008	1008	1008	Deleted:	"All Models except 23079-..."
1008	1008	1008	Replaced:	"200" With: "700"
1009	1009	1009	Inserted:	"Compounding (Models 23079-..."
1018	1018	1018	Inserted:	"Dielectric testing is accomplished during..."
1018	1018	1018	Deleted:	"Dielectric Testing causes a..."
1020	1020	1020	Deleted:	"Dielectric Testing causes a cumulative..."
1020	1020	1020	Deleted:	"

Documents

	Name	Modification Date
Newer Document:	03001-Disassembly.fm	March 21, 2002 9:08 am
Older Document:	03001-Disassembly.fm	December 18, 2000 2:12 pm

Text Content Changes

Page Numbers

Composite	Newer	Older	Change	Text
3002	3002	3002	Replaced:	"000" With: "002"
3002	3002	3002	Deleted:	"23079"
3002	3002	3002	Replaced:	"000" With: "002"
3002	3002	3002	Replaced:	"23079" With: "-003,"
3002	3002	3002	Replaced:	"005" With: "004"
3002	3002	3002	Deleted:	"23079"
3002	3002	3002	Replaced:	"009" With: "006"
3002	3002	3002	Deleted:	"23079"
3002	3002	3002	Replaced:	"010" With: "007"
3002	3002	3002	Replaced:	"25" With: "50"
3003	3003	3003	Deleted:	"On models 23079-006..."
3004	3004	3004	Replaced:	"Remove Air Inlet Adapter (...)" With: "Remove air inlet adapter (...)"
3004	3004	3004	Replaced:	"inlet" With: "inlet adapter"
3007	3007	3007	Inserted:	"(145)"
3011	3011	3011	Replaced:	"160" With: "185"

HIGHLIGHTS



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

3015	3015	3015	Replaced:"20" With: "35"
3015	3015	3015	Replaced:"30" With: "45"
3015	3015	3015	Replaced:"25" With: "40"
3015	3015	3015	Replaced:"40" With: "55"
3015	3015	3015	Replaced:"35" With: "50"
3015	3015	3015	Replaced:"60" With: "75"
3016	3016	3016	Replaced:"55" With: "70"
3016	3016	3016	Replaced:"45" With: "60"
3016	3016	3016	Replaced:"50" With: "65"

Documents

	Name	Modification Date
Newer Document:	05001-Check.fm	March 21, 2002 2:43 pm
Older Document:	05001-Check.fm	December 18, 2000 2:12 pm

Text Content Changes

Page Numbers

Composite	Newer	Older	Change	Text
5008	5008	5008	Inserted:	"Check blind rivet nut on..."
5013	5013	5012	Inserted:	"Visually examine Rivnuts for..."
5018	5018	5016	Inserted:	"Use of Glyptol on either..."
5021	5021	5019	Inserted:	"Use of Glyptol on either..."
5021	5021	5019	Replaced:	"Use an ohmmeter to check..." With: "Use an ohmmeter to check..."

Documents

	Name	Modification Date
Newer Document:	06001-Repair.fm	March 21, 2002 10:05 am
Older Document:	06001-Repair.fm	December 18, 2000 2:12 pm

Text Content Changes

Page Numbers

Composite	Newer	Older	Change	Text
6011	6011	6011	Replaced:	"Rivet Nut" With: "Rivnut"
6011	6011	6011	Deleted:	"in Brush Access Cover"
6011	6011	6011	Replaced:	"Remove the rivet nut (...)" With: "Remove the fastener by drilling..."
6011	6011	6011	Replaced:	"rivet nut" With: "rivnut"
6011	6011	6011	Deleted:	"Making sure that key of..."
6014	6014	6012	Inserted:	"Brush Holder Rivet Alignment"

Anchored Frame Changes

Page Numbers

Composite	Newer	Older	Change
6011	6011	6011	Changed Content

HIGHLIGHTS



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

6012 6012 6011 Inserted

Documents

	Name	Modification Date
Newer Document:	07001-Assembly.fm	March 25, 2002 7:41 am
Older Document:	07001-Assembly.fm	December 18, 2000 2:12 pm

Text Content Changes

Page Numbers

Composite	Newer	Older	Change Text
7004	7004	7006	Replaced:"60" With: "75"
7004	7004	7006	Replaced:"7004" With: "7002"
7005	7005	7007	Replaced:"7005" With: "7003"
7005	7005	7007	Replaced:"35" With: "50"
7005	7005	7007	Replaced:"40" With: "55"
7005	7005	7007	Replaced:"25" With: "40"
7005	7005	7007	Replaced:"30" With: "45"
7005	7005	7007	Replaced:"20" With: "35"
7007	7007	7009	Replaced:"7A" With: "8.A"
7008	7008	7009	Replaced:"7006" With: "7004"
7008	7008	7010	Deleted: "Apply lubricating and assembly paste..."
7009	7009	7010	Replaced:"7007" With: "7005"
7010	7010	7011	Replaced:"7008" With: "7006"
7010	7010	7011	Replaced:"bearing" With: "inside"
7011	7011	7012	Replaced:"7009" With: "7007)"
7012	7012	7013	Replaced:"185" With: "160"
7012	7012	7013	Replaced:"7010" With: "7008)." "
7013	7013	7014	Replaced:"7011." With: "7009)." "
7016	7016	7017	Inserted: "Apply lubricating and assembly paste..."
7018	7018	7019	Replaced:"7014" With: "7012"
7019	7019	7020	Replaced:"7015" With: "7013"
7019	7019	7020	Replaced:"7016" With: "7014"
7020	7020	7021	Replaced:"7017" With: "7015"
7021	7021	7022	Replaced:"Maintenance Manual 23700 or" With: ""
7021	7021	7022	Replaced:"7018" With: "7016"
7022	7022	7023	Replaced:"7019" With: "7017"
7023	7023	7024	Replaced:"7020" With: "7018"
7024	7024	7025	Replaced:"in using one of two..." With: "in"
7024	7024	7027	Replaced:")" With: ")", if applicable,"
7027	7027	7030	Replaced:"5J" With: "5L"

Documents

	Name	Modification Date
Newer Document:	08001-Fits.fm	March 21, 2002 10:47 am
Older Document:	08001-Fits.fm	December 18, 2000 2:12 pm

HIGHLIGHTS



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

Text Content Changes

Page Numbers

Composite	Newer	Older	Change	Text
8004	8004	8004	Replaced:	"Nut, Blind Rivet" With: "Rivnut"
8004	8004	8004	Replaced:	"20" With: "35"

Documents

	Name	Modification Date
Newer Document:	10001-PartsList.fm	March 21, 2002 1:22 pm
Older Document:	10001-PartsList.fm	December 18, 2000 2:12 pm

Text Content Changes

Page Numbers

Composite	Newer	Older	Change	Text
10004	10004	10004	Inserted:	"23079-000-2"

Table Changes

Page Numbers

Composite	Newer	Older	Change	Table Title
10003	10003	10003	Changed:	No Title
10007	10007	10007	Deleted:	No Title
10007	10007	10007	Inserted:	No Title
10015	10015	10017	Deleted:	No Title
10015	10015	10017	Inserted:	No Title
10019	10019	10021	Deleted:	No Title
10019	10019	10021	Inserted:	No Title
10021	10021	10023	Deleted:	No Title
10021	10021	10023	Inserted:	No Title
10023	10023	10025	Deleted:	No Title
10023	10023	10025	Inserted:	No Title

Anchored Frame Changes

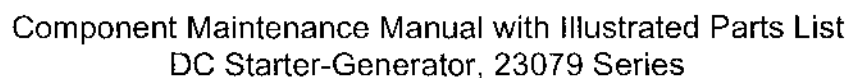
Page Numbers

Composite	Newer	Older	Change
10014	10014	10016	Changed Content
10020	10020	10022	Changed Content



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

THIS PAGE INTENTIONALLY LEFT BLANK

[illegible]

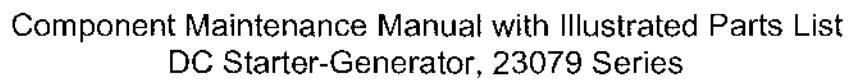
RECORD OF REVISIONS



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

THIS PAGE INTENTIONALLY LEFT BLANK

RECORD OF REVISIONS

[illegible]

Retain this record in the front of the CMM. On receipt of temporary revisions, insert affected pages into the manual. Record the temporary revision number, page number, date issued, date inserted into the CMM, and initials on this page. When temporary revisions are removed from the CMM, record the date removed and initials.

© 2003 Goodrich Corporation

24-30-08



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

THIS PAGE INTENTIONALLY LEFT BLANK

RECORD OF TEMPORARY REVISIONS



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

SERVICE BULLETIN LIST

Service Bulletin Number	Status	Rev	Date Issued	Service Bulletin Number	Status	Rev	Date Issued
23079-01	Incorporated		2/20/79				
23079-02	Invalid	1	11/30/94				
23079-0XX-24-01	Invalid	1	4/29/94				
23079-0XX-24-02	Incorporated	3	12/23/98				
23079-0XX-24-03	Incorporated	1	12/23/98				
23079-0XX-24-04	Incorporated	1	12/23/98				
23079-0XX-24-05	Incorporated	1	12/23/98				
23079-0XX-24-06	Incorporated		11/24/97				
23079-005-24-01	Incorporated		11/30/94				
23079-009-24-01	Invalid	1	4/29/94				
23079-009-24-02	Incorporated		9/28/92				
23079-009-24-03	Invalid	1	4/29/94				
23079-009-24-04	Incorporated		11/29/94				
23079-009-24-05	Incorporated	1	12/23/98				
23079-009-24-06	Incorporated	1	12/23/98				
23079-009-24-07	Incorporated		1/23/97				
23079-010-24-01	Incorporated	1	10/6/95				
23079-010-24-02	Incorporated		2/8/95				
Retain this record in the front of the CMM. On receipt of service bulletins, record the service bulletin number, status (valid or invalid), revision number (if applicable) and the date issued. Unless invalidated, all service bulletins affecting the 23079 Series of DC Starter-Generators shall be kept with the CMM.							



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

SERVICE INFORMATION LETTER LIST

Service Information Letter Number	Status	Rev	Date Issued	Service Information Letter Number	Status	Rev	Date Issued
23079-000-1-01	Limited		02/26/96				
23079-009-01	Invalid	1	04/29/94				
23079-009-02	Limited		01/09/95				
23079-009-03	Limited		08/28/95				
23079-010-01	Valid		02/13/95				
23079-010-02	Limited		02/26/96				
23079-010-03	Limited		02/05/95				
23080-3110-01	Valid		03/15/94				
23701	Valid		03/05/96				
23703	Valid		01/24/97				
23075-1230-01	Valid		02/13/92				
03-6010-XX-01	Valid		05/02/95				
02-5600-01	Valid	1	08/17/92				

Retain this record in the front of the CMM. On receipt of Service Information Letters, record the Service Information Letter number, status, revision number (if applicable) and the date issued. Unless invalidated, all Service Information Letters affecting the 23079 Series of DC Starter-Generators shall be kept with the CMM.

SERVICE INFORMATION LETTER LIST



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

LIST OF EFFECTIVE PAGES

<u>Title</u>	<u>Page</u>	<u>Date</u>
EXCLUSIVE DISTRIBUTORS	ii	Feb 28/03
HIGHLIGHTS	1	Feb 28/03
	2	Feb 28/03
	3	Feb 28/03
	4	Feb 28/03
	5	Feb 28/03
	6	Blank
RECORD OF REVISIONS	iii	Feb 28/03
	iv	Blank
RECORD OF TEMPORARY REVISIONS	v	Feb 28/03
	vi	Blank
SERVICE BULLETIN LIST	vii	Feb 28/03
	viii	Blank
SERVICE INFORMATION LETTER LIST	ix	Feb 28/03
	x	Blank
LIST OF EFFECTIVE PAGES	xi	Feb 28/03
	xii	Feb 28/03
	xiii	Feb 28/03
	xiv	Feb 28/03
	xv	Feb 28/03
	xvi	Feb 28/03
TABLE OF CONTENTS	xvii	Feb 28/03
	xviii	Feb 28/03
LIST OF ILLUSTRATIONS	xix	Feb 28/03
	xx	Feb 28/03
	xxi	Feb 28/03
	xxii	Blank
LIST OF TABLES	xxiii	Feb 28/03
	xxiv	Blank
INTRODUCTION	xxv	Feb 28/03
	xxvi	Feb 28/03
	xxvii	Feb 28/03
	xxviii	Feb 28/03
	xxix	Feb 28/03
	xxx	Blank
DESCRIPTION AND OPERATION	1	Feb 28/03
	2	Feb 28/03
	3	Feb 28/03
	4	Feb 28/03
	5	Feb 28/03
	6	Feb 28/03
	7	Feb 28/03

LIST OF EFFECTIVE PAGES



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

LIST OF EFFECTIVE PAGES (Continued)

<u>Title</u>	<u>Page</u>	<u>Date</u>
	8	Feb 28/03
	9	Feb 28/03
	10	Feb 28/03
TESTING AND FAULT ISOLATION.....	1001	Feb 28/03
	1002	Feb 28/03
	1003	Feb 28/03
	1004	Feb 28/03
	1005	Feb 28/03
	1006	Feb 28/03
	1007	Feb 28/03
	1008	Feb 28/03
	1009	Feb 28/03
	1010	Feb 28/03
	1011	Feb 28/03
	1012	Feb 28/03
	1013	Feb 28/03
	1014	Feb 28/03
	1015	Feb 28/03
	1016	Feb 28/03
	1017	Feb 28/03
	1018	Feb 28/03
SCHEMATICS AND WIRING DIAGRAMS.....	2001	Feb 28/03
	2002	Blank
DISASSEMBLY	3001	Feb 28/03
	3002	Feb 28/03
	3003	Feb 28/03
	3004	Feb 28/03
	3005	Feb 28/03
	3006	Feb 28/03
	3007	Feb 28/03
	3008	Feb 28/03
	3009	Feb 28/03
	3010	Feb 28/03
	3011	Feb 28/03
	3012	Feb 28/03
	3013	Feb 28/03
	3014	Feb 28/03
	3015	Feb 28/03
	3016	Feb 28/03

LIST OF EFFECTIVE PAGES

LIST OF EFFECTIVE PAGES (Continued)

<u>Title</u>	<u>Page</u>	<u>Date</u>
	3017	Feb 28/03
	3018	Feb 28/03
	3019	Feb 28/03
	3020	Feb 28/03
CLEANING	4001	Feb 28/03
	4002	Feb 28/03
	4003	Feb 28/03
	4004	Feb 28/03
CHECK	5001	Feb 28/03
	5002	Feb 28/03
	5003	Feb 28/03
	5004	Feb 28/03
	5005	Feb 28/03
	5006	Feb 28/03
	5007	Feb 28/03
	5008	Feb 28/03
	5009	Feb 28/03
	5010	Feb 28/03
	5011	Feb 28/03
	5012	Feb 28/03
	5013	Feb 28/03
	5014	Feb 28/03
	5015	Feb 28/03
	5016	Feb 28/03
	5017	Feb 28/03
	5018	Feb 28/03
	5019	Feb 28/03
	5020	Feb 28/03
	5021	Feb 28/03
	5022	Feb 28/03
	5023	Feb 28/03
	5024	Feb 28/03
	5025	Feb 28/03
	5026	Feb 28/03
	5027	Feb 28/03
	5028	Blank
REPAIR	6001	Feb 28/03
	6002	Feb 28/03
	6003	Feb 28/03
	6004	Feb 28/03
	6005	Feb 28/03
	6006	Feb 28/03

LIST OF EFFECTIVE PAGES



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

LIST OF EFFECTIVE PAGES (Continued)

<u>Title</u>	<u>Page</u>	<u>Date</u>
	6007	Feb 28/03
	6008	Feb 28/03
	6009	Feb 28/03
	6010	Feb 28/03
	6011	Feb 28/03
	6012	Feb 28/03
	6013	Feb 28/03
	6014	Feb 28/03
ASSEMBLY	7001	Feb 28/03
	7002	Feb 28/03
	7003	Feb 28/03
	7004	Feb 28/03
	7005	Feb 28/03
	7006	Feb 28/03
	7007	Feb 28/03
	7008	Feb 28/03
	7009	Feb 28/03
	7010	Feb 28/03
	7011	Feb 28/03
	7012	Feb 28/03
	7013	Feb 28/03
	7014	Feb 28/03
	7015	Feb 28/03
	7016	Feb 28/03
	7017	Feb 28/03
	7018	Feb 28/03
	7019	Feb 28/03
	7020	Feb 28/03
	7021	Feb 28/03
	7022	Feb 28/03
	7023	Feb 28/03
	7024	Feb 28/03
	7025	Feb 28/03
	7026	Feb 28/03
	7027	Feb 28/03
	7028	Feb 28/03
	7029	Feb 28/03
	7030	Blank
FITS AND CLEARANCES	8001	Feb 28/03
	8002	Feb 28/03
	8003	Feb 28/03
	8004	Feb 28/03

LIST OF EFFECTIVE PAGES



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

LIST OF EFFECTIVE PAGES (Continued)

<u>Title</u>	<u>Page</u>	<u>Date</u>
SPECIAL TOOLS, FIXTURES, AND EQUIPMENT	9001	Feb 28/03
	9002	Feb 28/03
	9003	Feb 28/03
	9004	Feb 28/03
	9005	Feb 28/03
	9006	Feb 28/03
	9007	Feb 28/03
	9008	Feb 28/03
	9009	Feb 28/03
	9010	Feb 28/03
	9011	Feb 28/03
	9012	Feb 28/03
	9013	Feb 28/03
	9014	Feb 28/03
	9015	Feb 28/03
	9016	Feb 28/03
	9017	Feb 28/03
	9018	Blank
ILLUSTRATED PARTS LIST	10001	Feb 28/03
	10002	Feb 28/03
	10003	Feb 28/03
	10004	Feb 28/03
	10005	Blank
	10006	Feb 28/03
	10007	Feb 28/03
	10008	Feb 28/03
	10009	Feb 28/03
	10010	Feb 28/03
	10011	Feb 28/03
	10012	Feb 28/03
	10013	Feb 28/03
	10014	Feb 28/03
	10015	Feb 28/03
	10016	Feb 28/03
	10017	Blank
	10018	Feb 28/03
	10019	Feb 28/03
	10020	Feb 28/03
	10021	Feb 28/03
	10022	Feb 28/03
	10023	Feb 28/03
	10024	Blank

LIST OF EFFECTIVE PAGES



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

LIST OF EFFECTIVE PAGES (Continued)

<u>Title</u>	<u>Page</u>	<u>Date</u>
SPECIAL PROCEDURES	11001	Feb 28/03
	11002	Blank
REMOVAL	12001	Feb 28/03
	12001	Feb 28/03
	12002	Blank
INSTALLATION	13001	Feb 28/03
	13002	Blank
SERVICING	14001	Feb 28/03
	14002	Blank
STORAGE	15001	Feb 28/03
	15002	Feb 28/03
	15003	Feb 28/03
	15004	Blank
REWORK	16001	Feb 28/03
	16002	Blank



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

TABLE OF CONTENTS

<u>Title</u>	<u>Page</u>
EXCLUSIVE DISTRIBUTORS	ii
HIGHLIGHTS	1
RECORD OF REVISIONS	iii
RECORD OF TEMPORARY REVISIONS	v
SERVICE BULLETIN LIST	vii
SERVICE INFORMATION LETTER LIST	ix
LIST OF EFFECTIVE PAGES	xi
TABLE OF CONTENTS	xvii
LIST OF ILLUSTRATIONS	xix
LIST OF TABLES	xxiii
INTRODUCTION	xxv
DESCRIPTION AND OPERATION	1
TESTING AND FAULT ISOLATION	1001
SCHEMATICS AND WIRING DIAGRAMS	2001
DISASSEMBLY	3001
CLEANING	4001
CHECK	5001
REPAIR	6001
ASSEMBLY	7001
FITS AND CLEARANCES	8001
SPECIAL TOOLS, FIXTURES, AND EQUIPMENT	9001
ILLUSTRATED PARTS LIST	10001
SPECIAL PROCEDURES	11001
REMOVAL	12001
INSTALLATION	13001
SERVICING	14001
STORAGE	15001
REWORK	16001



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

LIST OF ILLUSTRATIONS

<u>Figure</u>	<u>Title</u>	<u>Page</u>
1	Model 23079 Series DC Starter-Generator Features	1
2	Models 23079-006, -007 Schematic Diagram	6
2a	Models 23079-000, -000-1, -000-2, -002, -002-1, -003, -004, -005, -009, -010 Schematic Diagram	6
3	Outline Drawing 23079-000, -000-1, -000-2, -005, -009, -010	7
4	Outline Drawing 23079-002, -002-1	8
5	Outline Drawing 23079-003, -004	9
6	Outline Drawing 23079-006, -007	10
1001	Shunt Start Generator Test Connection Diagram, Closed Loop	1003
1002	Series Start Generator Test Connection Diagram, Closed Loop	1004
1003	Shunt Start Locked Rotor Test Connection Diagram	1004
1004	Series Start Locked Rotor Test Connection Diagram	1005
1005	Models 23079-006 and -007 Equalizing Voltage Acceptance Limits	1007
1006	All Models except 23079-006 and -007 Equalizing Voltage Acceptance Limits	1008
1007	Commutation Viewing Angle	1010
3001	Removing Terminal Block Cover	3003
3002	Removing Air Inlet Cover	3004
3003	Removing Air Inlet Adapter	3005
3004	Removing Brush Set	3006
3005	Removing Thermal Switch	3007
3006	Drive Shaft Removal	3008
3007	Dampener Hub Removal	3009
3008	Removing Dampener Backplate	3010
3009	Removing Anti-drive End End Bell Assembly	3011
3010	Removing Armature from Anti-Drive End End Bell Assembly	3012
3011	Removing Bearing from Anti-Drive End End Bell Assembly	3013
3012	Removing Bearing from Drive End of Armature Shaft	3014
3013	Removing Brush Holder Assemblies	3015
3014	Removing Brush Springs from Brush Holder Assembly	3016
3015	Disassembling Brush Holder Assembly	3016
3016	Removal of Thermal Switch Lead	3018
3017	Miscellaneous Hardware Removal	3019
3018	Removal of Terminal Block from Stator Housing	3020
5001	Inspecting Mounting Adapter	5006
5002	Inspecting Air Inlet	5007
5003	Inspecting Brush Access Cover	5008
5004	Inspecting Drive Shaft and Dampener Hub	5010
5005	Inspecting Friction Ring	5011
5006	Inspecting Dampener Plate	5012
5007	Inspecting Dampener Backplate	5012

LIST OF ILLUSTRATIONS



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

LIST OF ILLUSTRATIONS (Continued)

<u>Figure</u>	<u>Title</u>	<u>Page</u>
5008	Inspecting Anti-Drive End End Bell	5015
5009	Checking Brush Holder Assembly	5016
5010	Testing Brush Spring Pressure	5016
5011	Inspecting Drive End End Bell Assembly	5017
5012	Drive-End End Bell Assembly Ribs	5018
5013	Inspecting Armature	5021
5014	Inspecting Bearing Retainer	5022
5015	Inspecting Baffle Disk	5022
5016	Inspecting Stator and Housing Assembly	5025
6001	Machining Anti-Drive End End Bell after Plating	6005
6002	Machining the Drive End End Bell Assembly Bearing Liner	6006
6003	Polarizing the Field	6010
6004	Brush Access Cover Repair	6011
6005	Attaching Rivnut® Fastener	6012
6006	Complete Brush Holder	6013
6007	Brush Holder Rivet Alignment	6014
7001	Terminal Block Attachment	7004
7002	Brush Spring Installation	7005
7003	Brush Holder Attachment to the Anti-drive End End Bell	7007
7004	Baffle Disc and Ball Bearing Installation on Drive End of Armature Shaft ...	7008
7005	Spacer/Baffle Disc and Retaining Ring Installation on Armature Shaft	7009
7006	Ball Bearing Installation into Anti-drive End End Bell Assembly	7011
7007	Attaching the Bearing Retainer	7012
7008	Attaching Anti-drive End End Bell Assembly and Armature	7013
7009	Attaching Drive End End Bell to Stator and Housing Assembly	7014
7010	Determining Shim Requirements	7015
7011	Attaching Anti-drive End End Bell with Attached Armature to Stator and Housing Assembly	7017
7012	Installing Drive Shaft into Armature Shaft	7018
7013	Installing the Fan	7019
7014	Attaching the Thermal Switch to the Stator and Housing Assembly	7020
7015	Installing Brush Sets	7021
7016	Attaching the Stator and Brush Leads	7022
7017	Attaching the Air Inlet Adapter with Air Inlet Attached	7023
7018	Attaching the Air Inlet	7024
8001	Measuring Liner Diameter	8002
8002	Measures the Journal Diameter	8003
8003	Terminal Block Schematics	8003
9001	Commutation Viewing Adapter	9002
9002	Drive End Armature Shaft Adapter	9003
9003	Anti-Drive End Armature Shaft Adapter	9004

LIST OF ILLUSTRATIONS



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

LIST OF ILLUSTRATIONS (Continued)

<u>Figure</u>	<u>Title</u>	<u>Page</u>
9004	Dampener Hub Driver	9005
9005	PlusNut™ Fastener Header	9006
9006	Inner Race Bearing Driver	9007
9007	Outer Race Bearing Driver	9008
9008	Commutator Turning Fixture	9009
9009	Rivet Alignment and Rivet Press Fixtures	9010
9010	Armature Support	9011
9011	Anti-drive End End Bell Assembly Support	9012
9012	Horizontal Stator Support Fixture	9013
9013	Vertical Stator Support Fixture	9014
9014	Drive End Hub Support	9015
9015	Anti-drive End Hub Support	9016
9016	Spline Wrench	9017
10001	DC Starter-Generator Assembly, 23079 Series	10006
10002	Anti-Drive End Endbell Assembly	10014
10003	Drive End End Bell Assembly	10018
10004	Stator and Housing Assembly	10020
10005	Quick Attach/Detach Mounting Kit	10022



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

LIST OF TABLES

<u>Table</u>	<u>Title</u>	<u>Page</u>
i	Hazardous Materials	xxix
1	Equipment Specifications	2
2	Overhung Moment and Weight	3
3	Mounting Flange and Drive Shaft Specifications	4
1001	Performance Test Conditions	1001
1002	Test Equipment	1002
3001	Disassembly Tools	3001
4001	Cleaning Materials and Equipment	4001
4002	Cleaning Procedures	4002
5001	Inspection Tools	5001
5002	Terminology	5026
6001	Repair Tools	6001
6002	Repair Materials	6002
6003	Surface Coating Specifications	6009
7001	Assembly tools	7001
7002	Assembly Materials	7002
8001	Acceptance Limits	8001
8002	Torque Limits	8004
9001	Special Tools, Fixtures, and Equipment	9001



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

INTRODUCTION

1. Purpose

This Component Maintenance Manual (CMM) provides detailed instructions for the service and overhaul of Goodrich Corporation Model 23079 Series DC Starter-Generators. Line maintenance is supported by Maintenance Manual (MM) 23700.

A. Overhaul Requirements

DC Starter-Generator overhaul occurs during regular scheduled maintenance to examine and, where necessary, repair or replace damaged or worn parts and assemblies. After overhaul and after the starter-generator has passed all acceptance tests, the starter-generator is assigned zero operating hours time since overhaul (TSO). Selective equipment repairs, without overhaul, do not affect the TSO. A DC starter-generator overhaul includes:

- (1) Replacement of non-reusable parts such as bearings, brushes, O-rings, and locking hardware. Replace all damaged, deformed, corroded or apparently defective hardware.
- (2) Cleaning of subassemblies.
- (3) Detailed inspection of parts, electrical check of all electromagnetic parts (including insulation integrity check), Non-Destructive Test (NDT) inspections, and complete inspection testing.

NOTE: Magnetic particle inspection of the parts as specified in the CHECK section is only required when an overhaul of the generator assembly is being done.

- (4) Reconditioning of housing finishes as required.
- (5) Repair/rework of parts as required.
- (6) Recertification, including final assembly, records, and release tags.

B. Repair Requirements

An illustrated parts list is included at the back of this CMM for part identification and ordering of approved replacement parts. The structure and content of this manual is in general accordance with the requirements established by the Air Transport Association of America (ATA) for aircraft CMMs. This manual is formatted to conform to ATA Specification No. 100.

All weights and measures used in this manual are in English units, followed by the Système Internationale (SI) equivalent in parentheses.

Give careful attention to applicable warnings and cautions. Before starting any overhaul procedure, become thoroughly familiar with the capabilities and limitations of the equipment. Make sure all necessary safety equipment, test equipment, repair materials, special tools, and fixtures are available before starting.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

If an error, omission, or other technical discrepancy is found in this CMM, please fill out a copy of the Technical Publication Comment Form found at the back of this CMM and send to Technical Publications Supervisor at Goodrich Corporation or provide information to Technical Publications e-mail at aurora.techpubs@goodrich.com.

2. User Qualification and Certification Requirements

The information contained in this CMM is intended for use by persons trained and certified in the repair and overhaul of aircraft electrical accessories. Eligibility and certification of overhaul technicians shall be in accordance with the guidelines established by the U.S. Federal Aviation Administration or an equivalent regulatory authority. Refer to U.S. Code of Federal Regulations 14 CFR, parts 65.101 through 65.103 and part 145.39.

Where special processes are included by reference to military or commercial standards, the qualifications and certification of personnel performing the process shall be the same as indicated in the cited standard. Make sure to follow the detailed requirements of all cited military or commercial standards used in the repair of this equipment.

3. Quality Assurance Requirements

The repair station's quality assurance activity is responsible for the correct performance of all tests and inspections specified in this CMM. The quality assurance activity shall also be responsible for maintaining all necessary tests, inspection, and maintenance records for each unit received for service.

All instrumentation and inspection equipment used shall be calibrated and controlled in accordance with ISO 10012-1 with all standards traceable to the National Bureau of Standards or an equivalent standards regulatory authority.

The Quality Assurance activity shall be responsible for certifying that personnel, skills, and material meet the requirements of the work to be performed. Components of the Starter-Generator undergoing overhaul, that are recovered as products of disassembly, shall be examined 100% to determine serviceability.

Documented evidence shall be maintained by the Quality Assurance activity that specifications applicable to special processes such as soldering, non-destructive testing, plating, etc. have been complied with during overhaul and/or repair of the starter-generator.

4. Safety Advisory

This CMM describes physical and chemical processes, which require the use of chemicals, or other commercially available materials that need precautionary attention.

Warnings, Cautions and Notes are used throughout the text to highlight and emphasize important points, as they become necessary. Warnings give information that must be followed precisely to avoid personal injury and/or possible death. Cautions contain information that must be followed to avoid damage to equipment. Notes assist the reader and make the technician's job easier.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

5. Non-Goodrich Corporation, Authorized Components and Processes Policy

Goodrich Corporation authorizes the use of genuine Goodrich Corporation spare parts which meet stringent engineering design specifications and quality standards, and have traceability to having been procured and certified to these specifications by Goodrich Corporation Quality Assurance incoming and in-process inspection systems. The Goodrich Corporation Customer Service Centers are the only authorized distributors of genuine Goodrich Corporation replacement parts and complete units.

All repair and overhaul facilities are obligated to provide the FAA, or any other in-county air authority, with proper traceability documentation indicating approval of all spare parts, materials and processes to ensure configuration compliance and continued air worthiness.

The use of any non-Goodrich Corporation unauthorized parts, or any parts not having been submitted to the Goodrich Corporation Quality Assurance inspection system will invalidate any and all factory warranties. All Goodrich Corporation warranties are automatically voided on any Goodrich Corporation designed unit that has been modified by the installation of any unauthorized parts, materials or unapproved processes supplied by other outside services. The repair station's quality assurance activity shall assume product liability for all units that have been modified in this fashion.

Damage resulting from the use of non-Goodrich Corporation replacement parts, materials or processes is not covered by the Goodrich Corporation warranty or service policy for any product or for any application.

6. Abbreviations

The following is a list of the abbreviations & acronyms used in this document.

A	Ampere
AC	Alternating Current
A/D	Analog to Digital
AR	As Required
ASC	Auxiliary Start Connector
CMM	Component Maintenance Manual
D/A	Digital to Analog
DC	Direct Current
FAA	Federal Aviation Administration
GCS	Generator Control Switch
GCU	Generator Control Unit
Hz	Hertz



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

IAW	In Accordance With
ISO	International Standardization Organization
KPa	Kilopascals
L	Liter
Lbs	Pounds
LRU	Line Replaceable Unit
MIL	Military Specification
MS	Millisecond
ND	Non-Destructive
O/C	Overcurrent
OSHA	Occupational Safety Health Administration
POR	Point of Regulation
PP	Peak to Peak
PSIG	Pounds per square inch gauge
PWB	Printed Wire Board
RMS	Root/Mean/Square
RPM	Revolutions Per Minute
SI	Systems Internationale
SPD	Standard Process Document
TIR	Total Indicator Runout/Total Indicator Reading
TSO	Time Since Overhaul
V	Volt or Voltage
VAC	Voltage Alternating Current
VDC	Voltage Direct Current
Vcc	Voltage, Power Supply
Ω	Ohm



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

7. Materials List

The materials listed in this section are required for processes throughout this manual. Warnings and/or cautions will proceed their use.

WARNING: BEFORE HANDLING ANY OF THE FOLLOWING MATERIALS, BE AWARE OF ALL HANDLING, STORAGE, AND DISPOSAL PRECAUTIONS RECOMMENDED BY THE MANUFACTURER OR SUPPLIER. FAILURE TO COMPLY WITH MANUFACTURER OR SUPPLIER RECOMMENDATIONS CAN RESULT IN SERIOUS INJURY, PHYSICAL DISORDER, OR DEATH.

Material	Used in...
Isopropyl Alcohol	CLEANING, CHECK, REPAIR, ASSEMBLY
Electrical Resin	REPAIR
Lubricating Paste	ASSEMBLY
Chemical Film Dichromate	REPAIR
Silicone Resin Conformal Coating	REPAIR
Ensolv TM	REPAIR, CLEANING

Table i - Hazardous Materials



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

DESCRIPTION AND OPERATION

1. Equipment Description

The 23079 Series DC Starter-Generator is an air-cooled DC Starter-Generator manufactured by Goodrich Corporation. The generator is designed for cable interface with a Goodrich Corporation Generator Control Unit (GCU) and a 28 volt DC control bus.

The 23079 Series DC Starter-Generator (See Figure 1) consists of a rotor assembly enclosed by and rotating in a stator and housing assembly. The rotor assembly rotor shaft is hollow and contains a spline-fitted generator drive shaft attached at the anti-drive end of the generator. The rotor assembly is supported by bearings mounted in the drive end bearing support and anti-drive end bearing support assembly which is part of the stator and housing assembly.

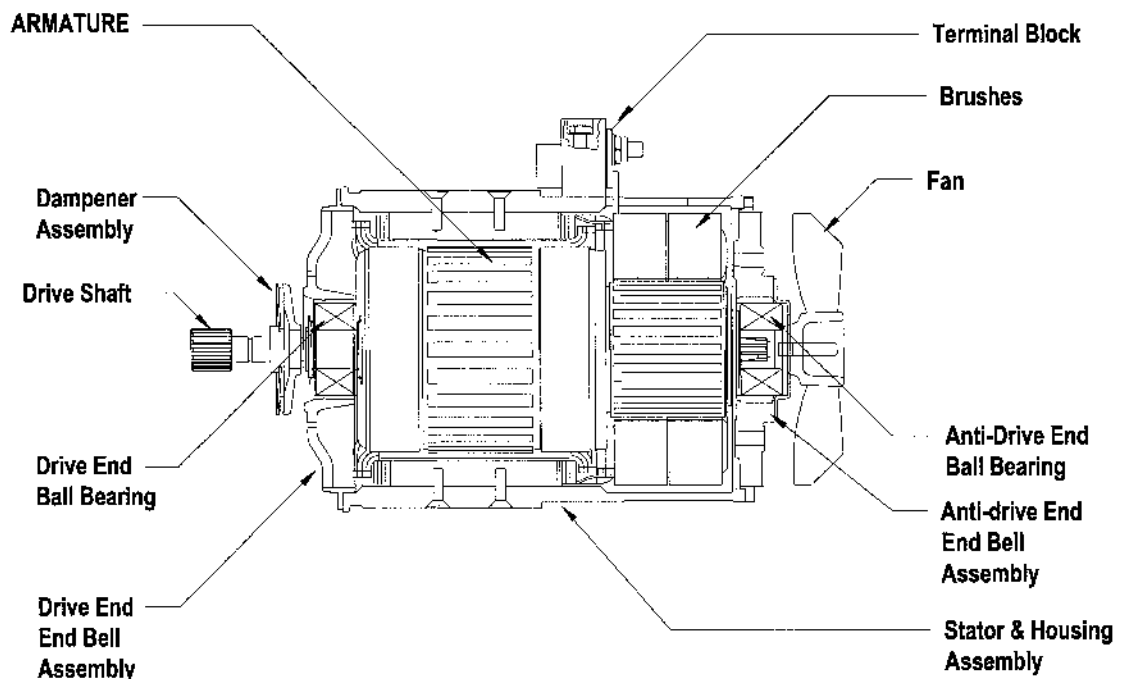


Figure 1 - Model 23079 Series DC Starter-Generator Features



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

At the drive end, a clutch-type dampener assembly absorbs torsional vibration generated by changes in engine gearbox speed and electrical load conditions. The dampener consists of a hub, backplate, friction ring, and dampener plate. A fan, installed on the anti-drive end of the drive shaft, draws in air to permit self-cooled operation at all rated conditions.

The unit includes electrical sensing devices for feeder fault, over-current, speed, and bearing failure. These devices interact with the GCU to protect and control power circuits and equipment. The generator's electrical rating is 30 volts DC at operating speeds from 7700 to 12175 rpm under a maximum 300A continuous load condition. A detailed listing of the 23079 Series DC Starter-Generator specifications follows in Table 1.

A quick-attach-detach (Q.A.D.) kit is required for installation but is not part of the generator configuration. The Q.A.D. kit consists of a mounting adapter, which attaches to the engine gearbox accessory drive pad, and a rim clenching clamp that secures the generator to the mounting adapter. All parts of the Goodrich Corporation 23079 Series DC Starter-Generator, including the Q.A.D. kit, are found in the ILLUSTRATED PARTS LIST.

2. Equipment Specifications

Details of equipment specifications are contained in this section

Characteristic	Model Number(s)	Specification
Rated Power Output	All models	9 kW
Rating for continuous load within speed range	All models	300A, 30 VDC, 7,700 to 12,175 RPM
External starting power supply limitations	23079-000 23079-002 23079-003	1,000 A maximum.
	23079-006 23079-007	30 V Max., 0.04 Ω resistor in starting circuit.
	23079-000-1 23079-000-2 23079-002-1 23079-004 23079-005 23079-009 23079-010	1,700 A maximum.
Direction of Rotation	All models	Counterclockwise (viewed from drive end)
Air Inlet Opening	23079-003 23079-004 23079-006 23079-007	3.00 in (76.2mm) diameter
	23079-000 23079-000-1 23079-000-2 23079-002 23079-002-1 23079-005 23079-009 23079-010	4.75 in (120.7 mm) diameter

Table 1 - Equipment Specifications



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

Characteristic	Model Number(s)	Specification
Cooling	All models	Self-cooled on ground. Combination cooling in flight.
Terminal Designations	23079-006 23079-007	Shunt Start: B+ Positive E- Negative A+ Shunt Field D Equalizer
	23079-000 23079-000-1 23079-000-2 23079-002 23079-002-1 23079-003 23079-004 23079-005 23079-009 23079-010	Series Start: B+ Positive E- Negative A+ Shunt Field D Equalizer C+ Starting

Table 1 - Equipment Specifications (Continued)

Characteristic	Model Number(s)	Specification
Starter-Generator Maximum Weight without QAD kit	23079-000-2	31.2 lbs. (14.15 kg) max.
	23079-006 23079-007	32.3 lbs. (14.53 kg) max.
	23079-000 23079-000-1 23079-002 23079-002-1 23079-003 23079-004	30.7 lbs. (13.81 kg) max.
	23079-005 23079-010	31.5 lbs. (14.17 kg) max.
	23079-009	31.6 lbs. (14.22 kg) max.
	All Models	1.4 lbs. (0.63 kg) max.
Overhung Moment with QAD kit	23079-006 23079-007	135 in-lbs. (15.24 N·m) max.
	23079-000 23079-000-1 23079-000-2 23079-002 23079-002-1 23079-003 23079-004 23079-005	134 in-lbs. (15.13 N·m) max.
	23079-009	140 in-lbs. (15.82 N·m) max.
	23079-010	143 in-lbs. (16.16 N·m) max.

Table 2 - Overhung Moment and Weight



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

Characteristic	Model Number(s)	Specification
Mounting Flange & Drive Shaft Specification	23079-006 23079-007	S/G's Spline conforms to MS18056 (.75 in. nominal). Mounting flange conforms to MS18054.
	23079-000 23079-000-1 23079-000-2 23079-002 23079-002-1 23079-003 23079-004 23079-005 23079-009 23079-010	S/G's Spline and QAD Mounting Features conform to AND10262-XII and mate with AND10262-XII-A drive pad.
Drive Shaft	23079-006 23079-007	Shear Torque: 700 in-lbs. (79.1 Nm max.) Number of Teeth: 16 Pitch Diameter: 0.8000 in. (20.320mm)
	23079-000 23079-000-1 23079-000-2 23079-002 23079-002-1 23079-003 23079-004 23079-005 23079-009 23079-010	Shear Torque: 2000 in-lbs. (226 Nm max.) Number of Teeth: 16 Pitch Diameter: 0.8000 in. (20.320 mm)

Table 3 - Mounting Flange and Drive Shaft Specifications

3. Equipment Improvements and Modifications

The 23079 Series DC Starter-Generator may incorporate one or more modifications. A modification is indicated by a letter in the MOD status block on the nameplate or modification status label. Information regarding modifications are detailed in the service bulletins found in the SERVICE BULLETIN LIST in the front of this manual.

4. DC Starter-Generator Operation

Electromagnetic interaction between the armature and stator produces output torque during engine starting and DC electrical power during engine operation. Figures 2 and 2a are schematic diagrams for the 23079 Series DC Starter-Generator models.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

A. Starter Operation

Dependent upon the starter-generator model, the 23079 Series of DC Starter-Generator operates as either a shunt starter motor or a series starter motor to start the engine.

(1) Shunt Start

- (a) During the engine start cycle, the aircraft power bus applies DC voltage across terminals B and E which supplies current to the armature and stator windings. At the same time, a voltage is applied to shunt field terminals A and D from a Goodrich Corporation Generator Control Unit (GCU). The starting power can be supplied by aircraft batteries or a ground power source. Figure 2 is a schematic diagram for a typical shunt start DC starter generator.
- (b) During the engine start cycle as the rotational speed of the armature increases, the magnetic flux generated by the shunt field windings creates a back electromotive force (EMF) which opposes the voltage supplied by the starting power source. This causes the armature current and the starter output torques to decrease. A feature known as field weakening reduces the back EMF which improves the starter-generator performance by allowing the starter-generator to provide assisting torque at higher speeds.

(2) Series Start

During the engine start cycle the aircraft power bus applies DC voltage across terminals C+ and E- supplying current to the armature and stator windings. The starting power can be supplied by aircraft batteries or a ground source. Figure 2a is a schematic diagram of a typical series start DC starter-generator.

B. Generator Operation

Following starter operation, the GCU will build up output voltage from the generator residual voltage. During generator operation, the machine provides a DC output (terminals B and E) regulated by the generator shunt field current (terminal A). By quickly reacting to changes in generator operating conditions such as engine speed or electrical load, the GCU regulates the current level provided to the generator shunt field (terminal A). By regulating the shunt field current, the system voltage is maintained at a nominal 28 VDC.

NOTE: Figures 3 through 6 are outline drawings of the various 23079 Series models.

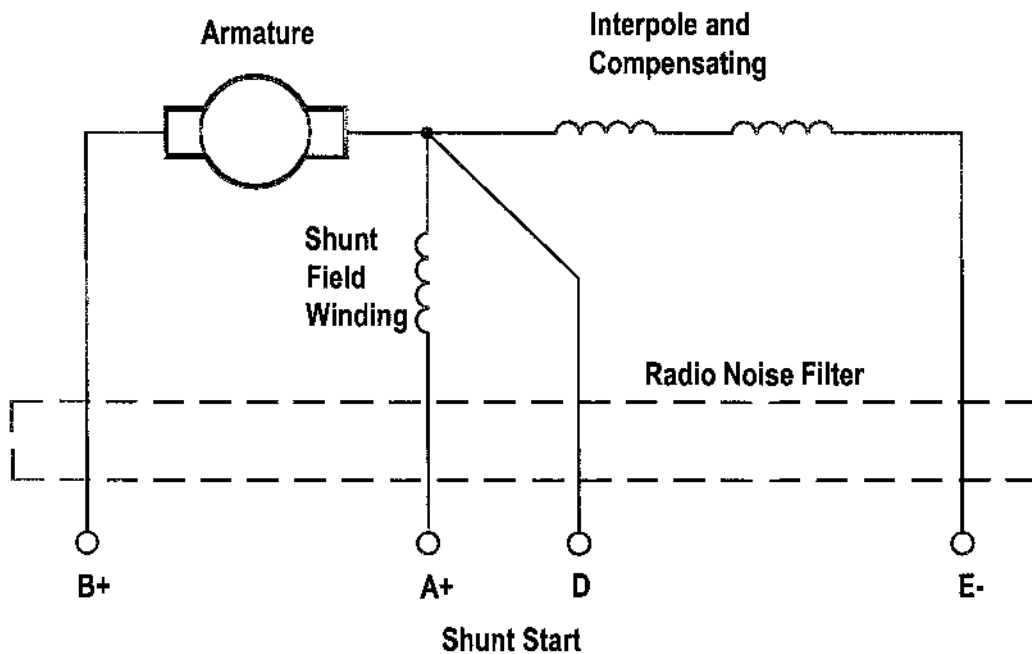


Figure 2 - Models 23079-006, -007 Schematic Diagram

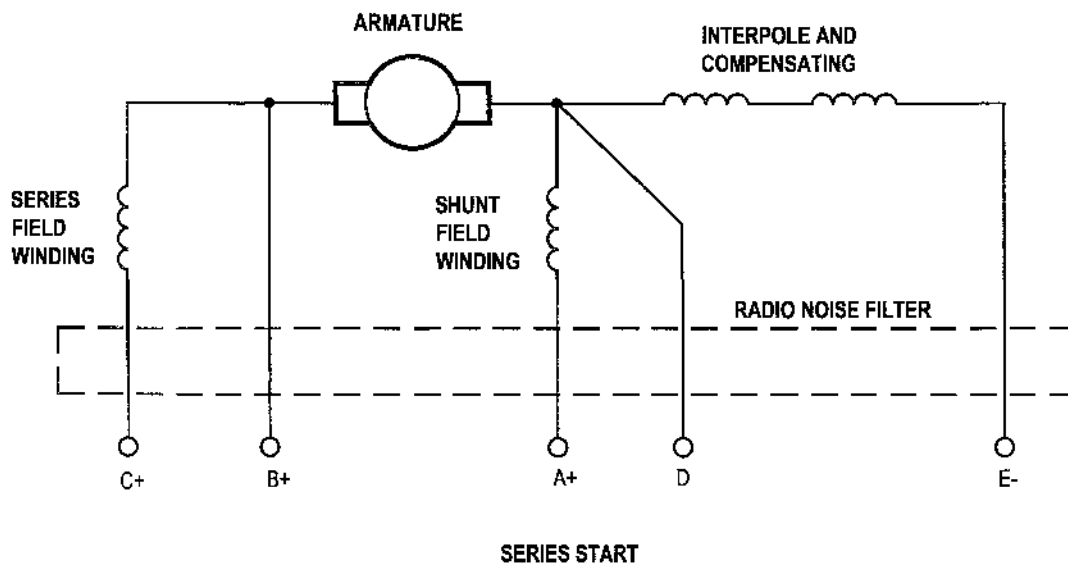


Figure 2a - Models 23079-000, -000-1, -000-2, -002, -002-1, -003, -004, -005, -009, -010
Schematic Diagram



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

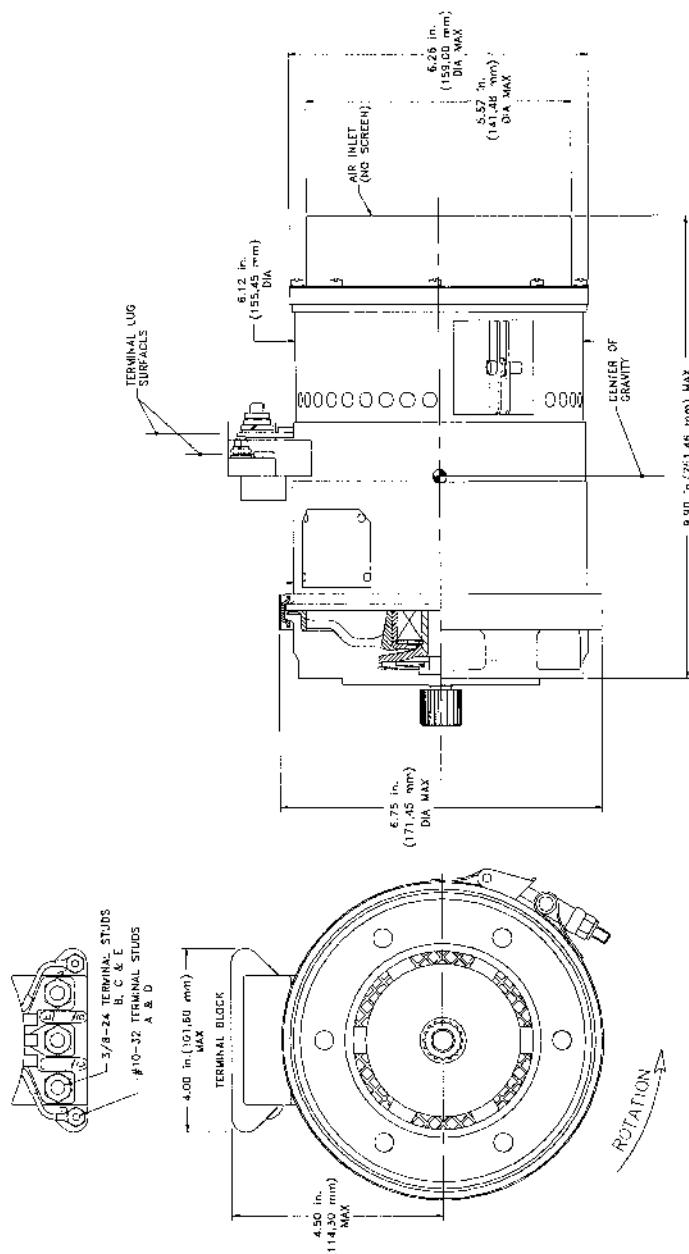


Figure 3 - Outline Drawing 23079-000, -000-1, -000-2, -005, -009, -010



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

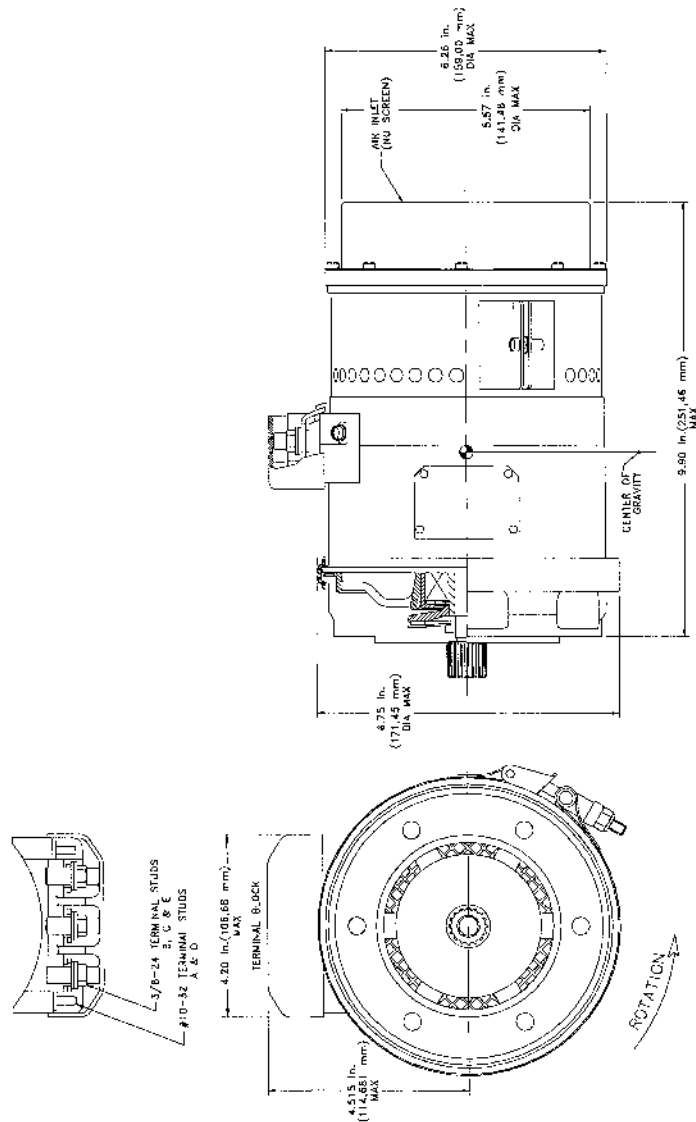


Figure 4 - Outline Drawing 23079-002, -002-1



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

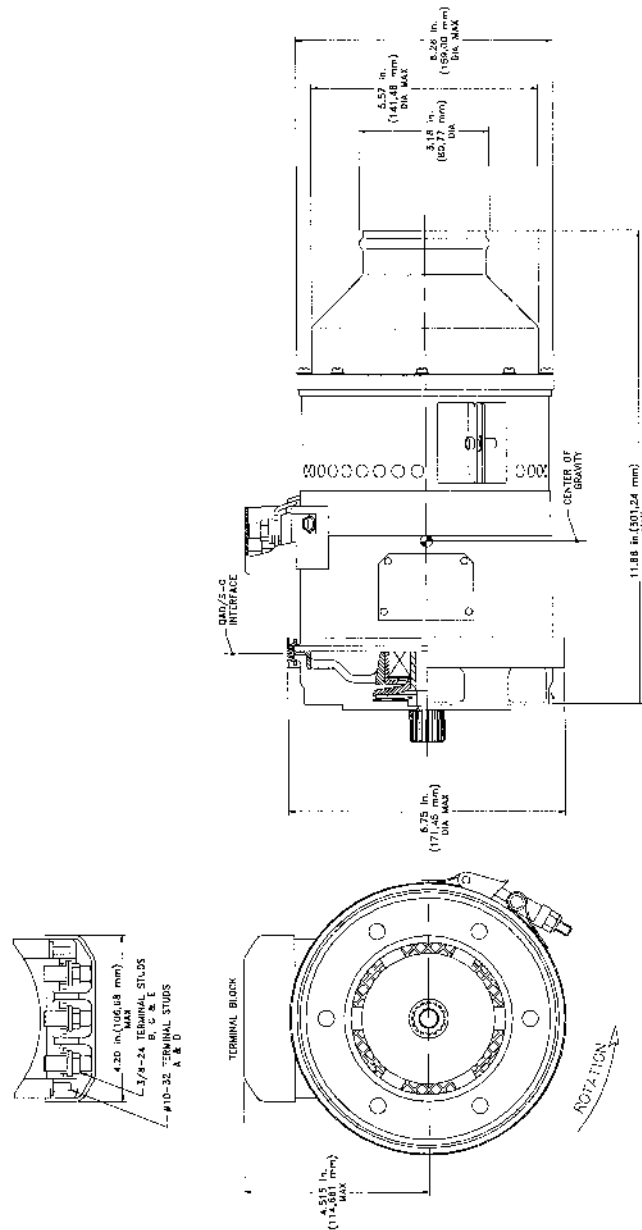


Figure 5 - Outline Drawing 23079-003, -004



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

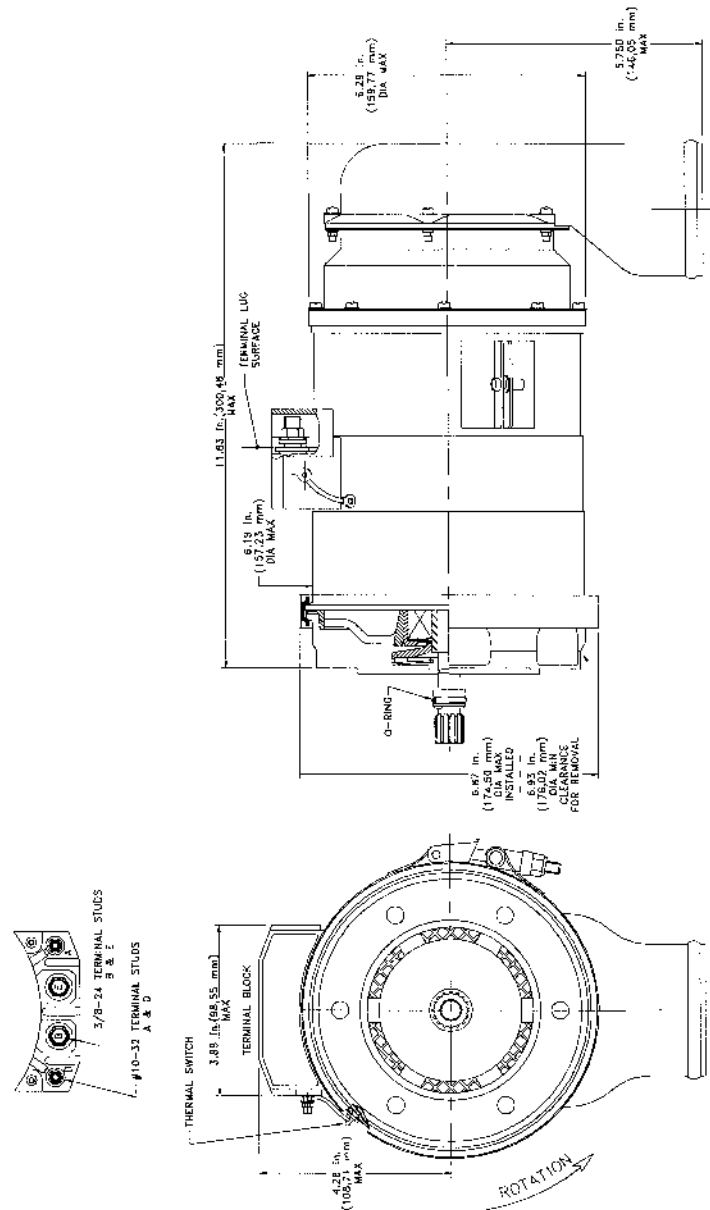


Figure 6 - Outline Drawing 23079-006, -007



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

TESTING AND FAULT ISOLATION

1. Introduction

WARNING: THE DC STARTER-GENERATOR CAN PRODUCE HIGH OUTPUT CURRENT CAPABLE OF CAUSING SEVERE SHOCK OR DEATH. ALL POWER MUST BE OFF BEFORE REMOVING OR REPLACING TEST EQUIPMENT, INSTRUMENTS OR ASSEMBLIES. TAKE EXTREME CARE WHEN PERFORMING "LIVE CIRCUIT" TESTING AND FAULT ISOLATION PROCEDURES.

Prior to performing any testing, confirm that the starter-generator is clean (Ref. CLEANING) and in good mechanical condition by inspection (Ref. CHECK).

The acceptance test constitutes a complete functional checkout of the starter-generator and should be performed in total if unit is disassembled, cleaned, and reassembled.

Verification Testing - Conducted to assist in fault isolation, confirm the cause for removal or qualify the unit for continued service.

Acceptance Testing - Conducted after overhaul or repair to confirm unit is ready to be returned to service and to verify minimum performance standards.

TSO - Only after a unit is overhauled and acceptance testing is completed (in the sequence given in this section) can zero operating hours "Time-Since-Overhaul" (TSO) be assigned.

Record all test results on a copy of the data sheet provided at end of this section.

2. Test Conditions

Unless otherwise specified, all of the following apply to a test.

CAUTION: IF AN ACCEPTANCE LIMIT IS EXCEEDED BY EVEN A SMALL MARGIN, DO NOT CONTINUE TESTING

Parameter	Operating Condition
Ambient Temperature	50° to 104°F (10° to 40°C)
Barometric Pressure	Ambient pressure at sea level, 27 to 33 in. (690 to 840 mm) of mercury
Load Current Tolerance	±5 Amps DC
Mounting	Drive shaft must be in horizontal position
Speed Tolerance	±50 RPM
Starter-Generator Mounting	Drive Shaft Horizontal
Terminal Voltage Tolerance	±0.2 VDC

Table 1001 - Performance Test Conditions

TESTING AND FAULT ISOLATION



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

Parameter	Operating Condition
Forced Cooling Air	Air pressure equal to 4" (10 cm) H ₂ O, measured 12" (30.5 cm) upstream of starter-generator air inlet.
	Air inlet duct to be straight, 3" dia. x 36" long (7.5 cm dia. x 91.5 cm long).
Brush Seating	100% in direction of rotation, 75% in axial direction. Refer to Goodrich Corporation Standard Practice Document SPD 1006 for Brush Installation, Seating, and Run-in procedures.

Table 1001 - Performance Test Conditions (Continued)

3. Test Equipment

Equivalent substitutes may be used for items listed. Use instruments with accuracy and calibration in compliance with ISO 10012-1, Part 1.

Equipment Type (Quantity)	Range and Accuracy or Equipment Rating	Ref. Des.
Commutation Viewing Adapter (1)	Special Tool (Figure 9001)	Not Illustrated
DC Ammeter(1)	Range: 0 to 10 Amps Accuracy: $\pm 1\%$ of reading	A ₁
DC Voltage Regulator	Range: 0 to 36 VDC	Not illustrated
DC Voltmeter (1)	Range: 0 to 750 mV Accuracy: $\pm 1\%$ of reading	V ₁
DC Voltmeter (2)	Range: 0 to 50 VDC Accuracy: $\pm 1\%$ of reading	V ₂ , V ₃
DC Voltmeter (1)	Range: 0 to 30 VDC Accuracy: $\pm 1\%$ of reading	V ₄
Generator Drive Stand (1)	Range: 5500 to 14,000 RPM at rated load and 15,000 at no load.	Not Illustrated
Generator Load Switch (1)	Rating: 30 V, 300 Amps	SW ₁
High Potential Tester (1)	Rating: 250 Volts RMS, at commercial frequency	Not Illustrated
Manometer (1)	Range: 0 to 10 in.(0 to 254 mm) H ₂ O Accuracy: $\pm 5\%$ of reading	Not Illustrated

Table 1002 - Test Equipment

Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

Equipment Type (Quantity)	Range and Accuracy or Equipment Rating	Ref. Des.
Oscilloscope, Dual Tone (1)	Digital: Bandwidth of 100 MHz, Sampling Rate of 100MS/Second Analog: Bandwidth of 60 to 100 MHz	Not Illustrated
Precision Shunt (1)	Rating: 500 Amps at 500 mVDC	SH ₁
Shunt Field Switch (1)	Rating: 30 VDC, 10 Amps	SW ₃
Terminal Block (1) (Non-serviceable)	Use a terminal block with blown capacitors during dielectric testing to avoid damaging a serviceable block.	Terminal Block
Thermometer (1) or equivalent temperature measuring device	Range: 65 to 300° F (18 to 150° C) Accuracy: $\pm 1^\circ$ F ($\pm 0.5^\circ$ C)	Not Illustrated
Variable Load Bank	Range: 30 VDC, 0 to 300 Amps	Variable Load Bank
Voltage Regulator Switch	Rating: 30 VDC, 10 Amps	SW ₂

Table 1002 - Test Equipment (Continued)

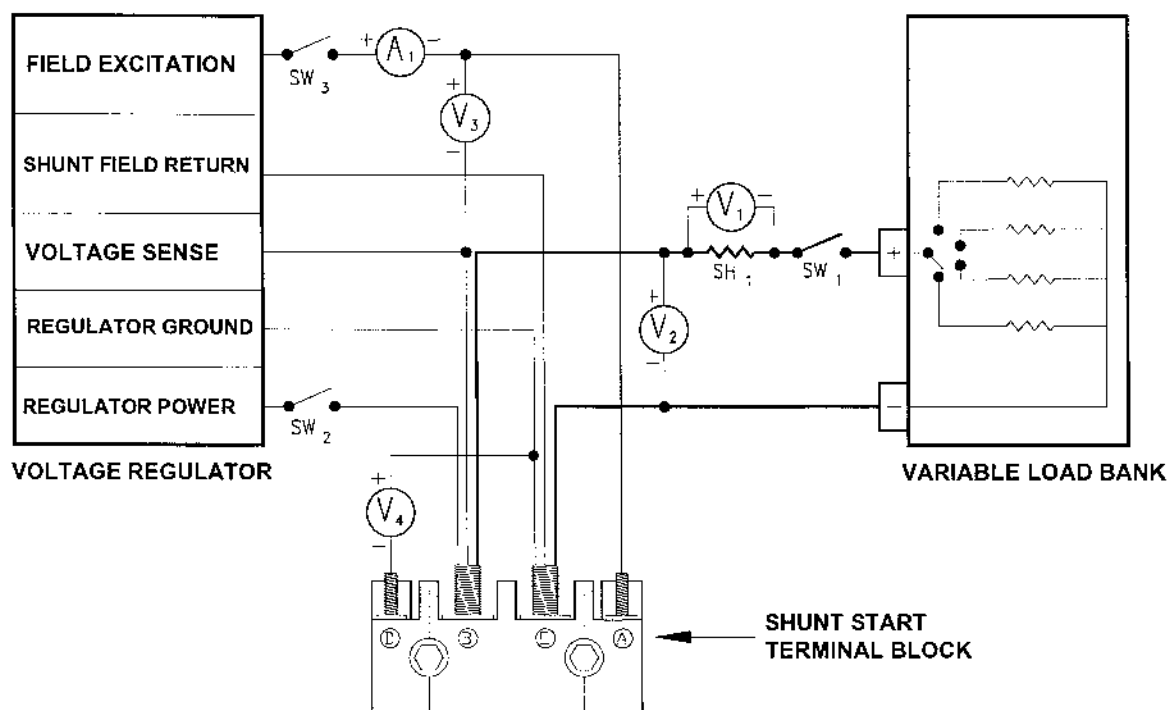
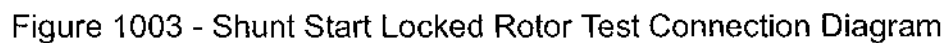


Figure 1001 - Shunt Start Generator Test Connection Diagram, Closed Loop



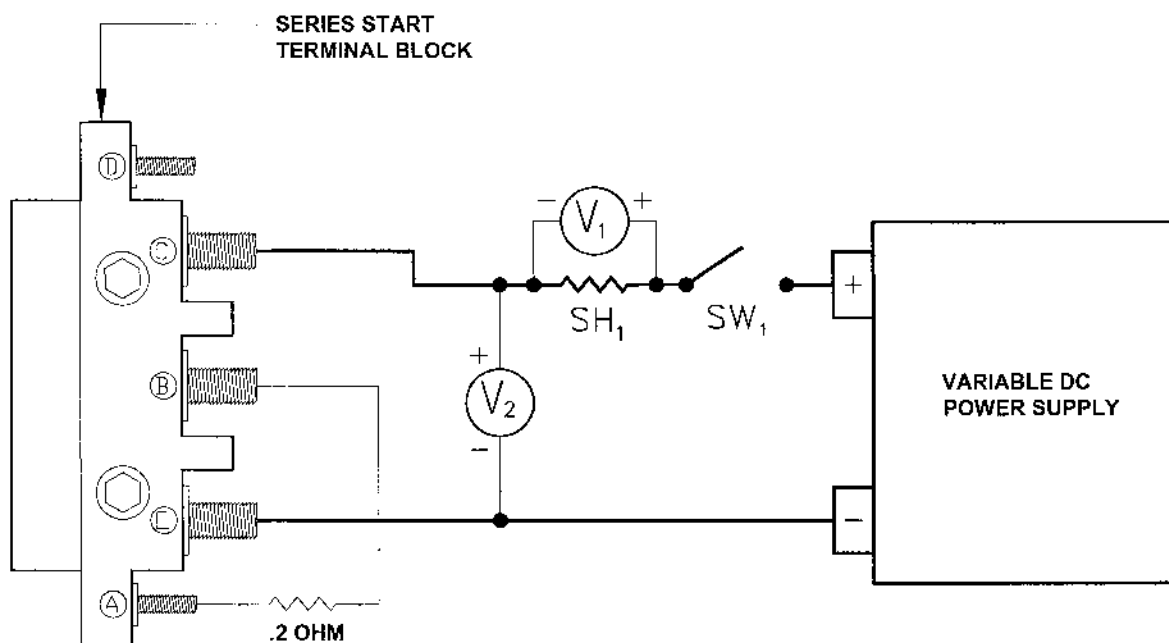


Figure 1004 - Series Start Locked Rotor Test Connection Diagram

4. **Test Set Up**

Unless otherwise specified, numbers in parentheses () refer to item numbers in Figure 10001 of the ILLUSTRATED PARTS LIST.

CAUTION: THE STARTER-GENERATOR MUST BE SUPPORTED AT ALL TIMES DURING INSTALLATION AND REMOVAL. DO NOT ALLOW THE UNIT TO HANG UNSUPPORTED. EXCESSIVE BENDING LOADS ON THE DRIVE SHAFT CAN DAMAGE THE SHEAR SECTION.

A. **Initial Inspection.**

- (1) Perform an initial inspection in accordance with (IAW) Paragraphs 3.A. through 4.A.(2) of the CHECK section. Work in a brightly lit work area to determine its overall condition.
- (2) Be sure that a dummy terminal block is attached to the stator and housing assembly (235) before proceeding with acceptance testing.

B. **Check each brush for proper brush seating.**

NOTE: Brushes installed in a starter generator being prepared for verification testing may or may not require brush seating.

Make sure all brushes (100) are correctly seated IAW procedures in the ASSEMBLY section.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

C. Install the commutation viewing adapter.

- (1) Remove air inlet (55) and, if applicable, air inlet adapter (70).
- (2) Remove brush access cover (85).
- (3) Install commutation viewing adapter

D. Install starter-generator on drive stand.

NOTE: The starter-generator mounts to a mounting adapter plate that is attached to the drive stand. A V-band clamp secures the starter-generator to the mounting adapter plate.

- (1) Install applicable spline adapter and mounting adapter plate onto drive stand.
- (2) Support anti-drive end of starter-generator, and align and install drive end to mounting adapter plate.
- (3) Ensure that drive stand and starter-generator mating splines are correctly engaged.
- (4) Install V-band clamp (10005-5) and torque to 70 in-lbs (7.9 N-m).

E. Connect starter-generator to electrical test circuit.

- (1) Turn all power off at starter-generator drive stand.
- (2) Connect starter-generator to test circuit as shown in Figures 1001 or 1002.
- (3) Assemble terminal block hardware to terminal block. See Paragraph 9.C. in ASSEMBLY section for instructions.

5. Acceptance Test Procedure

This section provides acceptance test procedures and the sequence in which they are performed.

CAUTION: IF AN ACCEPTANCE LIMIT IS EXCEEDED BY EVEN A SMALL MARGIN, DO NOT CONTINUE FURTHER TESTING OF STARTER-GENERATOR. BE AWARE THAT TESTING OF A DC STARTER-GENERATOR BY ANY LIMIT AFTER IT HAS EXCEEDED THAT LIMIT CAN DAMAGE THE STARTER-GENERATOR.

A. Maximum Speed for Regulation Test (Models 23079-006, -007 only)

- (1) Adjust starter-generator to 13,000 RPM, 30 VDC, with no load.

NOTE: Stabilization is not required.

- (2) Measure and record the voltage between terminals B and A (V_3) and the field current at Terminal A (A_1).

- (3) Calculate the resistance between terminals B and A by dividing the B and A voltage (V_3) by the field current (A_1).

The calculated external field circuit resistance must not be greater than 35 ohms.

B. Continuous Operating Speed and Equalizing Voltage Test

- (1) Install a thermocouple (or other temperature measuring device) to the outside of the stator and housing assembly in an area adjacent to a stator main pole.
- (2) Operate starter-generator at 12,175 RPM, 30 VDC at voltmeter V_2 and 300 A (V_1) until the temperature is stabilized.

NOTE: Stabilization is reached when the temperature of the shunt field winding, as determined by its resistance, or the temperature of the frame rises no more than 2° F (1° C) in five minutes.

- (3) Measure and record voltage between terminals D and E on voltmeter V_4 and the air inlet temperature.
 - (a) The voltage at voltmeter V_4 must be within voltage limits for the measured air inlet temperature. See Figures 1005 and 1006.

NOTE: Continue immediately with the minimum speed test.

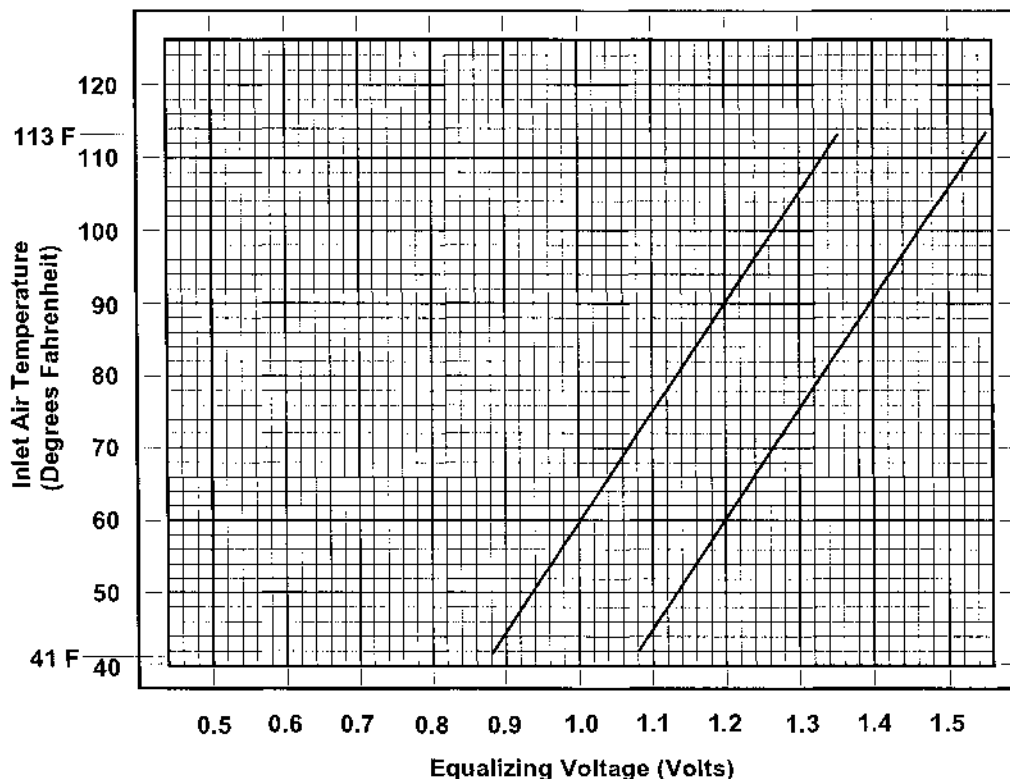


Figure 1005 - Models 23079-006 and -007 Equalizing Voltage Acceptance Limits

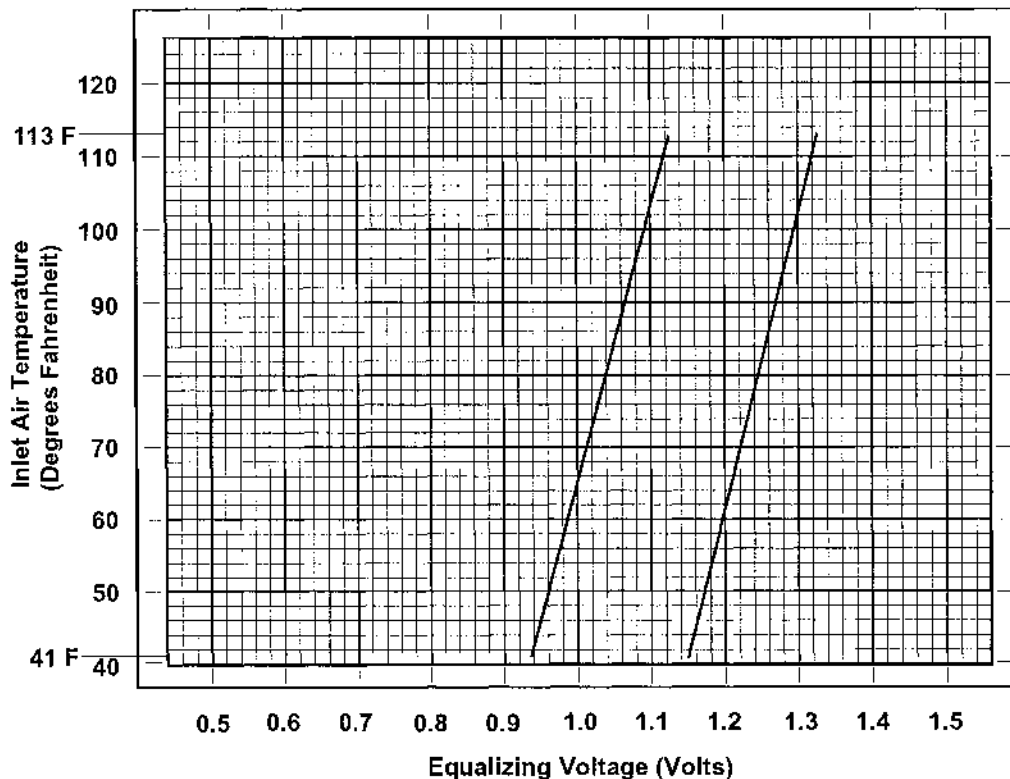


Figure 1006 - All Models except 23079-006 and -007 Equalizing Voltage Acceptance Limits

C. Minimum Speed for Regulation Test.

- (1) Adjust starter-generator to 7,700 RPM, 30 VDC, and 300 Amperes.

NOTE: Stabilization is not required.

- (2) Measure and record the voltage between terminals B and A (V_3) and the field current at Terminal A (A_1).

The field current must not be greater than 10 amperes.

- (3) Calculate the resistance between terminals B and A by dividing the B and A voltage (V_3) by the field current (A_1).

The calculated external field circuit resistance must not be less than 0.5 ohm.

D. Minimum Speed Test.

- (1) Disconnect the air supply tube.
- (2) Operate starter-generator to 7,200 RPM, 26 VDC, and 300 Amperes for 15 minutes.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

- (3) Measure and record the voltage between terminals B and A (V_3) and the field current at Terminal A (A_1).

The field current must not be greater than 10 amperes.

- (4) Calculate the resistance between terminals B and A by dividing the B and A voltage (V_3) by the field current (A_1).

The calculated external field circuit resistance must not be less than 0.5 ohm.

E. Overspeed Test

With starter-generator hot as a result of the previous test, operate at 14,000 RPM, no load with the field switch open for five minutes.

There must be no evidence of mechanical degradation (excessive noise, vibration, or loosening of parts).

F. Compounding (Models 23079-006, -007 only)

- (1) Operate starter-generator at 12,175 RPM, 30 VDC.
- (2) Apply loads of 0, 75, 150, 225, and 300 Amperes.
- (3) Measure and record the field current at Terminal A (A_1).

Field current must rise with the increasing load.

G. Commutation Check.

- (1) Operate starter-generator at 12,175 RPM, 30 VDC, and 300 Amperes.
- (2) Visually examine the condition of commutation.
 - (a) See Figure 1007 for proper viewing angle (Arrow A), Arrow B shows the viewing angle for determining whether the commutation is black or pin point.
 - (b) Acceptable commutation is defined as: Continuous pinpoint sparking not to exceed approximately 0.12 in. (3 mm) beyond the trailing edge of each brush with no more than 10 single sparks extending beyond 0.25 in. (6 mm) in any one minute period. Sparks beyond 0.25 in. (6 mm) must be considered "arcing".

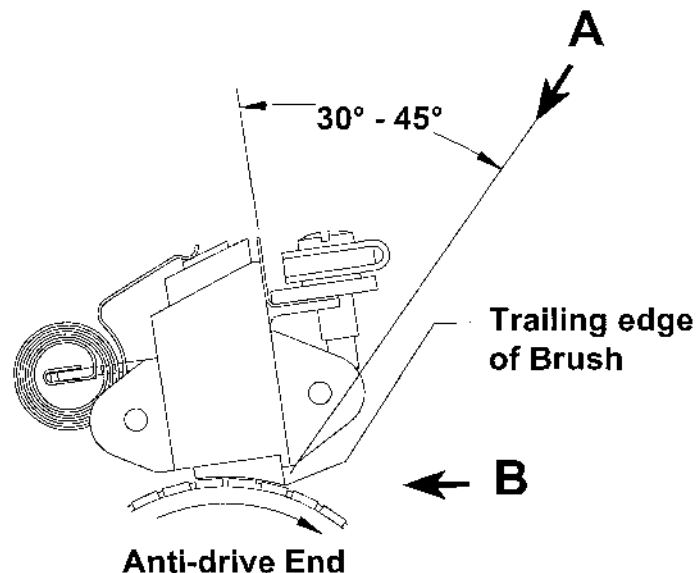


Figure 1007 - Commutation Viewing Angle

H. Dielectric Strength Test

CAUTION: DO NOT PERFORM DIELECTRIC TESTING DURING MAINTENANCE INSPECTION OR TROUBLESHOOTING OF THE UNIT AS IT MAY CAUSE CUMULATIVE DEGRADATION OF THE ELECTRICAL CIRCUIT INSULATION. THIS TEST IS RECOMMENDED ONLY FOR ACCEPTANCE TESTING OF AN OVERHAULED UNIT THAT HAS BEEN THOROUGHLY CLEANED.

WARNING: BEFORE USING HIGH VOLTAGE TESTER, MAKE CERTAIN TESTER HAS BEEN TURNED "OFF" AND THAT NO ONE IS TOUCHING ANY PORTION OF CONNECTION POINTS OR PROBE LEAD WIRES. FAILURE TO COMPLY WITH THIS WARNING CAN RESULT IN SEVERE ELECTRIC SHOCK TO THE OPERATOR OR ASSISTANTS.

While the starter-generator is still hot as a result of testing:

- (1) Turn OFF all power to test stand. Disconnect starter-generator from test circuit.
- (2) Remove commutator viewing adapter (Ref. Figure 9001).
- (3) Connect all terminals of dummy terminal block together.
- (4) Attach positive cable of high potential test unit to shorted terminal leads.
- (5) Attach negative cable of high potential test unit to frame of the starter-generator.
- (6) Turn high potential test unit power ON.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

CAUTION: INCREASE AND DECREASE TEST VOLTAGE SLOWLY (100 VOLTS PER SECOND, MAXIMUM). INCREASING AND DECREASING VOLTAGES TOO QUICKLY MAY CAUSE DAMAGE TO THE STARTER-GENERATOR.

- (7) Slowly increase high potential voltage from 0 to 250 VAC RMS at commercial frequency. Maintain test voltage for one minute. Measure and record leakage current in each circuit.

No arc over, flash over, or evidence of insulation breakdown is permitted during this test. Failure may be evidenced by fluctuations and/or a steady increase in leakage current. Leakage current greater than 5.0 mA constitutes failure.

- (8) Slowly reduce high potential voltage to zero and turn OFF power supply.

I. Starter No Load Speed and Radial Vibration Test.

NOTE: This test is applicable to all series start models except 23079-006 and -007.

- (1) Mount starter-generator in a resilient cradle (foam rubber cushioned) with drive shaft supported (centered). Reference Figure 9012.
- (2) Connect a 0.2 Ohm resistor between terminals A and B.
- (3) Apply 26 VDC between terminals C and E.
- (4) Record speed of starter-generator.

No load speed must be greater than 4,500 RPM.

- (5) Record amount of radial vibration deflection.

Total radial vibration must not exceed 0.001 in. (0.03 mm) deflection.

J. Shunt Start Locked Rotor Test.

NOTE: This procedure is applicable only to shunt start models 23079-006 and -007.

- (1) Rigidly mount starter-generator to starter test stand mounting flange.
- (2) Connect a 0.2 Ohm resistor between terminals A and B.
- (3) Connect a DC power supply, ammeter, and voltmeter to starter-generator terminals B and E as shown in Figure 1003.

CAUTION: SHUT POWER OFF IMMEDIATELY IF THE SUPPLY EXCEEDS 700 AMPERES OR 12.0 VDC WITH ARMATURE IN A LOCKED CONDITION.

- (4) With voltage output of DC power supply set to zero, turn power supply ON.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

CAUTION: DO NOT APPLY VOLTAGE FOR MORE THAN 3 SECONDS. THE STARTER- GENERATOR CAN BE DAMAGED BY HIGH TEMPERATURES CAUSED BY A HIGH LOCKED ROTOR CURRENT.

- (5) Quickly increase voltage between terminals B and E until output torque is 15.0 ft-lbs (20.3 Nm).
- (6) Turn DC power supply OFF.
- (7) Disconnect electrical test lead and remove starter-generator from test stand.
- (8) Record B to E terminal voltage and line current.
 - (a) The current is not to exceed 700 Amperes and the voltage must be less than or equal to 12.0 VDC
 - (b) If the current level is exceeded, repeat this test with the armature rotated to a different position.
 - (c) If the starter-generator fails this test a second time, refer to the fault isolation tables in this section.

K. Series Start Locked Rotor Test.

NOTE: This procedure is applicable only to series start models 23079-000, 23079-000-1, 23079-002, 23079-002-1, 23079-003, 23079-004, 23079-005, 23079-009 and 23079-010.

- (1) Rigidly mount starter-generator to starter test stand mounting flange.
- (2) Connect a 0.2 Ohm resistor between terminals A and B.
- (3) Connect a DC power supply, ammeter, and voltmeter to starter-generator terminals C and E as shown in Figure 1004.

CAUTION: SHUT POWER OFF IMMEDIATELY IF SUPPLY EXCEEDS 630 AMPERES OR 9.5 VDC WITH THE ARMATURE IN A LOCKED CONDITION.

- (4) With voltage output of the DC power supply set to zero, turn power ON.

CAUTION: DO NOT APPLY VOLTAGE FOR MORE THAN 3 SECONDS. THE STARTER- GENERATOR CAN BE DAMAGED BY HIGH TEMPERATURES CAUSED BY A HIGH LOCKED ROTOR CURRENT.

- (5) Quickly increase voltage between terminals C and E until output torque is 20.0 ft-lbs (27.1 Nm).
- (6) Record C to E terminal voltage and line current.
- (7) Turn DC power supply OFF.
- (8) Disconnect electrical test lead and remove starter-generator from the test stand.

TESTING AND FAULT ISOLATION



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

- (a) The current is not to exceed 630 Amperes and the voltage must be less than or equal to 9.5 VDC.
- (b) If the current level is exceeded, repeat this test with the armature rotated to a different position.
- (c) If the starter-generator fails this test a second time, refer to fault isolation tables in this section.

L. Commutator Runout Check.

CAUTION: RUN THIS CHECK ONLY DURING ACCEPTANCE TESTING AND ONLY FOR STARTER- GENERATORS THAT HAVE BEEN OVERHAULED AND HAVE AN ARMATURE WITH A RESURFACED COMMUTATOR.

CAUTION: DO NOT TOUCH THE POLISHED COMMUTATOR SURFACE WITH BARE HANDS. SKIN ACIDS AND OILS CAN CONTAMINATE CONDUCTING SURFACES, CAUSING CORROSION OR POOR ELECTRICAL CONTACT.

- (1) Firmly support starter-generator on an inspection plate.
- (2) Rotate armature on its own bearings.
- (3) Measure commutator runout with a dial indicator
- (4) Record maximum bar-to-bar runout between any two adjacent commutator bars.
 - (a) Runout must not exceed 0.0002 in. (0.005 mm) for any two adjacent bars.
 - (b) If this test is failed, refer to fault isolation tables in this section.
- (5) Record Total Indicator Runout (TIR) for the entire commutator.
 - (a) TIR must not exceed 0.0008 in. (0.020 mm).
 - (b) If this test is failed, refer to fault isolation tables in this section.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

6. Fault Isolation Tables

The following tables list faults that may occur during acceptance testing, performance testing, or when in service. For each fault, probable causes are listed along with related corrective actions. When a fault is detected, perform the corrective actions necessary to return the starter-generator to a serviceable condition. A complete acceptance test is required after the corrective action is accomplished.

NOTE: Total or partial starter-generator disassembly may be required for all or most of the tests described in this section.

Fault	Probable Cause	Corrective Action
A. Continuous Operating Speed and Equalizing Voltage Test		
High equalizing voltage (V_4).	Cooling air flow is low.	Check entire air flow path for obstructions. Remove any obstructions if present and re-test starter-generator.
	The armature is shorted or grounded	Clean the armature IAW procedure in the CLEANING section. Dielectric test components of armature assembly IAW Paragraph 8.B. Replace rotor assembly if it fails dielectric test.
	The stator and housing assembly is grounded or shorted	Clean stator and housing assembly IAW procedure in the CLEANING section. Dielectric test stator and housing assembly IAW Paragraph 8.C. Replace stator and housing assembly if it fails the dielectric test.
	The brushes are incorrectly seated or installed.	Repeat test and view commutation. If commutation is more than pinpoints, seat brush again according to the procedure in the ASSEMBLY section.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

Fault	Probable Cause	Corrective Action
High equalizing voltage (V_4) - Continued	One of the brush leads is loose.	Remove brush cover. Inspect all brush leads for a loose connection. Secure lead to brush holder if a loose lead is found and there is no evidence of arcing. Test starter-generator again.
	One of the brush circuits is open.	Remove brush cover. Inspect all brush leads for an open circuit. REPLACE defective brush (new brush only) if an open lead is found. If brushes are not new, all brushes must be replaced as a set. Perform brush run-in IAW procedure in ASSEMBLY section. Test starter-generator again.
B. Output voltage reversed.	Field is incorrectly polarized.	Polarize field IAW procedure II in REPAIR section.
C. Minimum Speed for Regulation Test		
The field current limit is exceeded (A_1).	An excessive load was applied during testing.	Check applied load at load bank. Apply correct load
	Stator windings are shorted or grounded.	Clean stator and housing assembly IAW procedure in CLEANING section. Dielectric test stator and housing assembly IAW Paragraph 8.C. Replace stator and housing assembly if it fails the dielectric test.
	Armature is shorted or grounded	Clean armature IAW procedure in CLEANING section. Dielectric test components of armature assembly IAW Paragraph 8.B. Replace armature if it fails the dielectric test.
D. Overspeed Test		
Noisy Operation.	Bearings are defective and/or installed incorrectly.	Replace bearings.

TESTING AND FAULT ISOLATION



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

Fault	Probable Cause	Corrective Action
Noisy Operation. (continued)	Cooling fan blades rubbing or out of balance.	Check fan blades and air inlet. Replace cooling fan if damaged. Repair or replace air inlet if damaged. NOTE: There are no authorized procedures to repair or balance a damaged fan.
	Drive shaft spline is worn.	Disassemble drive shaft from starter-generator. Inspect drive shaft IAW Paragraph 8.I. of the CHECK section. Repair or replace drive shaft as necessary.
	Rotor assembly striking against the stator and housing assembly.	Disassemble starter-generator. Check both parts for physical damage. Repair or replace damaged parts as necessary.
Starter-generator vibrates.	Bearings are defective and/or installed incorrectly.	Replace bearings.
	Armature assembly out of balance.	Disassemble starter-generator. Check armature assembly balance. Rebalance armature assembly IAW paragraph 9 of REPAIR section.
E. Dielectric Strength Test		
Leakage current fluctuation, increase and/or over limit.	Stator and housing assembly grounded or defective.	Clean stator and housing assembly IAW procedures in CLEANING section Dielectric test stator and housing assembly IAW Paragraph 8.C. Replace stator and housing assembly if it fails the dielectric test.



**Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series**

VERIFICATION / FINAL ACCEPTANCE DATA SHEET

Model Number: 23079 Inspected by: _____
 Serial Number: _____ Date: _____
 Repair Shop: _____

Inspection/Test	Requirement/Limit	Meas.	Accept	Reject
Visual inspection	Check for physical damage			
Maximum Speed for Regulation	Calculated B to A resistance (0.35 Ohm Max.)	Ohms		
Continuous Operating Speed and Equalizing Voltage Test	Air Inlet temperature	°F		
	D to E voltage (See Fig. 1005) (Models 23079-006 and -007)	VDC		
	D to E voltage (See Fig. 1006) (All other models)	VDC		
Minimum Speed for Regulation	B to A Voltage (V_2)	VDC		
	Field current (A_1) (10.0 amps Max.)	Amps		
	Calculated B to A resistance (0.5 Ohm Min.)	Ohms		
Minimum Speed Test	B to A voltage (V_2)	VDC		
	Field current (A_1) (10 amps Max.)	Amps		
	Calculated B to A resistance (0.5 Ohm Min.)	Ohms		
Overspeed Test	No sign of electrical or mechanical failure			
Compounding	Field current rises with increasing load			
Commutation Test	Not to exceed pinpoints			
Dielectric Test	Leakage 5 mA Max.	mA		
Locked Rotor Model 23079-006 and 23079-007	B-E voltage (12 VDC Max.)	VDC		
	Line current (700 Amps Max.)	Amps		
Locked Rotor - All other models	C-E voltage (9.5 VDC Max.)	VDC		
	Line current (630 Amps Max.)	Amps		

TESTING AND FAULT ISOLATION



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

Inspection/Test	Requirement/Limit	Meas.	Accept	Reject
Commutator Runout Test	Bar-to-bar (0.0002 in. Max.)	Inch		
	(0.0051 mm Max.)	mm		
	Total indicated runout (0.0008 in. Max.)	Inch		
	(0.0203 mm Max.)	mm		



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

SCHEMATICS AND WIRING DIAGRAMS

1. Introduction

This section is unassigned.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

DISASSEMBLY

1. Introduction

This section provides disassembly procedures for the 23079 DC Starter-Generator. No further disassembly beyond that given in this section is necessary unless further disassembly is indicated in the CHECK or the TESTING AND FAULT ISOLATION sections.

NOTE: Do not open or desolder permanent connections unless you need to replace worn, damaged, or defective parts

2. Disassembly Tools

Other than normal shop tools, those listed in Table 3001 are needed for starter-generator disassembly.

Tools	Reference
#40 Drill Bit	Commercially Available
Anti-drive End Hub Support	Figure 9015
Armature Shaft Adapter, Anti-drive End	Figure 9003
Armature Shaft Adapter, Drive End	Figure 9002
Armature Support	Figure 9010
Anti-drive End End Bell Assembly Support	Figure 9011
Bearing Puller	Commercially Available
Dampener Hub Driver	Figure 9004
Drive End Hub Support	Figure 9014
Horizontal Stator Support	Figure 9012
Inner Race Bearing Driver	Figure 9006
Outer Race Bearing Driver	Figure 9007
Spline Wrench	Figure 9016
Vertical Stator Support	Figure 9013
Wire Hook Tool	Not Illustrated

Table 3001 - Disassembly Tools

3. Disassembly of Starter-Generator

Numbers in parentheses () refer to item numbers in Figure 10001 of the ILLUSTRATED PARTS LIST, unless otherwise specified.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

A. Remove the Q.A.D. mounting kit from the starter-generator if present.

- (1) Loosen the self-locking hex nut (10005-10) from the T-bolt (10005-15) and remove the V-band clamp (10005-5).
- (2) Remove the mounting adapter (10005-20).

B. Remove nameplate (10), modification status label (30) (if present), and caution label (35) from stator and housing assembly (235).

NOTE: Do not remove these items unless they are damaged or unreadable.

- (1) Remove and discard the four screws (15) that attach the nameplate to the stator and housing assembly.
- (2) Remove the nameplate from the stator and housing assembly. Keep it for reference when transferring the information to a replacement nameplate.
- (3) Remove the modification label (if present) from the stator and housing assembly. Keep it for reference when transferring the information to a replacement modification status label or nameplate.
- (4) Remove and discard the caution label from the stator and housing assembly.

C. Remove information plate (20) from stator and housing assembly.

NOTE: For models 23079-006 and 23079-007 only. Do not remove this item unless it is damaged or unreadable.

- (1) Remove and discard two drive screws (25) that attach the information plate to the stator and housing assembly.
- (2) Remove the information plate. Keep the information plate for reference when transferring the information to a replacement plate.

D. Remove terminal block cover (45) from starter-generator if present.

NOTE: For models 23079-002, -002-1, -003, -004, -006, and -007 only.

- (1) Remove two screws (50) that attach the terminal block cover to the terminal block. (See Figure 3001.)
- (2) Remove the terminal block cover by pulling it off the terminal block.

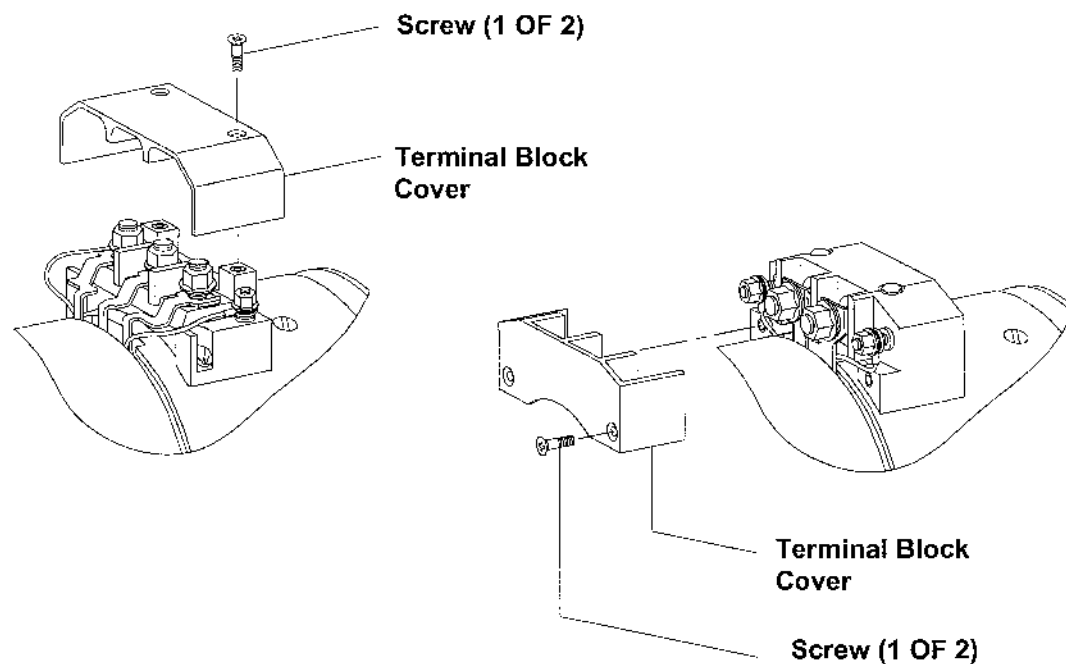


Figure 3001 - Removing Terminal Block Cover

E. Remove air inlet (55) from starter-generator.

- (1) Remove the screws (60) and flat washers (65) that attach the air inlet to the starter-generator. (See Figure 3002.)
- (2) Remove air inlet.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

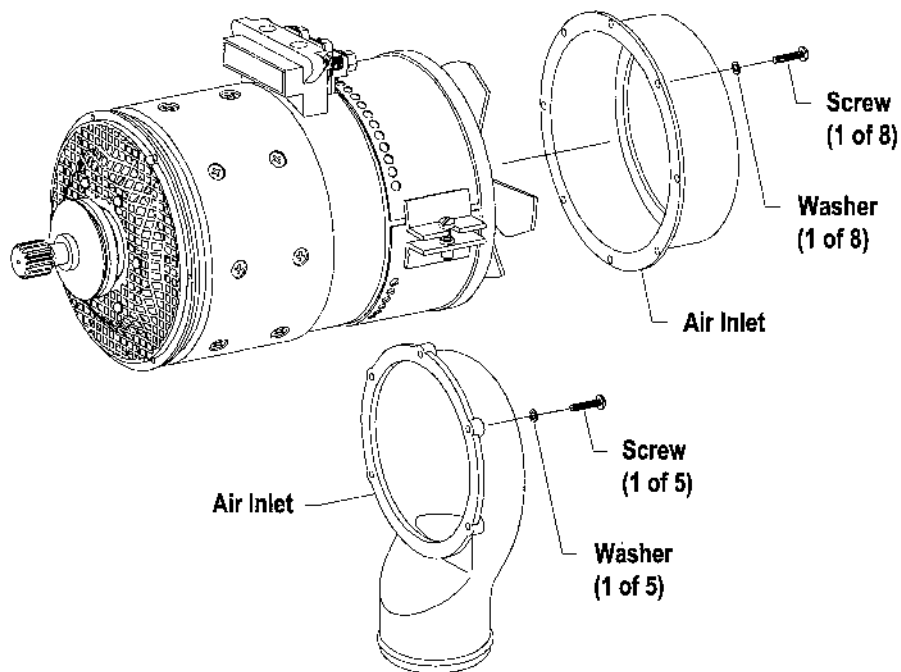


Figure 3002 - Removing Air Inlet Cover

F. Remove air inlet adapter (70) from starter-generator (Models 23079-006, -007 only).

- (1) Remove the screws (75) and flat washers (80) that attach air inlet adapter to starter-generator. (See Figure 3003.)
- (2) Remove air inlet adapter.

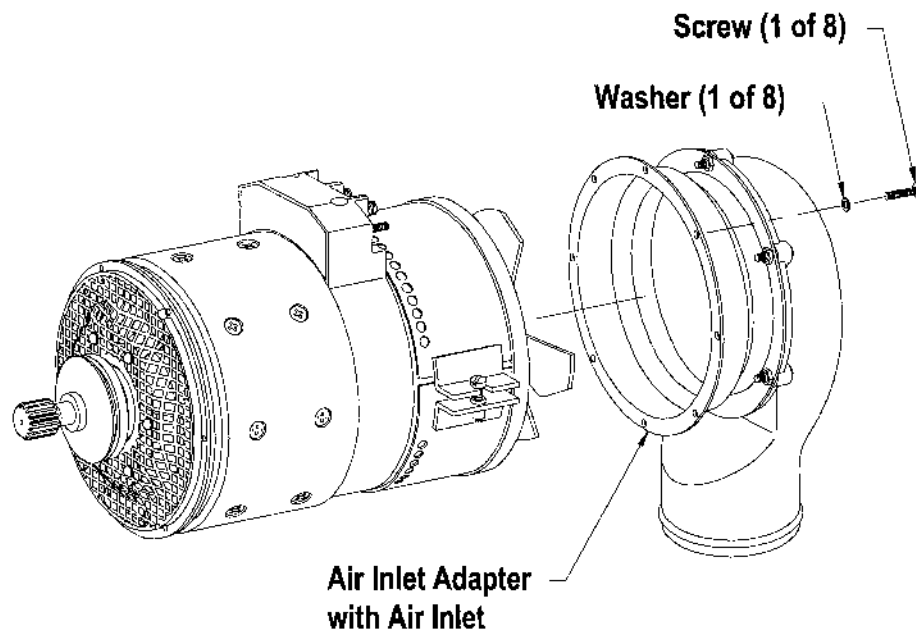


Figure 3003 - Removing Air Inlet Adapter

G. Remove brush access cover (85) from starter-generator.

- (1) Remove or sufficiently loosen screw (95) that secures the brush access cover to the stator and housing assembly.
- (2) Remove brush access cover.

H. Remove the brushes (100) from anti-drive end end bell assembly (160).

NOTE: Identification of brush holder assemblies and brush sets is not necessary during overhaul since all used brushes are discarded.

- (1) Remove the screw (105) that secure the brush leads to the brush holder assembly.

CAUTION: RAISE AND LOWER THE BRUSH SPRINGS SLOWLY. DO NOT LIFT BRUSH SPRINGS MORE THAN NEEDED TO REMOVE THE BRUSH SETS FROM THE BRUSH HOLDER.

- (2) Use a wire hook tool to lift the two brush springs away from a brush set. Remove the brush set from the brush holder assembly. (See Figure 3004.)
- (3) Slowly return brush springs to a resting position on the brush holder assembly.
- (4) Identify the brush set with the number on the brush holder from which it was removed.
- (5) Repeat steps H(1) through H(4) for each remaining brush holder assembly.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

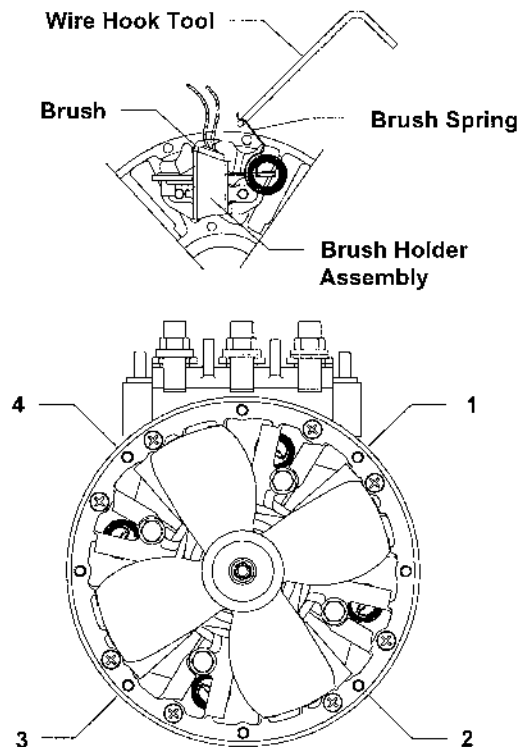


Figure 3004 - Removing Brush Set

I. Remove thermal switch (170) from stator and housing assembly (235).

NOTE: For models 23079-006 and 23079-007 only. Do not remove the thermal switch unless it is damaged.

- (1) Remove self-locking nut (175) that attaches thermal switch lead to terminal block.
- (2) Remove the thermal switch lead from the terminal block.
- (3) Remove and discard thermal switch from stator and housing assembly. (See Figure 3005.)
- (4) Remove and discard lockwasher (180).

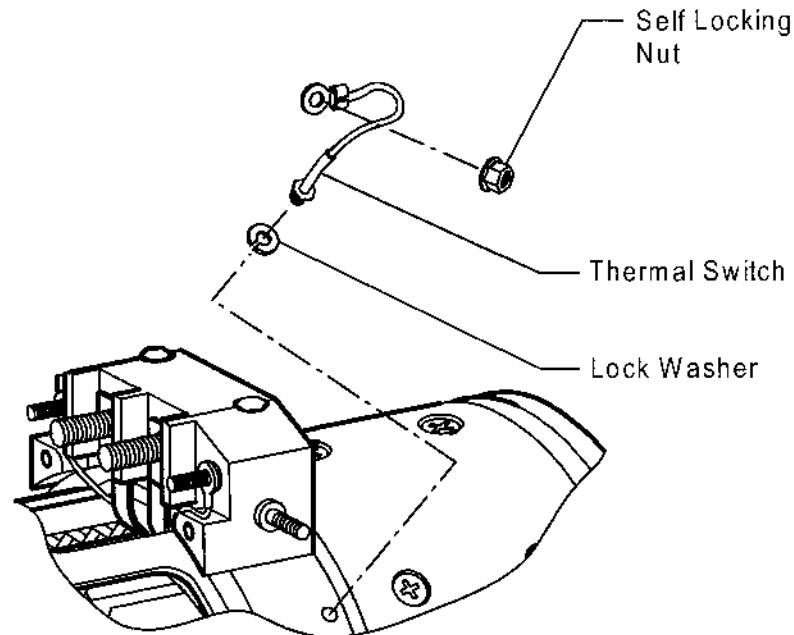


Figure 3005 - Removing Thermal Switch

J. Remove the fan (110) from the drive shaft (125).

NOTE: Put a spline wrench on the drive spline to prevent the drive shaft from turning while removing the self-locking nut.

- (1) Remove self-locking nut (115).
- (2) Remove flat washer (120).
- (3) Slide the fan off of the drive shaft

K. Remove the drive shaft (125) with the attached dampener hub (135), dampener plate (140), and friction ring (145) from armature shaft.

CAUTION: EXCESSIVE FORCE WHEN TAPPING THE DRIVE SHAFT MAY DAMAGE THE THREADS.

- (1) Lightly tap the anti-drive end of the drive shaft with a plastic or leather mallet to disengage it from the armature shaft. (See Figure 3006)
- (2) Pull the drive shaft out of the drive end of the starter-generator.
- (3) Remove the friction ring from the drive shaft.
- (4) Remove the dampener plate from the dampener hub.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

- (5) On models 23079-006 and 23079-007 remove and discard the O-ring (130) from the drive shaft.

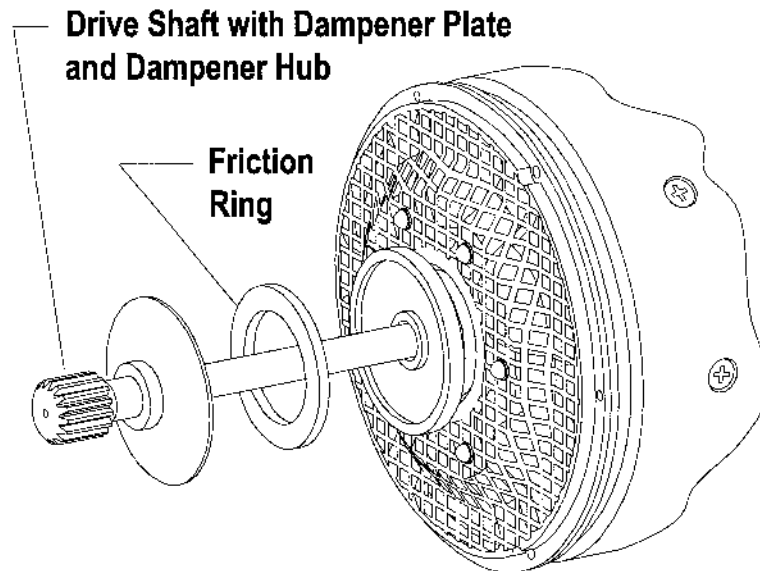


Figure 3006 - Drive Shaft Removal

- L. Remove the dampener hub (135) from the drive shaft (125). See Figure 3007.

NOTE: Do not remove the dampener hub unless it is damaged.

CAUTION: FAILURE TO USE A FOAM CUSHION UNDER THE DRIVE SHAFT WHEN REMOVING THE DAMPENER HUB CAN DAMAGE THE DRIVE SHAFT.

- (1) Set a foam cushion and anti-drive end hub support on an arbor press table.
- (2) Set a dampener hub driver on the drive shaft on the flat side of dampener hub.
- (3) Insert the drive shaft through the center of the anti-drive end hub support until the dampener hub driver is seated on the anti-drive end hub support.
- (4) Press the drive shaft down into the anti-drive hub support until the dampener hub disengages from the drive shaft

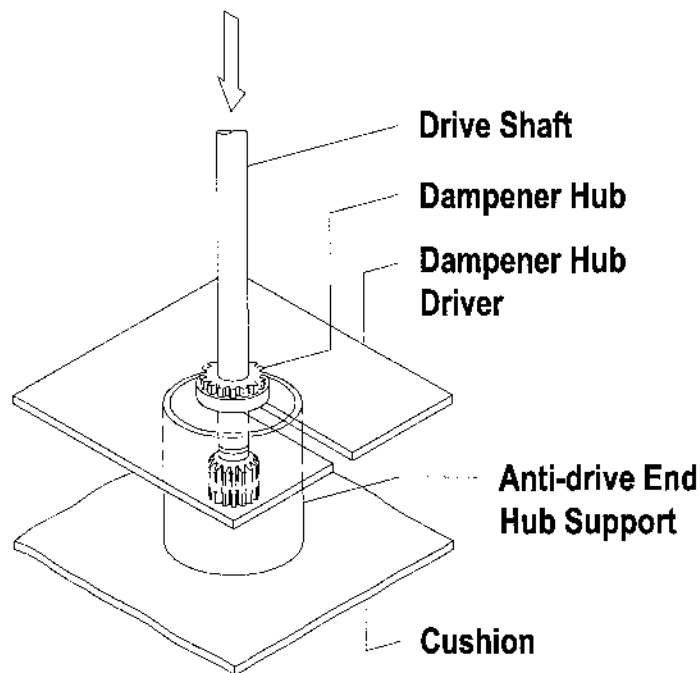


Figure 3007 - Dampener Hub Removal

M. Remove dampener backplate (150) and retaining ring (155) from armature shaft (205). See Figure 3008.

CAUTION: FAILURE TO USE A DRIVE END ARMATURE SHAFT ADAPTER WHEN REMOVING THE DAMPENER BACKPLATE CAN PERMANENTLY DAMAGE THE ARMATURE SHAFT.

- (1) Put an armature shaft adapter in the drive end of the armature shaft.

CAUTION: TAPE THE JAWS OF THE BEARING PULLER TO PREVENT DAMAGING THE DAMPENER BACKPLATE DURING REMOVAL.

- (2) Remove the dampener backplate from the armature shaft using a suitable bearing puller.
- (3) Remove and discard the retaining ring using external snap ring pliers.

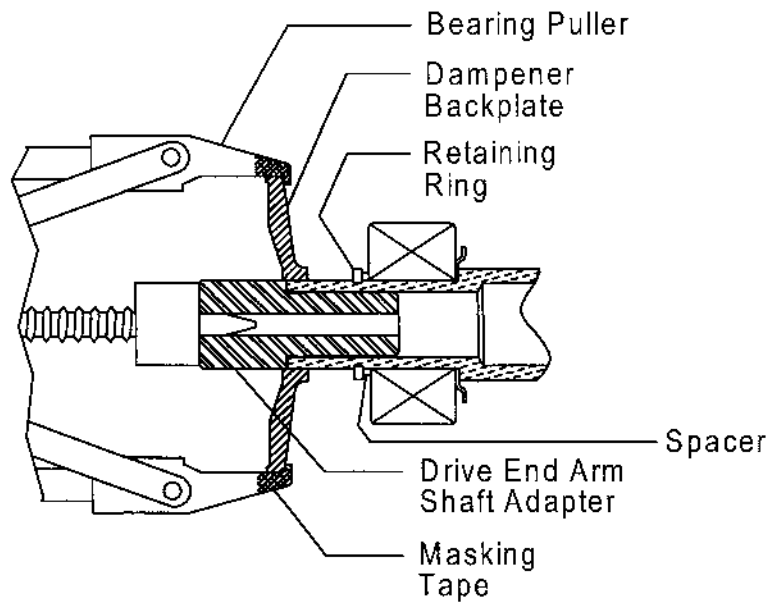


Figure 3008 - Removing Dampener Backplate

N. Remove the anti-drive end bell assembly (160) and the attached armature (205) from the stator and housing assembly (235).

- (1) Set starter-generator drive end down on a vertical stator support. (See Figure 3009)
- (2) Remove eight screws (165) that attach the anti-drive end end bell assembly to the stator and housing assembly.
- (3) Carefully lift the anti-drive end end bell assembly, with the attached armature, from the stator and housing assembly.

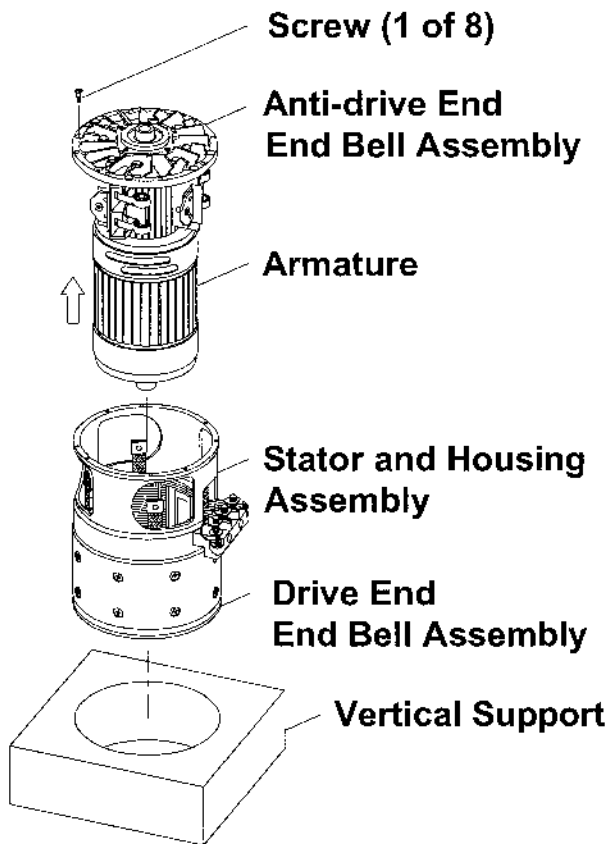


Figure 3009 - Removing Anti-drive End End Bell Assembly

O. Remove drive end end bell assembly (185) from stator and housing assembly (235).

- (1) Set the stator and housing assembly on a horizontal stator support.
- (2) Remove three screws (190) that attach the drive end end bell assembly to the stator and housing assembly.

CAUTION: DO NOT USE EXCESSIVE FORCE WHEN TAPPING DRIVE END END BELL ASSEMBLY OR DAMAGE TO THE MATING FLANGE OF THE STATOR AND HOUSING ASSEMBLY MAY RESULT.

- (3) Remove the drive end end bell assembly from the stator and housing assembly.

P. Remove the armature (205) from the anti-drive end end bell assembly (160). See Figure 3010.

- (1) Set a foam cushion and an anti-drive end end bell assembly support on an arbor press table. The foam cushion will help protect the armature shaft.

CAUTION: USE ONLY THE ANTI-DRIVE END ARMATURE SHAFT ADAPTER WHEN PRESSING THE ARMATURE FROM THE ANTI-DRIVE END END BELL ASSEMBLY TO PREVENT DAMAGE TO THE ARMATURE SHAFT.

- (2) Set the anti-drive end end bell assembly with the attached armature onto an anti-drive end end bell assembly support.
- (3) Insert an anti-drive end armature shaft adapter into the end of the armature shaft.

CAUTION: HOLD THE ARMATURE SECURELY WHEN IT IS PRESSED OUT OF THE ANTI-DRIVE END END BELL ASSEMBLY TO PREVENT POSSIBLE DAMAGE TO THE ARMATURE SHAFT.

- (4) While supporting the armature assembly securely with one hand, carefully press the armature shaft down and away from the anti-drive end bell assembly.
- (5) Carefully remove the armature from inside of the anti-drive end end bell assembly support.

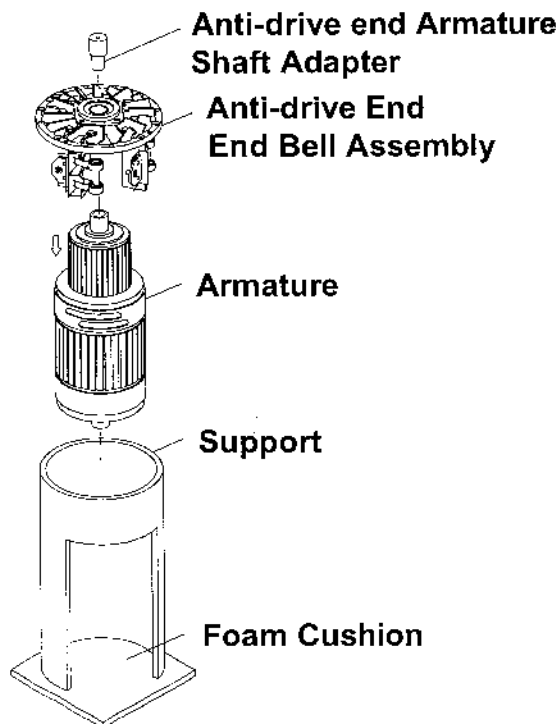


Figure 3010 - Removing Armature from Anti-Drive End End Bell Assembly

Q. Remove bearing retainer (210) from the anti-drive end end bell assembly (160).

- (1) Remove four screws (215) that attach the bearing retainer to the anti-drive end bell assembly.
- (2) Remove the bearing retainer.

R. Remove the ball bearing (220) from the anti-drive end end bell assembly, see Figure 3011.

CAUTION: USE A BEARING AND HUB SUPPORT WHEN PRESSING THE BEARING FROM THE ANTI-DRIVE END END BELL ASSEMBLY TO PREVENT DAMAGE TO THE ASSEMBLY.

- (1) Set an anti-drive end hub support on an arbor press table.
- (2) Set the anti-drive end end bell assembly on the anti-drive end hub support assembly with the outboard side down.
- (3) Set an inner race bearing driver on the inner race of the ball bearing installed in the anti-drive end end bell assembly.
- (4) Carefully press the bearing from the anti-drive end end bell assembly.
- (5) Remove the bearing from inside the anti-drive end hub support. Discard the bearing.

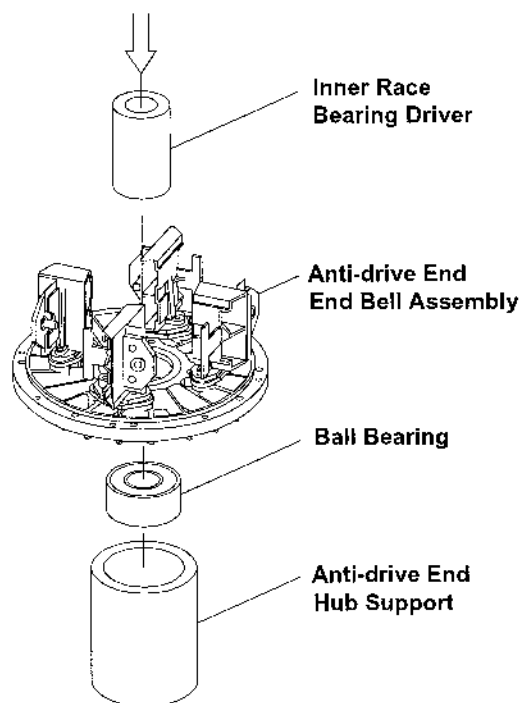


Figure 3011 - Removing Bearing from Anti-Drive End End Bell Assembly

S. Remove the ball bearing (220) and baffle disk (225) from the drive end of the armature shaft. See Figure 3012.

CAUTION: USE THE ARMATURE SHAFT ADAPTER WHEN REMOVING THE BEARING TO PREVENT PERMANENT DAMAGE TO THE ARMATURE SHAFT.

- (1) Remove the spacer (230) from the armature shaft.
- (2) Put a drive end armature shaft adapter into the drive end of the armature shaft.
- (3) Remove the bearing from the armature shaft using a suitable bearing puller.
- (4) Discard the bearing.
- (5) Remove the baffle disk from the armature shaft.

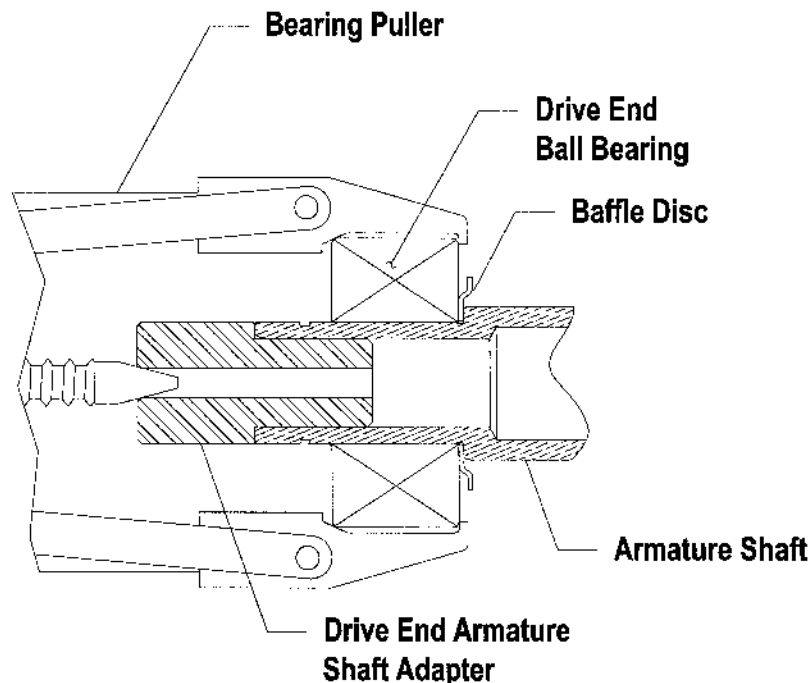


Figure 3012 - Removing Bearing from Drive End of Armature Shaft

4. Disassembly of the anti-drive end end bell Assembly.

Unless otherwise specified, numbers in parentheses () refer to the item numbers in Figure 10002 of the ILLUSTRATED PARTS LIST.

NOTE: Do not disassemble the anti-drive end end bell assembly further than required to examine, repair, or replace parts determined to be unserviceable.

A. Remove brush holder assemblies (15) from anti-drive end end bell (5).

- (1) Remove eight bolts (35) and flat washers (45) from the anti-drive end end bell. (See Figure 3013.)
- (2) Remove eight insulating washers (40) and four insulating boards (55) from the anti-drive end end bell.
- (3) Remove the insulating sleeves (50) from the through holes in the anti-drive end end bell.
- (4) Discard the insulating washers, insulating boards and insulating sleeves.

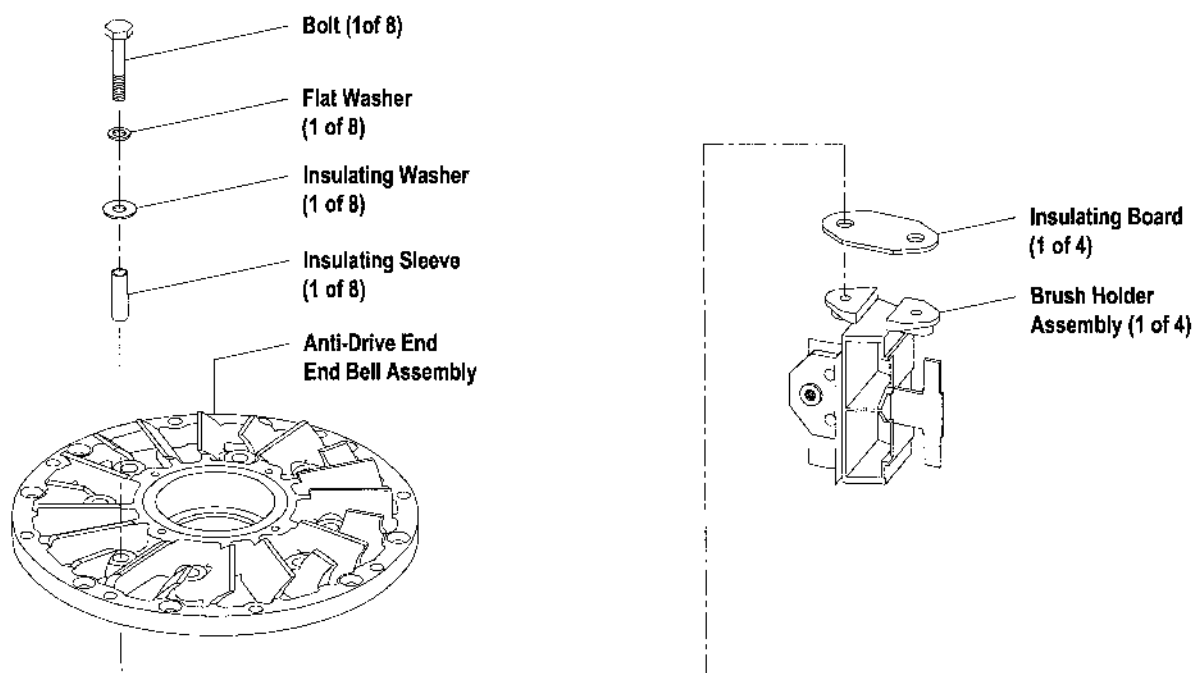


Figure 3013 - Removing Brush Holder Assemblies.

B. Remove the brush springs (75) from the brush holder assemblies (15).

- (1) Remove any of the eight brush springs that need to be replaced from the brush holder assemblies. (See Figure 3014.)
- (2) Discard the damaged brush springs

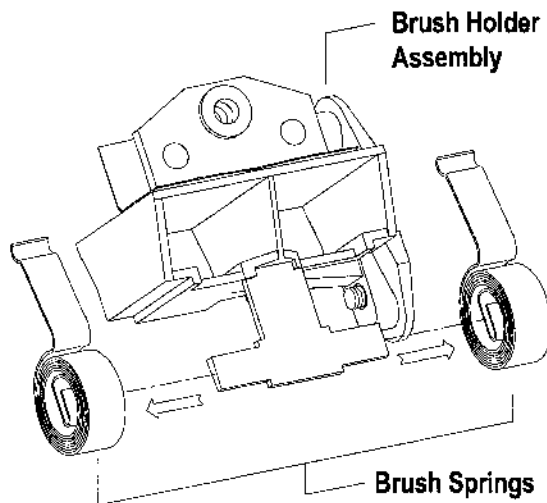


Figure 3014 - Removing Brush Springs from Brush Holder Assembly

C. Disassemble each brush holder assembly (15)

- (1) Use a drill with a #40 bit to remove and discard the two rivets (70) from the brush lead terminal board (60). (See Figure 3015.)
- (2) Remove and discard the brush lead terminal board and insulating sheet (65) from the brush holder.
- (3) Repeat Steps (1) and (2) on each brush holder assembly.

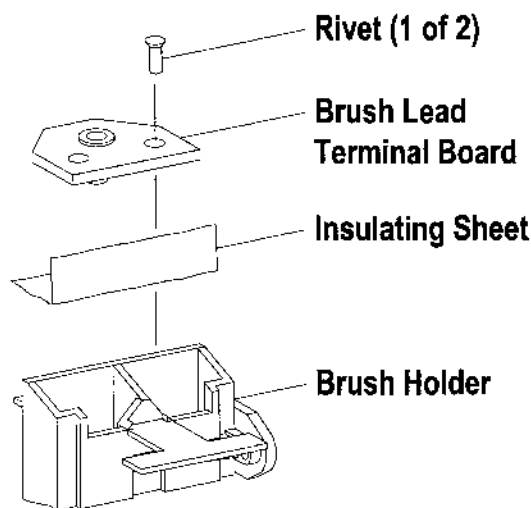


Figure 3015 - Disassembling Brush Holder Assembly



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

5. Disassembly of the Drive End End Bell Assembly.

Unless otherwise specified, numbers in parentheses () refer to the item numbers in Figure 10003 in the ILLUSTRATED PARTS LIST.

NOTE: Do not disassemble the drive end end bell further than required to examine, repair or replace parts determined to be unserviceable.

A. Remove the screen (15) from the drive end end bell (5).

- (1) Remove and discard the attaching drive screws (20) and flat washers (25).
- (2) Remove and discard the screen.

6. Disassembly of the Stator and Housing Assembly.

Unless otherwise specified, numbers in parentheses () refer to the item numbers in Figure 10004 in the ILLUSTRATED PARTS LIST.

NOTE: Do not disassemble the stator and housing further than removal of the terminal block for examination, repair or replacement of parts determined to be unserviceable.

A. Remove the lead of the thermal switch (10001-170) from the terminal block (30). (See Figure 3016)

NOTE: For models 23079-006 and 23079-007 only. The thermal switch may already have been removed if it was damaged.

- (1) Remove the self-locking nut (10001-175) from the terminal block.
- (2) Remove the thermal switch lead from the terminal block. (See Figure 3016.)
- (3) Isolate the thermal switch lead with electrical insulating tape or equivalent.
- (4) Secure the thermal switch lead to the stator housing (5) so it is not damaged.

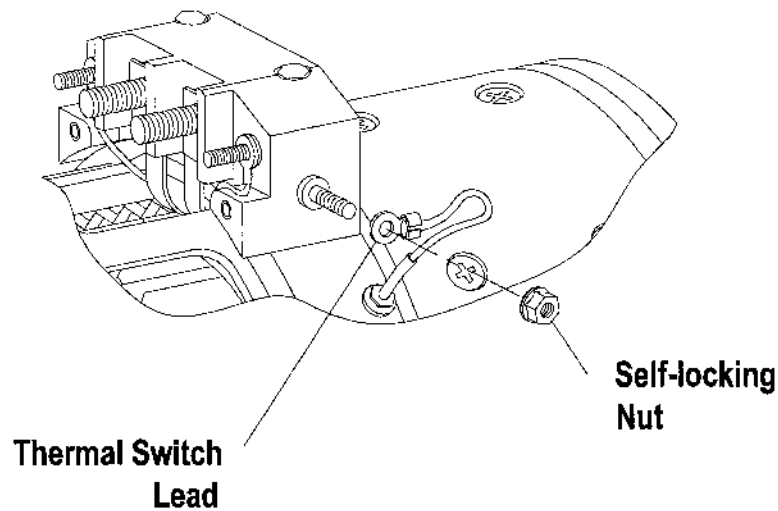


Figure 3016 - Removal of Thermal Switch Lead

B. Removal of Miscellaneous Hardware from the Terminal Block (30) if Present.

- (1) Remove the self-locking nuts (10) and flat washers (15) from the terminal studs B, E, and, if present C. (See Figure 3017.)
- (2) Remove the self-locking nuts (20) and flat washers (25) from terminal studs A and D.
- (3) Discard all self-locking nuts and flat washers.

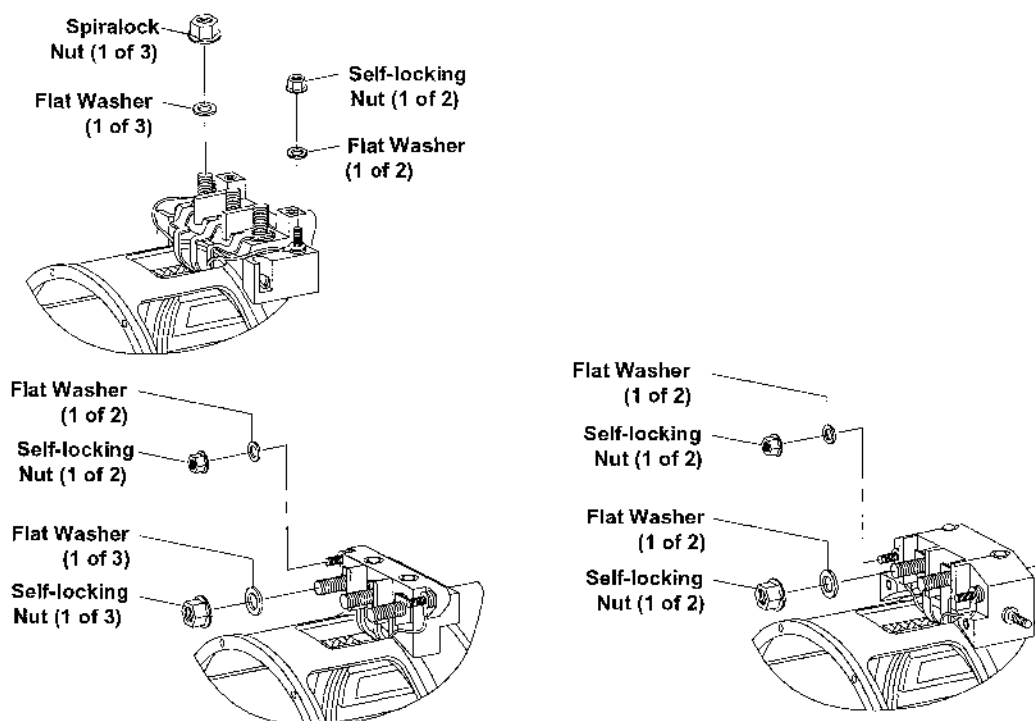


Figure 3017 - Miscellaneous Hardware Removal

C. Remove the terminal block (30) from the stator housing (5). See Figure 3018.

- (1) Remove the screws (35), spring lockwashers (40) and flat washers (45) that attach the terminal block to the stator housing. (See Figure 3018.)
- (2) Remove the small stator leads from terminal studs A and D.
- (3) On models 23079-002, 23079-002-1, 23079-003, and 23079-004:
 - (a) Carefully bend back the large stator leads just enough to release terminal studs B, E, and C from the large stator leads.
 - (b) Slide the terminal block out from under the large stator leads.
- (4) On all other models, slide the terminal block out from under the large stator leads.

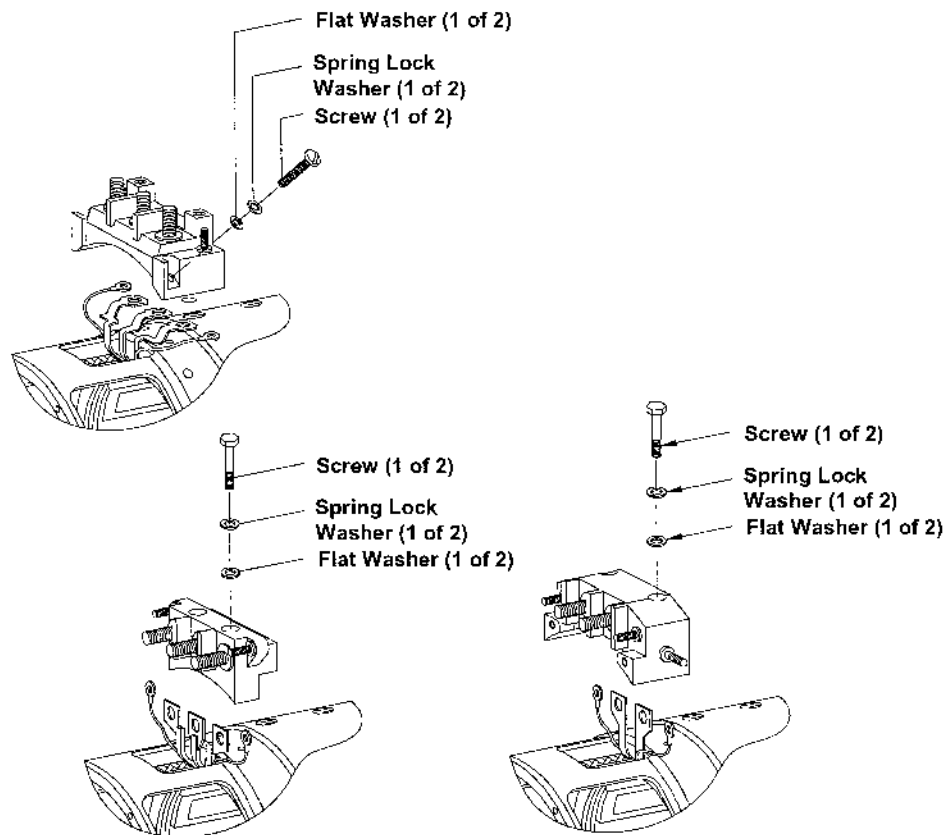


Figure 3018 - Removal of Terminal Block from Stator Housing



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

CLEANING

1. Introduction

This section provides the cleaning procedures for the 23079 Series DC Starter-Generator.

2. Cleaning Materials

Table 4001 Lists the material and equipment required to clean the generator.

NOTE: Equivalent substitutes may be used for the materials listed in Table 4001.

Material	Description/Specification	Source/CAGE Code
Brush	Nonmetallic, soft bristle	Commercially Available
Cleaning Pads/Wiping Cloths	Lint-free, soft fabric	Commercially Available
Corrosion Preventive	Cold application, water displacing soft film MIL-C-16173, Grade 3.	Commercially Available
Detergent <u>NOTE:</u> The detergent used to clean the generator must not remove any base finishes or metal on any parts.	Formula 815 GD Applications: Hot tanks - aluminum safe Steam Cleaning Pressure Sprayers Ultrasonic Cleaning	The Brulin Corporation Indianapolis, IN (V94058)
	Formula 1990 GD Application: Spray Wash Cleaning	The Brulin Corporation Indianapolis, IN (V94058)
Isopropyl Alcohol	TT-I-735, Grade A Flash Point: 53°F (12°C), Flammable Refer to the Material Safety Data (MSD) Sheet for the material for additional safety information	Commercially Available

Table 4001 - Cleaning Materials and Equipment



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

3. Cleaning Procedures

Before performing the CHECK, REPAIR, ASSEMBLY, TESTING AND FAULT ISOLATION, and final inspection procedures, the components of the generator must be cleaned, rinsed, and dried. All signs of moisture, oil, and other contamination must be removed from the internal structure of the generator to make sure all parts and circuits function correctly. Clean the generator according to the following procedures.

WARNING: COMPRESSED AIR USED FOR CLEANING MUST BE FREE OF OIL AND WATER. COMPRESSED AIR USED FOR CLEANING CAN CREATE AIRBORNE PARTICLES THAT MAY ENTER THE EYES. PRESSURE MUST NOT EXCEED 30 PSIG (207 kPa). EYE PROTECTION IS REQUIRED.

A. Blow out dirt particles.

- (1) Put the air inlet, metal dampener components, drive shaft, anti-drive end end bell, rotor assembly, drive end end bell, and the stator and housing assembly into a ventilated air spray booth.
- (2) Blow away all evidence of dirt with clean, dry, compressed air.

B. Clean the components.

Wash the parts using one or more of the approved cleaning procedures outlined in Table 4002.

Procedure	Instructions
Spray	Put the parts into a high pressure spray wash booth.
	<u>CAUTION:</u> LIMIT THE PRESSURE USED TO CLEAN PARTS. HIGH PRESSURE MAY REMOVE OR DAMAGE SURFACE COATINGS, CONNECTIONS AND INSULATION.
	Spray the parts with a solution of detergent and water. Refer to the MSD sheets for the recommended dilution of water and detergent and solution temperature.
Ultrasonic	<u>CAUTION:</u> ULTRASONIC CLEANING CAN DAMAGE PLATING, BLACK OXIDE COATINGS, AND, IN ALUMINUM PARTS, DIMENSIONAL TOLERANCES. LIMIT EXPOSURE TO TWO OR THREE MINUTES.
	Put the parts into the ultrasonic cleaning tank.
	Allow the parts to soak for 15 to 20 minutes maximum.
	Remove all evidence of dirt with a soft bristle brush or cleaning rag.
Agitation	Put all parts into an agitation tank.
	Agitate the parts until all evidence of dirt can be removed with a soft bristle brush or cleaning rag.

Table 4002 - Cleaning Procedures



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

Procedure	Instructions
Soak	Put all parts into a soaking tank.
	Soak the parts until all evidence of dirt can be removed with a soft bristle brush or cleaning rag.

Table 4002 - Cleaning Procedures

C. Rinse the parts.

NOTE: Use minimal pressure to rinse the parts.

- (1) Put the parts into a high pressure spray booth.
- (2) Flush the part with hot water or steam.

D. Repeat the applicable cleaning procedure as necessary until the parts are clean and free of dirt and carbon residue.

E. Dry the parts.

- (1) Remove any remaining moisture from the parts with a dry lint-free cloth.

WARNING: WEAR INSULATED GLOVES AND OBSERVE ALL SAFETY PRECAUTIONS WHEN HANDLING HEATED PARTS TO AVOID PERSONAL INJURY.

- (2) Put the cleaned parts in an oven at a temperature of 250°F (120°C) for 2-1/2 hours.

WARNING: ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. DO NOT USE NEAR OPEN FLAMES, WELDING AREAS, OR ON HOT SURFACES. INHALATION OF VAPORS CAN CAUSE DROWSINESS, DIZZINESS, AND HEADACHE. CONTACT WITH SKIN CAN CAUSE IRRITATION. USE IN A WELL-VENTILATED AREA

- (3) Check each part for carbon particles by wiping with isopropyl alcohol on a clean lint-free cloth.

4. Corrosion Preventive.

If the drive shaft is not expected to be re-installed into a starter-generator immediately after cleaning, apply a corrosion preventive compound to its black oxide coating (identified by a very dark finish).

NOTE: The drive shaft must be completely dry before applying the corrosion preventive compound.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

WARNING: THE CORROSION PREVENTIVE COMPOUND IS FLAMMABLE AND TOXIC. EXTENDED EXPOSURE CAN CAUSE SKIN IRRITATION, DIZZINESS, AND HEADACHE. WHEN EXTENDED EXPOSURE IS NECESSARY, USE RESPIRATORY PROTECTION AND PROTECTIVE CLOTHING.

- A. Apply a coating of MIL-C-16173, grade 3, corrosion preventive compound to the drive shaft.
- B. Allow the coating to air dry for four hours. The coating will remain soft after drying.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

CHECK

1. Introduction

This section provides initial and detailed component inspection procedures for starter-generators returning from service for scheduled and unscheduled maintenance. Complete an initial inspection of the starter-generator before beginning any testing or disassembly. The results of this inspection will determine if a starter-generator is in satisfactory condition for a verification test or if it will need further disassembly, inspection, and/or replacement of parts.

This section provides the information that is needed to perform a satisfactory inspection of the starter-generator components. During an overhaul of the starter-generator, you must examine all primary components to find out if they are serviceable.

Reject a part if wear or damage is outside the acceptance limits or is not serviceable.

2. Necessary Tools and Materials

The following tools and materials are needed to inspect the starter-generator and its components.

Tool Description	Reference
Alcohol, Isopropyl	Table 4001
Brush, non-metallic, soft bristle	Commercially Available
Cleaning pads/wiping cloths, lint-free, soft fabric	Commercially Available
Dial Indicator	Commercially Available
Marposs Gauge	Commercially Available NOTE: This equipment shall have an accuracy of 0.00004 inch (1 micron)
Dynamic Balancer	Goodrich Corporation SPD1001
Growler	Commercially Available
High Potential (Dielectric) Tester	Table 1002
Ohmmeter	Table 1002
Tensiometer (Pull Scale)	Commercially Available

Table 5001 - Inspection Tools

3. General Information

Check Acceptance Limit Measurements for individual components are found in FITS AND CLEARANCES section of this CMM. If part does not meet acceptance limits, discard. Use the following procedures for all inspections.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

A. Examine the Starter-Generator and its components visually for:

- signs of corrosion
- chipped-off enamel
- nicks, cracks, dents, scratches
- gouging, scoring or glazing mating surfaces
- fretting corrosion on mating surfaces
- loose or defective attaching parts
- distortion
- discoloration
- crossed or stripped threads
- condensation at drain holes
- torn or cracked seals
- galling or glazing of friction lining residue
- bent or distorted springs
- rounding, stripping or uneven wear on gear teeth

B. Examine electrical components visually for:

- loose or defective attaching parts
- damage caused by excessive heat
- electrical arcing paths
- corroded contacts or terminals
- loose or defective electrical connector, contacts or terminals
- corroded electrical pins or connector
- bent connector pins
- loose, broken or shorted terminations

4. Initial Inspection

NOTE: If damage is found during inspection, stop the inspection immediately and begin further disassembly of the starter-generator using the instructions in the DISASSEMBLY section.

CHECK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

A. Visual Examinations

- (1) Visually examine the starter-generator in a brightly lit work area in accordance with (IAW) sections 3A and 3B.
 - (a) If shipping or handling damage exists, stop the inspection and notify supervisor.
 - (b) If operational damage exists, write down the components that need to be replaced or repaired and continue the inspection.
- (2) Examine the seating between the housing and both end bell assemblies. Make sure that all attaching hardware is in place.

B. Clean the exterior of the starter-generator.

WARNING: ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE; DO NOT USE NEAR OPEN FLAMES, WELDING AREAS, OR ON HOT SURFACES. INHALATION OF VAPORS CAN CAUSE DROWSINESS, DIZZINESS, AND HEADACHE. CONTACT WITH SKIN CAN CAUSE IRRITATION. USE IN A WELL-VENTILATED AREA.

- (1) Clean the surface of the starter-generator using a lint-free cleaning cloth moistened with isopropyl alcohol.
- (2) If needed, loosen any dust particles or grease with a soft bristle brush.

5. Non-Destructive Test (NDT) Inspections.

Inspection Requirements

NOTE: Dimensional, electrical, and NDT inspections are to be performed only if a part or assembly is damaged.

- (1) Magnetic particle inspection shall be done per ASTM-E1444, unless otherwise instructed.
- (2) Liquid penetrant inspection shall be done per ASTM-E1417, Method A or B, Sensitivity Level 2 or higher, unless otherwise instructed. Acceptance criteria shall be per MIL-STD-1907, Table II, Grade C.

6. Replacing Service Parts and Common Hardware

A. Overhaul

During overhaul, all bearings, brushes, O-rings, and locking hardware shall be discarded regardless of condition. Replace all damaged, deformed, corroded or apparently defective hardware.

B. Service Repair

- (1) During repair, inspection or servicing, replace all locking hardware regardless of condition if removed during disassembly. Replace all damaged, deformed, corroded or apparently defective hardware.

CHECK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

- (2) Replace bearings if removed during disassembly.
- (3) Replace brushes if inspection determines necessary.

7. Inspection of starter-generator Parts and Assemblies

A. Repair or Service Inspection

Examine only those components that have been disassembled for repair or service. Do not disassemble any component for inspections unless given instructions to do so in this section or in the TESTING AND FAULT ISOLATION section.

B. The following terms are referenced in this section.

• DISCARD -

If damage to the part is found or the part is outside of the acceptance limits in the FITS AND CLEARANCES section, the part must be removed and replaced with a new one.

• REPAIR -

If damage to the part is found or the part is outside of the acceptance limits in the FITS AND CLEARANCES section, the part can be repaired by a procedure found in the REPAIR section.

• DISASSEMBLE -

If damage is found to a part in an assembly, or part of the assembly is outside of the acceptance limits in the FITS AND CLEARANCES section, disassemble the part from the assembly and repair or replace the damaged part(s).

• REPLACE -

If damage occurs that causes a part to be outside of the acceptance limits in the FITS AND CLEARANCES section or there is no repair procedure for that part, replace the part. Retain the following parts to submit for remanufacture: rotor assembly and stator and housing assembly.

C. Dimensions and Points

Letters (i.e., A,B,C,D, etc.) used in an illustration in this section indicate dimensions or points.

D. Surfaces

Numbers (i.e., 1,2,3,4, etc.) used in an illustration indicate surfaces.

CHECK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

8. Inspection of Parts and Assemblies

Unless otherwise specified item numbers in parentheses () refer to items found in figure 10001 in the ILLUSTRATED PARTS LIST.

NOTE: In cases where a repair is authorized but the repair does not work, discard and replace the item in question regardless of whether or not it is so spelled out in this manual.

A. V-band Clamp (10005-5)

- (1) Examine the part IAW the procedures found in section 3A.
 - (a) DISCARD the clamp if cracks, corrosion, dents, nicks, or scratches exist.
 - (b) REPAIR T-bolt with two or less damaged thread turns. DISCARD if damage exceeds that.
 - (c) DISCARD the self-locking nut if removed.

B. Mounting Adapter (10005-20). (See Figure 5001)

- (1) Examine the part IAW the procedures found in Section 3A.
- (2) DISCARD the part if cracks are found.
- (3) REPAIR the part if minor dents, nicks and scratches or mating surface gouging, scoring, or glazing exist.
- (4) Visually examine mating surfaces for eroded surface coatings and pitting.

DISCARD mounting adapter if pilot flange and/or pilot bore diameter is not within the limits in the FITS AND CLEARANCES section.
- (5) Examine the adapter for damaged or missing guide pins.
 - (a) REPLACE guide pin(s) if damaged or missing.
 - (b) REPLACE guide pin(s) if guide pin height is not within the limits in the FITS AND CLEARANCES section.

CHECK

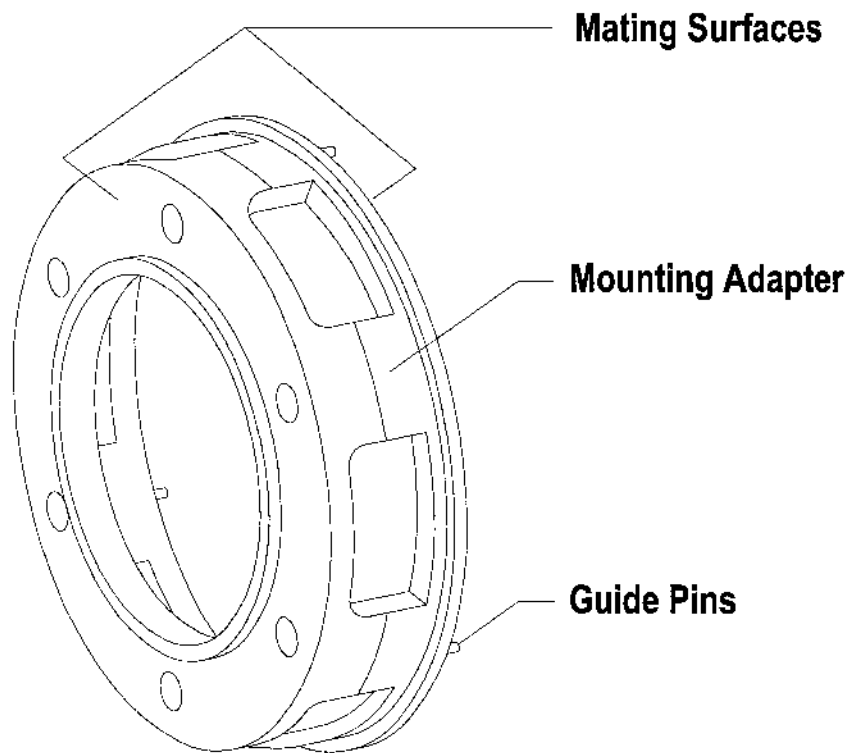


Figure 5001 - Inspecting Mounting Adapter

C. Nameplate (10) and Information Plate (20).

- (1) Examine the parts IAW the procedure found in section 3A.
 - (a) If damaged, transfer information from the old plate to a replacement plate.
 - (b) DISCARD the old plate.
- (2) Make sure the information on the plate is legible.
 - (a) If illegible, transfer information from the old plate to a replacement plate.
 - (b) DISCARD the old plate.
- (3) Make sure drive screws are in place and tight.
 - (a) REPLACE and DISCARD loose drive screws.
 - (b) REPLACE missing drive screws.

D. Terminal Block Cover (45).

- (1) Examine the part IAW the procedures found in section 3A.
- (2) DISCARD the terminal block cover if it is damaged.

E. Air Inlet (55). (See Figure 5002)

Examine the part IAW the procedures found in section 3A.

- (1) DISCARD the part if cracks exists, if damage extends to the mounting or inlet flange, or fretting corrosion exists.
- (2) REPAIR the part if minor nicks, dints, scratches, gouging, scoring or glazing is found.

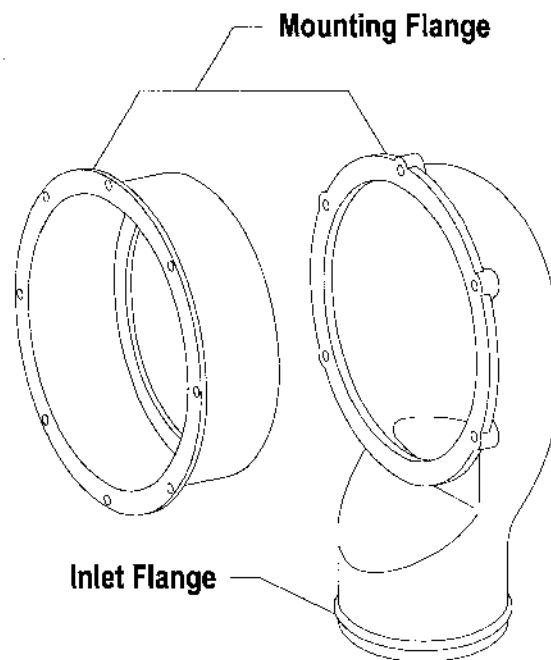


Figure 5002 - Inspecting Air Inlet

F. Air Inlet Adapter (70).

Examine the part IAW the procedures found in section 3A.

- (1) DISCARD the part if cracks, fretting corrosion, gouging, scoring, or glazing on the mating surfaces exist.
- (2) DISCARD the air inlet adapter if more than two threads are damaged on any of the threaded nuts. REPAIR if damage is two or less threads.

G. Brush Access Cover (85). (See Figure 5003)

- (1) Examine the part IAW the procedures found in section 3A.
 - (a) DISCARD the part if cracks are found.
 - (b) REPAIR the part if damage is minor.
- (2) Examine the insulating tape for tears or loose edges.
Repair if damage is found.
- (3) Check blind rivet nut on bracket for tightness or damaged threads.
Replace if damage is found

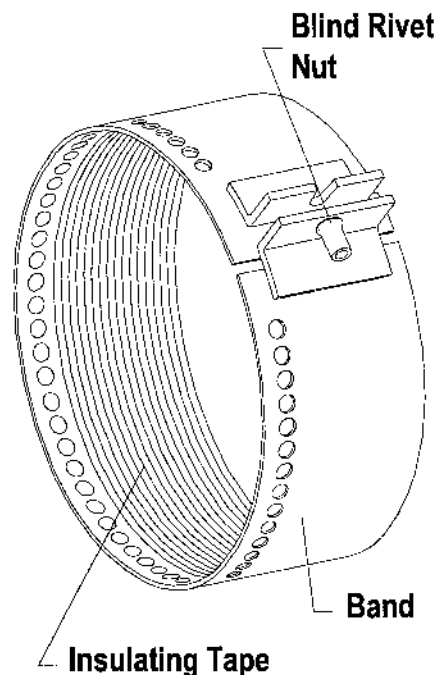


Figure 5003 - Inspecting Brush Access Cover

H. Fan (110)

- (1) Examine the part IAW the procedures found in section 3A.
DISCARD the part if damage is found.
- (2) Visually examine all blades for bending, nicks and scratches.
DISCARD the part if damaged.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

I. Drive Shaft (125) and Dampener Hub (135). (See Figure 5004)

- (1) Examine the part IAW the procedures found in section 3A.
 - (a) **DISASSEMBLE** the drive shaft and dampener hub if cracks or thread damage beyond two threads exist. **DISCARD** the damaged part.
 - (b) **REPAIR** the drive shaft and dampener hub if dents, nicks, and scratches exist.
- (2) Examine armature shaft mating spline, drive spline, and dampener hub for rounding stripping, or uneven wear.

DISASSEMBLE drive shaft and dampener hub if damaged. **DISCARD** the damaged part.
- (3) Measure the drive spline diameter over two gage pins.

DISASSEMBLE the drive shaft and dampener hub if the diameter is not within the limits in the **FITS AND CLEARANCES** section. **DISCARD** the drive shaft.
- (4) Perform a magnetic particle inspection per ASTM-E1444.

DISCARD the drive shaft and dampener hub if damage is found.

CHECK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

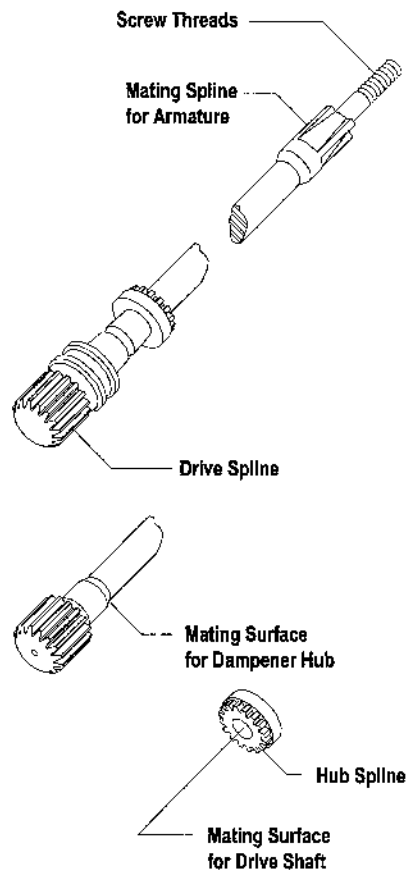


Figure 5004 - Inspecting Drive Shaft and Dampener Hub

J. Friction Ring (145). (See Figure 5005)

- (1) Visually examine all surfaces for cracks.

DISCARD the part if damage is found.

- (2) Measure the Friction Ring

DISCARD part if thickness is not within limits in FITS AND CLEARANCES section.

CHECK

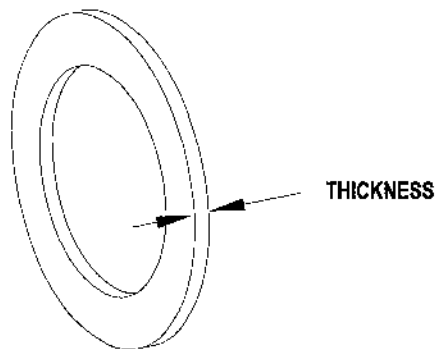


Figure 5005 - Inspecting Friction Ring

K. Dampener Plate (140). (See Figure 5006)

- (1) Examine the part IAW the procedures found in section 3A.
 - (a) DISCARD the part if cracks exist.
 - (b) REPAIR if minor nicks, scratches, gouging, scoring, or glazing is found.
- (2) Visually examine the spline teeth for rounding, stripping, or uneven wear.
DISCARD the part if damaged.
- (3) Measure the dampener plate thickness.

DISCARD if thickness is below the acceptance limit in the FITS AND CLEARANCES section.
- (4) Measure the distance between pins of the spline teeth.

DISCARD if distance is above the acceptance limit in the FITS AND CLEARANCES section.
- (5) Perform a magnetic particle inspection IAW ASTM-E1444.

DISCARD the part if damaged.

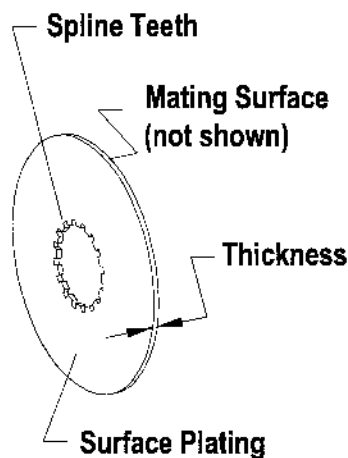


Figure 5006 - Inspecting Dampener Plate

L. Dampener Backplate (150) (See Figure 5007)

Examine the part IAW the procedures found in section 3A.

- (1) DISCARD the part if cracks exist.
- (2) REPAIR if minor nicks or scratches exist. DISCARD if gouging, scoring, or glazing exists on mating surfaces.

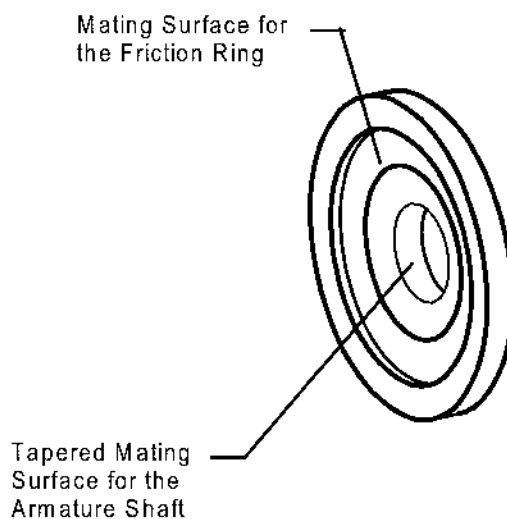


Figure 5007 - Inspecting Dampener Backplate

M. Anti-Drive End End Bell Assembly (Figure 10002)

- (1) Examine the part IAW the procedures found in section 3A. (See Figures 5008 and 5009.)



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

- (a) DISCARD the part if cracks or fretting and/or corrosion on the mating surface are found.
 - (b) REPAIR if minor dents, scratches and nicks or gouging scoring or glazing on the mating surfaces (marked 1, 2, 3, and 6 on figure) are found.
- (2) Using equipment with an accuracy of 0.00004 in. (1 μ m), measure bearing liner diameter "A." (See Figure 5008)

Repair if diameter does not meet the limits in the FITS AND CLEARANCES section. Refer to Goodrich Corporation SPD 1000. See Figure 6001 for machining dimensions.

- (3) Examine the helicoil inserts (10002-10) at locations 3 and 4 (see Figure 5008) for damage.

DISCARD if damage is found.

NOTE: When measuring brush spring pressure, readings can vary widely depending on where the measuring device touches the spring. Measurements must be taken from the curved tip of the spring at the point where it touches the brush.

- (4) Measure the brush spring pressure using a pull scale with harness (See Figure 5010).
- (a) Take six measurements
 - (b) Calculate the average from the measurements.
 - (c) DISCARD any brush spring that is not within the limits in the FITS AND CLEARANCES section.

CAUTION: DIELECTRIC TESTING IS ACCOMPLISHED DURING ACCEPTANCE TESTING AND ONLY FOR STARTER-GENERATORS THAT HAVE BEEN FULLY CLEANED.

- (5) Perform a dielectric test.

CHECK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

WARNING: FAILURE TO USE NECESSARY SAFETY PRECAUTIONS WHEN HANDLING HIGH VOLTAGE ELECTRICAL EQUIPMENT DURING HIGH POTENTIAL TESTING CAN CAUSE SERIOUS INJURY OR DEATH TO THE OPERATOR.

WARNING: BEFORE USING HIGH VOLTAGE TESTER, MAKE CERTAIN THAT TESTER HAS BEEN TURNED "OFF" AND THAT NO ONE IS TOUCHING ANY PORTION OF CONNECTION POINTS OR PROBE LEAD WIRES. FAILURE TO COMPLY WITH THIS WARNING CAN RESULT IN SEVERE ELECTRICAL SHOCK TO THE OPERATOR OR ASSISTANTS.

- (a) With power OFF, connect negative (black) test lead of high potential tester to an uncoated surface of anti-drive end end bell.
 - (b) Attach positive (red) test lead to metal surface of one of the brush holders.
 - (c) Set output voltage to zero and turn high potential power tester ON.
 - (d) At a rate not to exceed 100 VAC per second, slowly adjust output voltage to 250 VAC RMS at commercial frequency. Check and record leakage rate while adjusting voltage back to 0.
 - 1 No evidence of insulation breakdown, arcing, or sparking is acceptable.
 - 2 The leakage rate shall not be higher than 2 milliamperes.
 - (e) Turn high potential tester power OFF.
 - (f) Move positive test lead to one of the remaining brush holders.
 - (g) Repeat steps (c) and (d) on the remaining lead.
 - (h) Repeat steps (f) and (g) until all brush holders have been tested.
 - (i) Turn high potential tester power OFF.
 - (j) Disconnect test leads.
 - (k) If anti-drive end end bell fails the test, CLEAN it and run test again.
 - (l) If anti-drive end end bell fails again, replace insulating washers, and filter board assemblies and run test again.
- (6) Visually examine Rivnuts® for stripped threads or looseness. Replace if damage is found.

CHECK

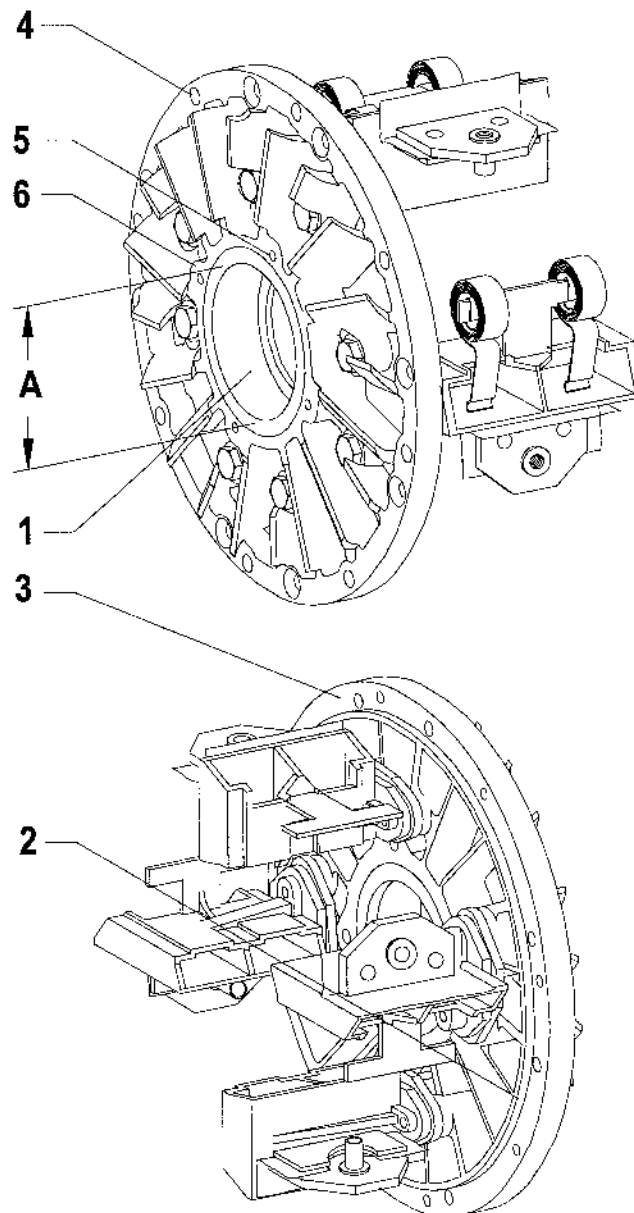


Figure 5008 - Inspecting Anti-Drive End End Bell

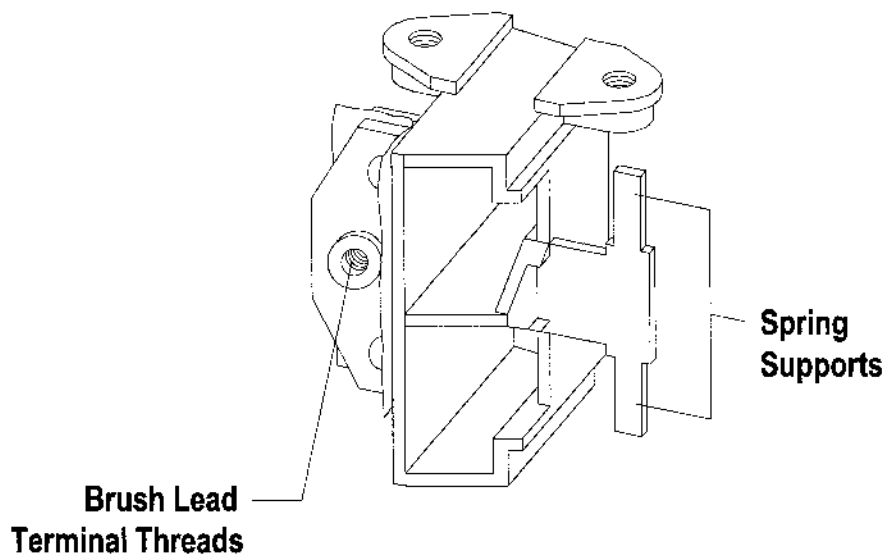


Figure 5009 - Checking Brush Holder Assembly

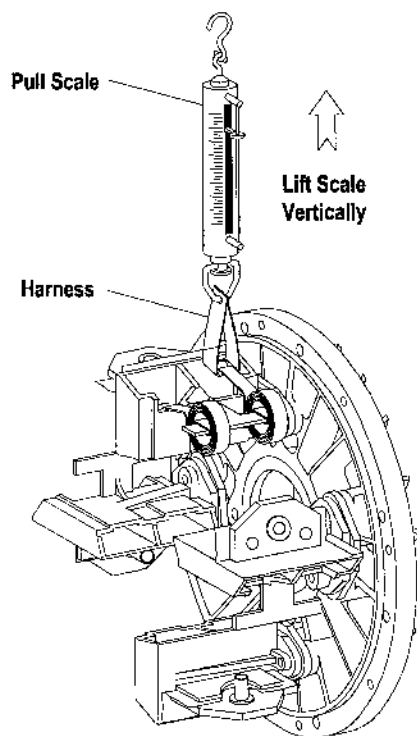


Figure 5010 - Testing Brush Spring Pressure

N. Drive End End Bell Assembly. (See Figure 5011)

- (1) Examine the part IAW procedures found in section 3.A.

CHECK

- (a) DISCARD the part if cracks or fretting and/or corrosion on the mating surface are found.
 - (b) REPAIR if minor dents, scratches and nicks or gouging scoring or glazing on the mating surfaces are found.
- (2) Measure the bearing liner diameter using a gage and equipment with an accuracy of 0.00004 in. (1 μ m).
- (a) REPAIR if part does not meet requirements of FITS AND CLEARANCES section.
 - (b) Modify end bell assembly to the pre-load configuration per REPAIR section if the diameter is not within limits given in the FITS AND CLEARANCES section.
- (3) Examine screen (10003-15) for tears.
- DISCARD screen if damage is found.

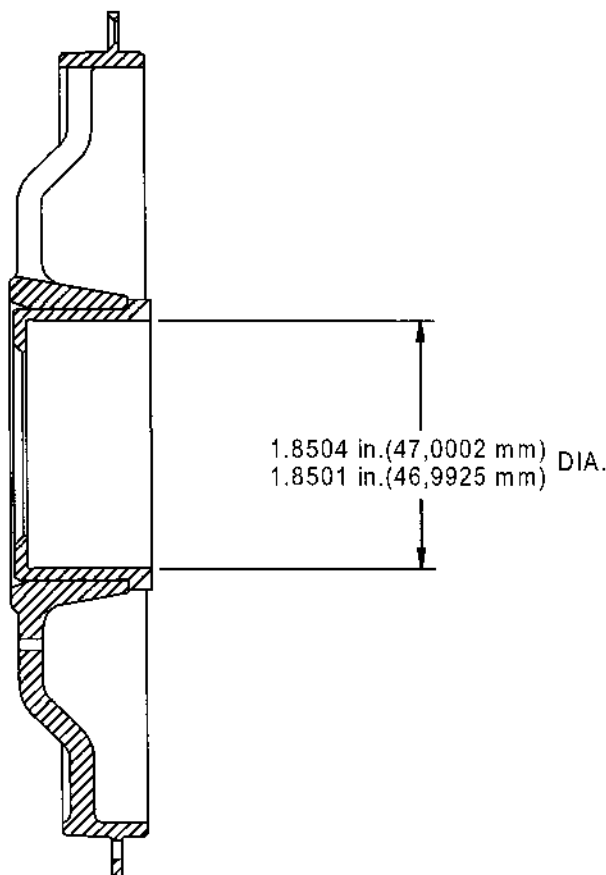


Figure 5011 - Inspecting Drive End End Bell Assembly

- (4) Inspect ribs for cracks.
 - (a) Remove screen (19) only if damage was noted during inspection.
 - (b) Position drive-end end bell assembly so that the screen rests towards the work surface.
 - (c) If drive-end end bell assembly is painted, remove paint from ribs on the underside surfaces as shown in Figure 5012

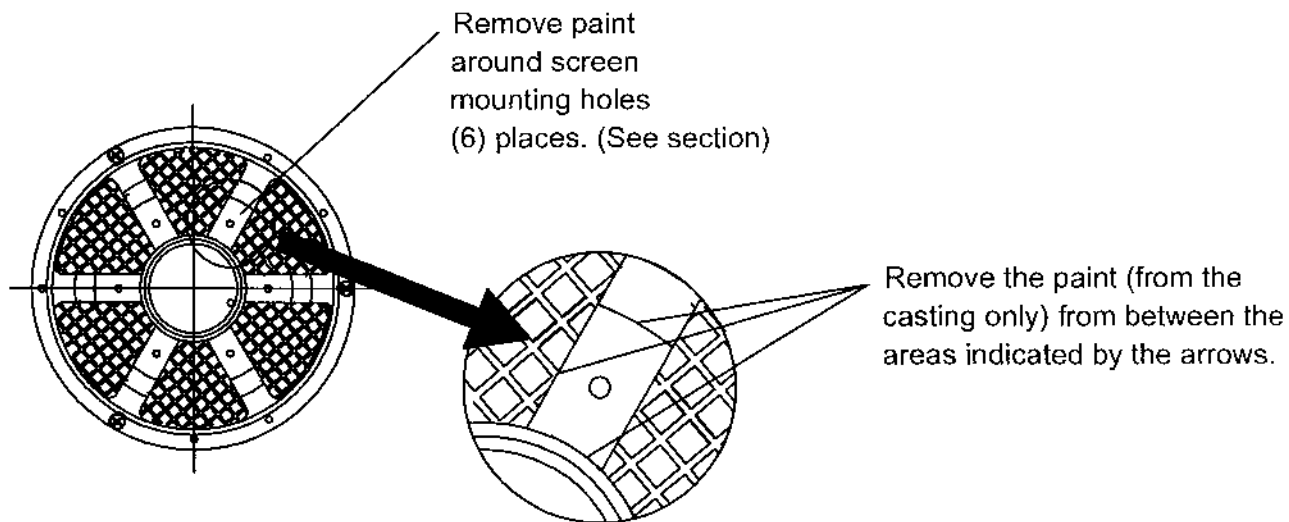


Figure 5012 - Drive-End End Bell Assembly Ribs

- (d) Inspect drive-end end bell assembly according to procedure in Section 4 of the CHECK section.
- (e) Liquid penetrant inspect the surfaces where paint was removed on the drive-end end bell assembly per ASTM-E1417, Method A or B, sensitivity level 2 or higher. Acceptance criteria per MIL-STD-1907, Table II, Grade C.
- (f) If cracking or damage is detected during inspection, discard damaged assembly and replace with a new assembly.
- (g) If no fault is found, repair the paint (if removed) as follows:
 - 1 Paint areas on drive-end end bell assembly where paint was removed for inspection.
 - 2 Restore end bell finish according to SPD 1002.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

O. Armature (205).

CAUTION: USE OF GLYPTOL ON EITHER THE ARMATURE OR STATOR WINDINGS IS STRICTLY PROHIBITED. UNITS HAVING GLYPTOL COATING ON EITHER OF THESE PARTS SHALL HAVE THESE PARTS REPLACED PRIOR TO RETURNING THE UNITS TO SERVICE. GLYPTOL MAY BE USED ON INTERNAL SURFACES OF THE HOUSING ONLY, AND ONLY WHERE OTHER COATINGS ARE NOT REQUIRED.

- (1) Examine the part IAW the procedures found in section 3A.
 - (a) DISCARD the part if cracks or fretting and/or corrosion on the mating surface are found.
 - (b) REPAIR if minor dents, scratches and nicks or gouging scoring or glazing on the mating surfaces are found.
- (2) Using a gauge and equipment with an accuracy of 0.00004 in. (1 μ m), measure journal diameters A and C (See Figure 5013).

REPAIR the bearing journals if they do not meet the limits in the FITS AND CLEARANCES section.
- (3) Measure the commutator diameter B.

Replace armature if the diameter does not meet limits in the FITS AND CLEARANCES section.
- (4) Examine the internal spline for rounding, stripping, or uneven wear.

Replace the armature if damage exists.
- (5) Measure the commutator mica undercut.

Repair the armature if it does not meet the limits of the FITS AND CLEARANCES section.
- (6) Measure bar-to-bar runout and total indicator runout (TIR) using a dial indicator and while supporting the armature on two "V" blocks.

Replace the armature if the measurements do not meet the limits in the FITS AND CLEARANCES section.
- (7) Check the armature balance in two planes using a dynamic balancer.

Repair the armature if it does not meet the limits of the FITS AND CLEARANCES section.

CHECK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

CAUTION: DIELECTRIC TESTING IS ACCOMPLISHED DURING ACCEPTANCE TESTING AND ONLY FOR STARTER-GENERATORS THAT HAVE BEEN FULLY CLEANED.

- (8) Perform a dielectric test.

WARNING: BEFORE USING HIGH VOLTAGE TESTER, MAKE CERTAIN THAT TESTER HAS BEEN TURNED "OFF" AND THAT NO ONE IS TOUCHING ANY PORTION OF CONNECTION POINTS OR PROBE LEAD WIRES. FAILURE TO COMPLY WITH THIS WARNING CAN RESULT IN SEVERE ELECTRICAL SHOCK TO THE OPERATOR OR ASSISTANTS.

- (a) With power OFF, connect negative (black) test lead to armature shaft.
 - (b) Attach positive (red) test lead to a commutator bar.
 - (c) Turn high potential power tester ON.
 - (d) At a rate not to exceed 100 volts per second, slowly adjust output voltage to 250 VAC RMS, commercial frequency. Check and record leakage rate while adjusting voltage back to 0.
 - 1 No evidence of insulation breakdown, arcing, or sparking is acceptable.
 - 2 The leakage rate shall not be higher than 2 milliamperes.
 - (e) Turn high potential tester power OFF.
 - (f) Move negative test lead to a retaining band.
 - (g) Repeat steps (c) and (d).
 - (h) Turn high potential tester power OFF.
 - (i) Move negative test lead to the remaining band.
 - (j) Repeat steps (c) and (d).
 - (k) Turn high potential tester power OFF.
 - (l) Disconnect tester leads.
 - (m) If armature fails the test, CLEAN it and run test again.
 - (n) If armature fails again, return it to Goodrich Corporation Repair and Overhaul for repair.
- (9) Test the armature windings for shorts using a growler and an iron feeler.

CHECK

When an alternating current is passed through a growler, it sets up a magnetic flux in the windings of the armature. As the flux passes through the windings, a potential is induced causing the current to flow if the winding is short-circuited. This current is detected with an iron feeler.

The most common way of using a growler is the "feeler method" in which the growler spans a slot containing a winding, and a "feeler" of iron, such as a hack-saw blade is held about 0.25 in. (6 mm) above the slot containing the other side of the same winding. If the winding is shorted, the feeler will be pulled down to the slot and will stick and vibrate. The feeler can also be used on the same side of the winding that is spanned by the growler.

REPLACE the armature if a short in the wiring exists.

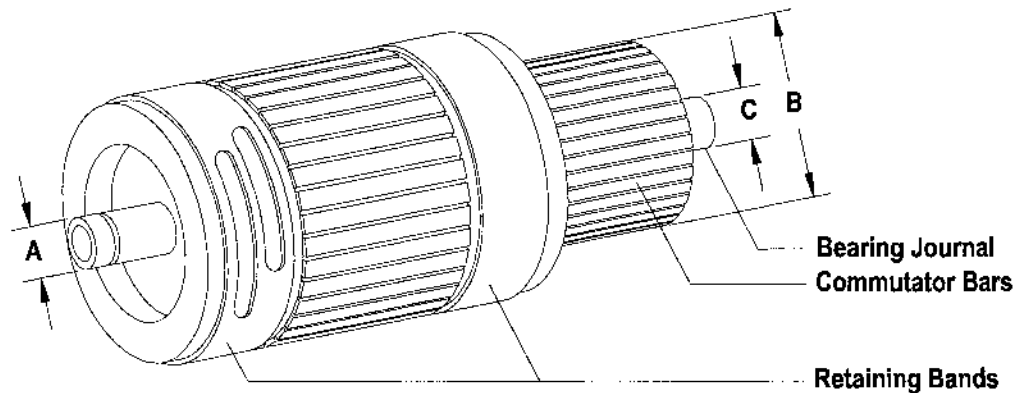


Figure 5013 - Inspecting Armature

P. Bearing Retainer (210)

- (1) Examine the part IAW the procedures found in section 3A.

DISCARD the part if damaged.

- (2) Carefully examine mating surfaces one and two for gouging, scoring, or glazing. (See Figure 5014)

DISCARD the part if damaged.

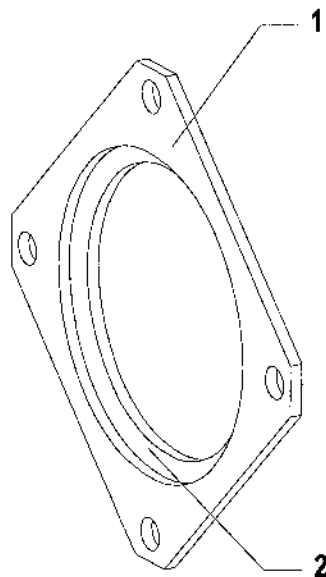


Figure 5014 - Inspecting Bearing Retainer

Q. Baffle Disk (225)

- (1) Examine the part IAW the procedures found in section 3A.
DISCARD the part if damaged.
- (2) Carefully examine mating surfaces 1, 2, and 3 for gouging, scoring, or glazing.
(See Figure 5015)
DISCARD the part if damaged.

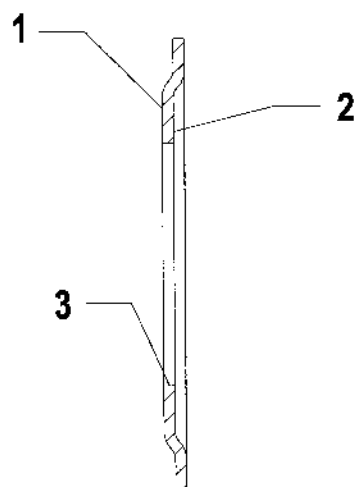


Figure 5015 - Inspecting Baffle Disk

CHECK

R. Spacer (230)

- (1) Examine the part IAW the procedures found in section 3A.

DISCARD the part if damaged.

S. Stator and Housing Assembly (10004-5). See Figure 5016.

CAUTION: USE OF GLYPTOL ON EITHER THE ARMATURE OR STATOR WINDINGS IS STRICTLY PROHIBITED. UNITS HAVING GLYPTOL COATING ON EITHER OF THESE PARTS SHALL HAVE THESE PARTS REPLACED PRIOR TO RETURNING THE UNITS TO SERVICE. GLYPTOL MAY BE USED ON INTERNAL SURFACES OF THE HOUSING ONLY, AND ONLY WHERE OTHER COATINGS ARE NOT REQUIRED.

- (1) Examine the part IAW the procedures found in section 3A and 3B.

- (a) DISCARD the part if cracks exist.
- (b) REPAIR the part if other damage exists.

CAUTION: DIELECTRIC TESTING IS ACCOMPLISHED DURING ACCEPTANCE TESTING AND ONLY FOR STARTER-GENERATORS THAT HAVE BEEN FULLY CLEANED.

- (2) Perform a dielectric test.

WARNING: BEFORE USING HIGH VOLTAGE TESTER, MAKE CERTAIN THAT TESTER HAS BEEN TURNED "OFF" AND THAT NO ONE IS TOUCHING ANY PORTION OF CONNECTION POINTS OR PROBE LEADWIRES. FAILURE TO COMPLY WITH THIS WARNING CAN RESULT IN SEVERE ELECTRICAL SHOCK TO THE OPERATOR OR ASSISTANTS.

- (a) Connect all stator leads together.
- (b) With power OFF, connect positive (red) hi-pot test lead to jumpered stator leads.
- (c) Connect negative (black) test lead to an uncoated surface of stator housing.
- (d) At a rate not to exceed 100 VAC per second, slowly adjust output voltage to 250 VAC RMS at commercial frequency. Check and record leakage rate while adjusting voltage back to 0.
- 1 No evidence of insulation breakdown, arcing, or sparking is acceptable.
 - 2 The leakage rate shall not be higher than 2 milliamperes.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

- (3) Use an ohmmeter to check for an open circuit between negative field lead E and positive terminal A.

REPLACE stator and housing assembly if continuity is not indicated.

T. Terminal Block (10004-30).

CAUTION: THE USE OF RE-MANUFACTURED TERMINAL BLOCKS IS NOT AUTHORIZED BY GOODRICH CORPORATION. DAMAGED TERMINAL BLOCKS (OTHER THAN THOSE WITH REPAIRABLE THREAD DAMAGE) MUST BE DISCARDED.

- (1) Examine the part IAW the procedures found in section 3A.
 - (a) REPAIR if two or less threads are damaged on the terminal studs.
 - (b) DISCARD if any other damage exists.
- (2) Measure the capacitance values between terminals B and E and between terminal B and the ground lead at a frequency of 110 to 130 Hz, 77°F (25°C). Refer to Figures 3, 4, 5 and 6 for terminal identification.

NOTE: Measurement of capacitance values is not necessary during servicing or overhaul unless a suspected fault causing EMI has been reported.

- (3) DISCARD the terminal block if capacitance values do not meet the requirements in the FITS AND CLEARANCES section.

U. Thermal Switch Assembly (170)

WARNING: USE NECESSARY PRECAUTIONS WHEN HANDLING HEATED PARTS TO AVOID INJURY TO PERSONNEL.

- (1) Use an oven to heat thermal switch to 340° F (171°C).
- (2) Use an ohmmeter to check for continuity of thermal switch assembly. Touch one lead of ohmmeter to switch assembly lead, and the other ohmmeter lead to sensing element in base of thermal switch. If continuity is not indicated, thermal switch is defective and must be replaced.

NOTE: Thermal switch assembly is a normally open switch with an operating temperature range of 0°F (-17°C) to 350° F (176.6°C). The switch closes when temperature reaches 330 ±8° F (165.5 ±4.4°C), and opens when temperature drops to 300 ±8° F (149 ±4.4°C).

CHECK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

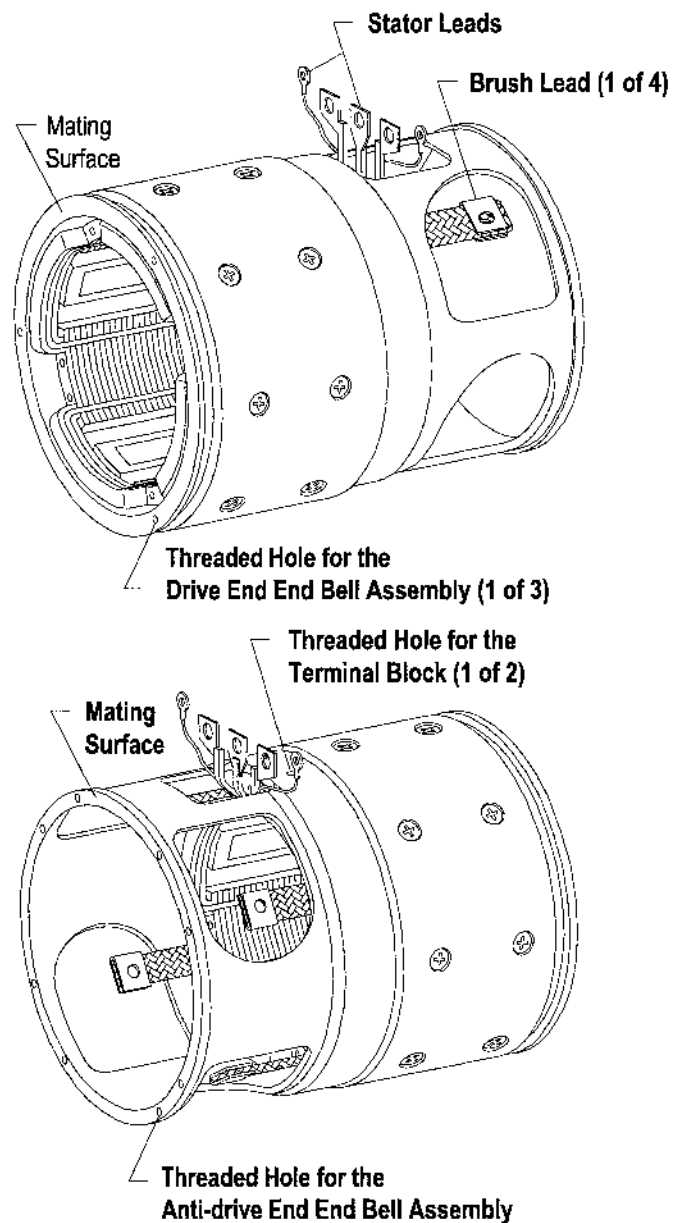


Figure 5016 - Inspecting Stator and Housing Assembly

CHECK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

9. Terminology

Table 5002 lists the definitions of terms used to describe damage conditions.

Term	Definition	Probable cause
Corrosion	The chemical or electrochemical reaction between a material, usually a metal, and its environment that produces a deterioration of the material and its properties	Environmental condition that causes deterioration.
Crack	A break in material.	Severe stress from overload or shock; possible extension of a scratch
Dent	A small, smoothly rounded depression.	A sharp blow or excessive pressure
Fretting	Wear that occurs between tight-fitting surfaces subjected to cyclical relative motion of extremely small amplitude. Usually, fretting is accompanied by corrosion, especially of the very fine wear debris.	Vibration between mating surfaces.
Fretting corrosion	The accelerated deterioration at the interface between contacting surfaces as the result of corrosion and slight oscillatory movement between the two surfaces.	Vibration between mating surfaces.
Galling	Transfer of metal from one surface to another.	Result of localized lubrication break-down between sliding surfaces.
Glazing	Smoothing and creep of a surface.	Result of localized lubrication break-down between sliding surfaces.
Gouging	Removal of surface material, typified by rough and deep depressions.	Protruding objects, incorrectly aligned.
Nick	A sharp bottomed depression that may have rough outer edges.	Dropping, banging.
Rounding	Removal of surface metal at corners or dulling of edges	Result of abrasion, vibration, or poor mating surfaces.
Scoring	A deep scratch following a path of part travel.	Result of localized lubrication break-down between sliding surfaces.

Table 5002 - Terminology

CHECK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

Term	Definition	Probable cause
Scratch	A very shallow furrow or irregularity, usually longer than wide.	Movement of a sharp object across the surface.
Short	Evidenced by sparking, arcing, or high current. Electrical condition where two conductors are connected by a path of very low resistance.	The result when two current-carrying conductors are connected by a path of negligible resistance.
Stripped Thread	Thread of a nut, stud, bolt, or screw damaged by tearing away parts of the thread.	Incorrect installation of threaded mating parts.
Tear	Parting of material	Excessive tension, caused by external force.
Wear	Slow removal of material. Wear may not be visible to the naked eye.	Result of abrasive substances contacting rolling surfaces and acting as a lapping compound.

Table 5002 - Terminology (Continued)

CHECK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

THIS PAGE INTENTIONALLY LEFT BLANK

CHECK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

REPAIR

1. Introduction

This section provides authorized repair instructions for Model 23079 Series Starter-Generators. Repairs are limited to:

- repairing damaged surfaces
- repairing damaged threads
- replacing helicoil inserts
- restoring bearing liners and journals
- commutator refinishing
- armature balancing
- restoring surface coatings of parts and assemblies
- replacement of brush access cover insulating tape
- polarizing the field

Authorized repairs include approved procedures referenced in this manual. Repair operations not referring to instructions given in this section are not authorized by Goodrich Corporation.

CAUTION: THE USE OF PARTS, MATERIALS, OR PROCESSES NOT AUTHORIZED BY GOODRICH CORPORATION FOR THE MAINTENANCE OR SERVICE OF THE DC STARTER- GENERATOR MAY AFFECT CONTINUED FLIGHT WORTHINESS OR INVALIDATE CERTIFICATION.

2. Repair Tools

Besides normal shop tools, the following are required to repair the DC Starter-Generator.

CAUTION: STEEL WOOL AND ABRASIVES CONTAINING IRON OR IRON OXIDE SHALL NOT BE USED AT ANY TIME DURING REPAIR OPERATIONS.

Tool description	Reference
Armature shaft adapters	Figure 9002
Commutator turning fixture	Commercially Available
Drive end hub support	Figure 9014
Flash field switch	Rating: 30 VDC, 10 Amperes
Helicoil insertion/removal tool	Commercially Available

Table 6001 - Repair Tools



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

Tool description	Reference
India Stone	Commercially Available
PlusNut® Fastener Header; P/N C1000-1032	Omni Fastener Inc. 9000 Sweet Valley Drive Valley View, OH 44125, USA
Six volt battery or equivalent DC power source	Commercially available
Thread chasers	Commercially available

Table 6001 - Repair Tools (Continued)

3. Repair materials

Table 6002 lists the materials required to repair the DC starter-generator. Where materials are specified as commercially available, equivalent materials can be used. Repair materials are not available from Goodrich Corporation. All items can be purchased commercially.

WARNING: BEFORE USING ANY OF THE FOLLOWING MATERIALS, BE AWARE OF ALL HANDLING, STORAGE AND DISPOSAL PRECAUTIONS RECOMMENDED BY THE MANUFACTURER OR SUPPLIER. FAILURE TO COMPLY WITH THE RECOMMENDATIONS MAY RESULT IN SERIOUS INJURY, PHYSICAL DISORDER, OR DEATH.

Material	Description/Specification	Source
Chemical film, chromate	For aluminum, MIL-C-5541, Class 1A or Class 3 (Alodine 1200) FLAMMABLE AND TOXIC. Read WARNING before using this material. See MSDS for more safety information.	Commercially available
INK, Marking, White	P/N: HM-7132 (Specify White when ordering)	Markem Machine Co. Keene, New Hampshire
OIL, Machine	(non-silicone oils only)	Commercially available
PRIMER, Zinc Chromate	TT-P-1757 Composition G, Color Yellow	Commercially available
Sandpaper	400/600 grit (Non-Aluminum oxide only)	Commercially Available
TAPE, Insulating	8 mil Fiberglass Fabric, PTFE-coated, Acrylic adhesive, Goodrich Corporation P/N 05-543209	CHR Industries New Haven, Connecticut

Table 6002 - Repair Materials

4. Surface Repair

Repair corroded or damaged surfaces of parts that have qualified for repair according to the CHECK section inspection guidelines.

A. Repair Procedure

- (1) Remove corrosion, sharp edges, burrs, nicks, or scratches from cast surfaces, machined mating surfaces, and sealing surfaces using an india stone or sandpaper. Remove only the sharp edges or burrs that are above the surface of the part.
- (2) Remove minor raised edges, burrs, nicks, or scratches on any polished or load-bearing surface. Remove only the material that extends above the polished surface and make sure that the dimensional tolerances are within limits.

5. Thread repair

CAUTION: DO NOT USE A THREAD CUTTING DIE. IRREPARABLE DAMAGE CAN RESULT FROM USING A THREAD-CUTTING DIE.

A. Procedure

- (1) Repair damaged threads with a thread chaser.
- (2) Remove any remaining sharp edges or burrs with an india stone.
- (3) Apply a light coating of machine oil to repaired threads to prevent corrosion.

6. Helicoil insert replacement**A. Procedure**

- (1) Remove the damaged helicoil insert with a helicoil insertion/removal tool.
- (2) Apply a thin coat of zinc chromate primer to the outer surface of the replacement helicoil insert.
- (3) Install the helicoil insert to a depth of 0.02 to 0.04 in. (0.5 to 1.0 mm) below the part surface while the primer is still wet.
- (4) Break off the helicoil installation tang.

7. Bearing Liner and Journal Restoration

If bearing journals or liners are worn beyond the acceptance limits in the FITS AND CLEARANCES section, restoration is possible. See Goodrich Corporation Standard Practice Document (SPD) 1000 for repair limitations and procedures for nickel plating.

NOTE: Service Bulletin SB23079-OXX-24-04 introduced the enhanced spring-loaded drive end end bell assembly and describes the drive end end bell assembly liner replacement.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

A. Machine the Anti-Drive End, End Bell after plating. See Figure 6001.

- (1) Machine the bearing liner diameter to the dimension in Figure 6001.
- (2) Remove all visible burrs from the newly machined surfaces of the bearing liner.
- (3) Inspect the drive end end bell assembly for compliance with dimensions in Figure 6001.
- (4) If the drive end end bell assembly is not in compliance with the dimensions shown, either rework or replace the drive end end bell.

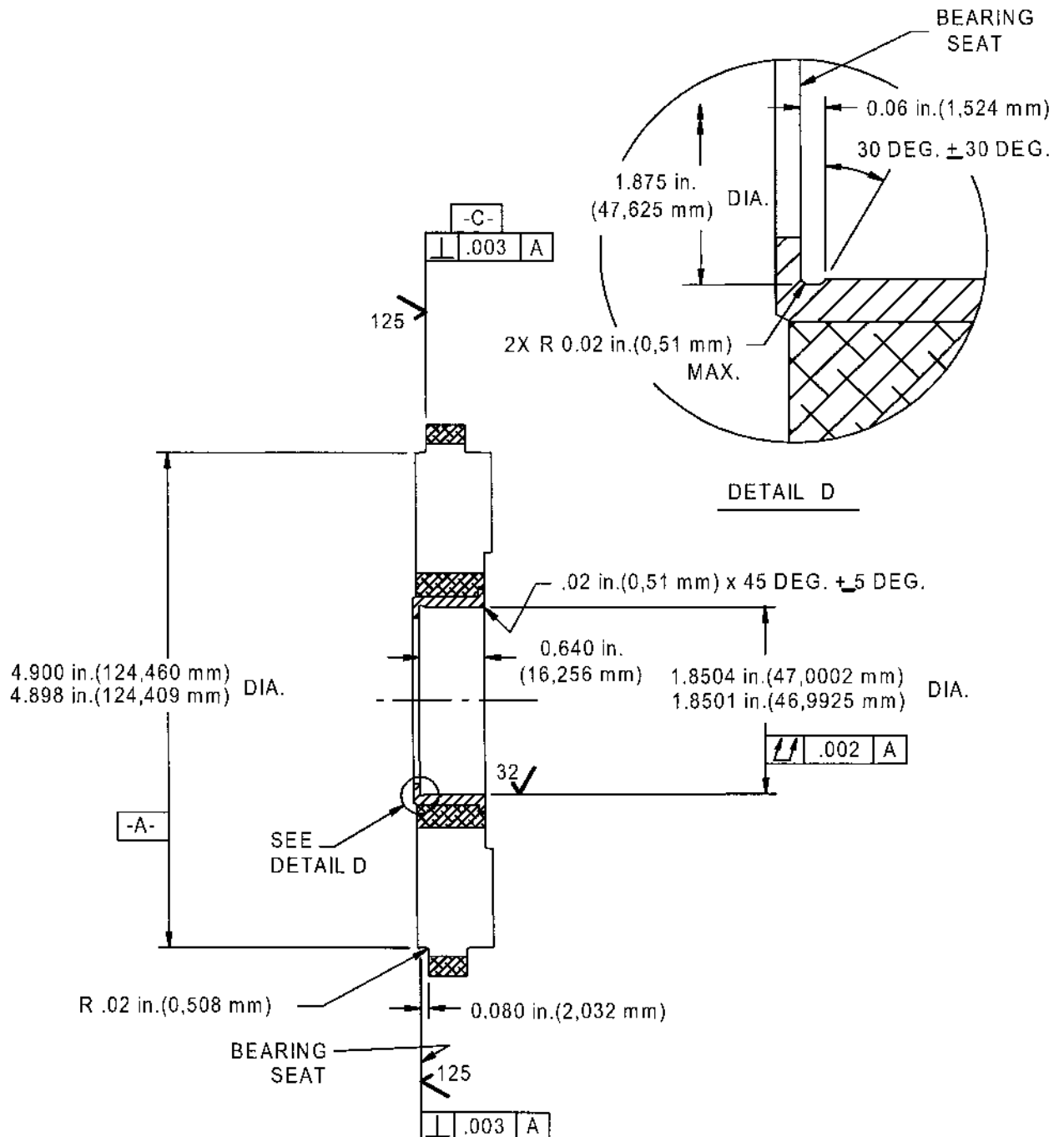


Figure 6001 - Machining Anti-Drive End End Bell after Plating

B. Machine the Drive End End Bell Bearing Liner after plating. See Figure 6002.

- (1) Machine the bearing liner diameter to the dimension in Figure 6002.
- (2) Remove all visible burrs from the newly machined surfaces of the bearing liner.
- (3) Inspect the drive end end bell assembly for compliance with the dimensions in Figure 6002.
- (4) If the drive end end bell is not in compliance with the dimensions shown, either rework or replace the drive end end bell.

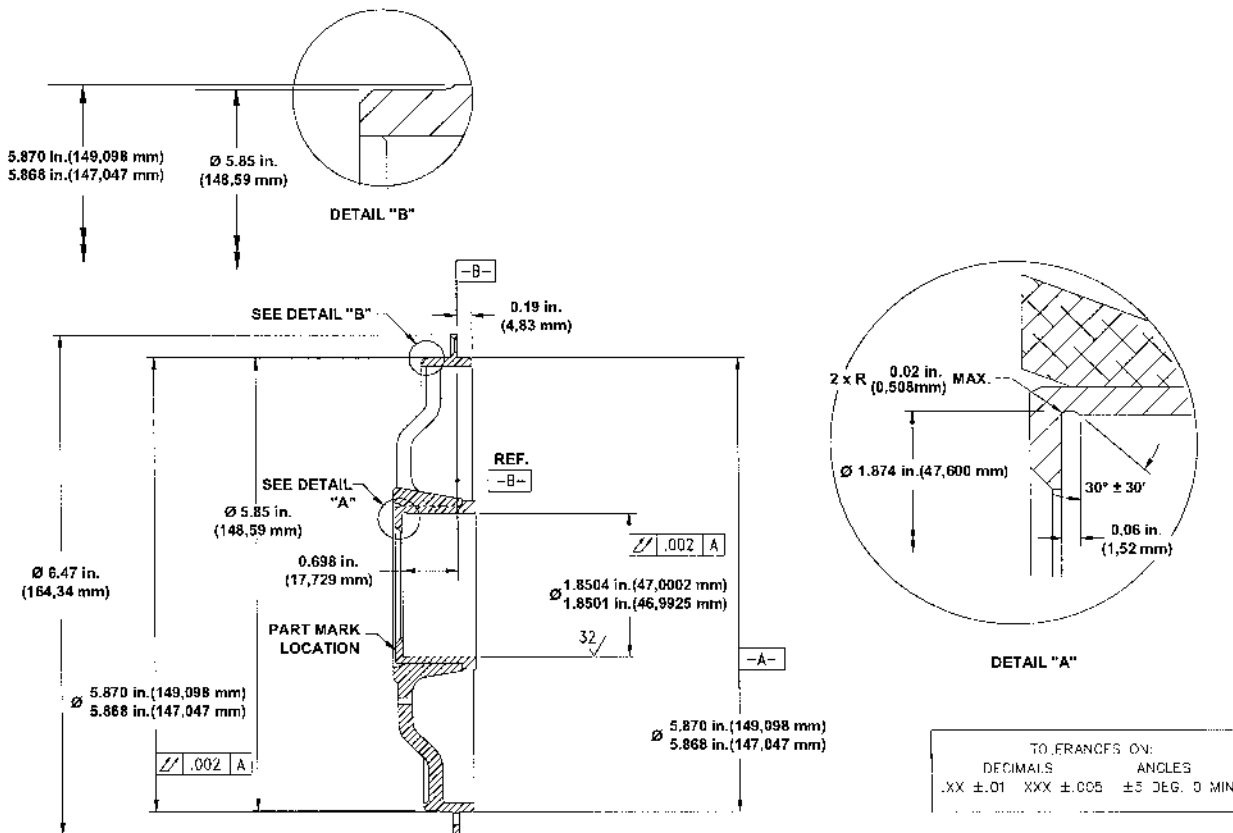


Figure 6002 - Machining the Drive End End Bell Assembly Bearing Liner

C. Machine Journal Diameter after Plating

Journal diameters are to be machined to meet the dimensions listed in the FITS AND CLEARANCES section.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

8. Commutator Refinishing.

A. Preparation

WARNING: MAKE SURE THE POWER TO THE LATHE IS SHUT OFF OR “LOCKED OUT” WHILE PREPARING THE MACHINE FOR COMMUTATOR RE-FINISHING. USE ALL SAFETY PRECAUTIONS WHILE WORKING WITH OR AROUND POWER MACHINE TOOLS. WEAR EYE PROTECTION.

- (1) Install a bearing into the commutator turning fixture.
- (2) Install the commutator turning fixture into the tail stock of the lathe.

CAUTION: AVOID TOUCHING THE COMMUTATOR WITH BARE HANDS. SKIN ACIDS AND OIL CAN CONTAMINATE CONDUCTING SURFACES CAUSING CORROSION AND POOR ELECTRICAL CONTACT.

- (3) Insert the commutator end of the armature into the commutator turning fixture.
- (4) Position the tail stock so that the drive end of the armature shaft slides into the head stock chuck.
- (5) Secure the tail stock.
- (6) Tighten the head stock chuck until the armature is secure.

B. Procedure

CAUTION: DO NOT REMOVE EXCESS MATERIAL FROM THE COMMUTATOR. SHORTENED COMMUTATOR LIFE AND THE REDUCTION OF THE COMMUTATOR DIAMETER CAN BE THE RESULT OF EXCESSIVE MATERIAL REMOVAL.

- (1) Cut the commutator to a surface finish to 64 to 100 micro inches (1.6 to 2.5 microns) RMS. The finish surface must extend from the end of the full undercut to within 0.06 inch (1.5 mm) of the outboard end of the commutator.

NOTE: The suggested feed rate is 0.005 inch (0.13 mm) per revolution at a surface speed of 1,000 to 1,500 surface feet (306 to 460 surface meters) per minute.

WARNING: COMPRESSED AIR USED FOR CLEANING MUST BE FREE OF OIL AND WATER. COMPRESSED AIR USED FOR CLEANING CAN CREATE AIRBORNE PARTICLES THAT MAY ENTER THE EYES. PRESSURE MUST NOT EXCEED 30 PSIG (207 kPa). EYE PROTECTION REQUIRED.

- (2) Clean the armature surfaces with compressed air, 30 PSIG (207kPa) maximum.
- (3) Measure the depth of the mica undercut between the commutator bars.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

- (4) If the undercut does not meet the limits in FITS AND CLEARANCES, use a 0.375 to 0.500 in. (9.52 to 12.69 mm) diameter cutter to undercut the mica to a maximum depth of 0.070 in. (1.78 mm) and a width of 0.035 to 0.045 in. (0.889 to 1.143 mm).

NOTE: The beginning of full mica undercut must be within 0.155 to 0.195 in. (3.94 to 4.95 mm) of the commutator riser. All cuts must be equally spaced and parallel with the centerline of the armature shaft within 0.005 in. (0.13 mm) of true position. All mica must be removed from the edges of the undercut.

- (5) Use a triangular scraping tool to remove sharp edges and burrs.
- (6) Remove the undercut residue from between the commutator bars with a soft bristle brush.

WARNING: COMPRESSED AIR USED FOR CLEANING MUST BE FREE OF OIL AND WATER. COMPRESSED AIR USED FOR CLEANING CAN CREATE AIRBORNE PARTICLES THAT MAY ENTER THE EYES. PRESSURE MUST NOT EXCEED 30 PSIG (207 kPa). EYE PROTECTION REQUIRED.

- (7) Clean the armature surfaces with compressed air, 30 PSIG (207kPa) maximum.
- (8) Examine the commutator according to the procedure in the CHECK section of this CMM.
- (9) Replace the armature if the commutator does not meet inspection requirements.

9. **Balancing the Armature.**

Balance the armature to the limits in FITS AND CLEARANCES and per Goodrich Corporation SPD 1001.

10. **Restoring surface coatings of parts and assemblies**

A. Restore surface coating of parts and subassemblies as follows:

- (1) Removal of exterior coatings must be accomplished according to the procedures in Goodrich Corporation Standard Practice Document (SPD) 1003.
- (2) Restore the surface coating of parts and assemblies according to the procedures in Goodrich Corporation SPD 1002. Use Table 6003 to find the correct primer and paint specification when using SPD 1002.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

Surface Coating	Surface Type	Coating Specification
Primer	Previously painted parts	Zinc chromate primer
	IVD coated parts	Epoxy primer
Paint	Previously painted parts	Gloss alkyd enamel
	IVD coated parts	Polyurethane enamel

Table 6003 - Surface Coating Specifications

B. FAA-PMA Identification.

On models 23079-000-1, 23079-002 and 23079-004 if the "FAA-PMA" marking has been covered with a new coat of paint, apply a new FAA-PMA marking using white marking ink with approximately 1/4 in. (6 mm) high letters about 1/4 in. (6 mm) above the caution label and reading in the same direction as the caution label.

Models 23079-005 and 23079-009 require a new FAA-PMA label to be applied above the caution label if the original label has been damaged.

11. Polarizing the Output Voltage.

A. General

If the polarity of the output voltage of a starter-generator has been reversed, this can be corrected by polarizing the field.

NOTE: If this procedure does not correct the polarity of the field, check for incorrect connections or a defective field circuit.

B. Procedure

- (1) Connect a six VDC, wet cell battery (or 6 VDC power supply) and bump switch in series with the starter-generator. See Figure 6003.

CAUTION: DO NOT CONNECT ANY OTHER POWER SUPPLY TO THE STARTER-GENERATOR. DO NOT POLARIZE THE FIELD WHILE THE STARTER-GENERATOR IS IN OPERATION.

- (a) Connect the negative power lead to terminal stud E.
 - (b) Connect the positive power lead to terminal stud A.
- (2) Press the bump switch for a maximum of five seconds to apply a DC voltage and polarize the field.
 - (3) Turn the power OFF.
 - (4) Disconnect the bump switch and the battery from the starter-generator.

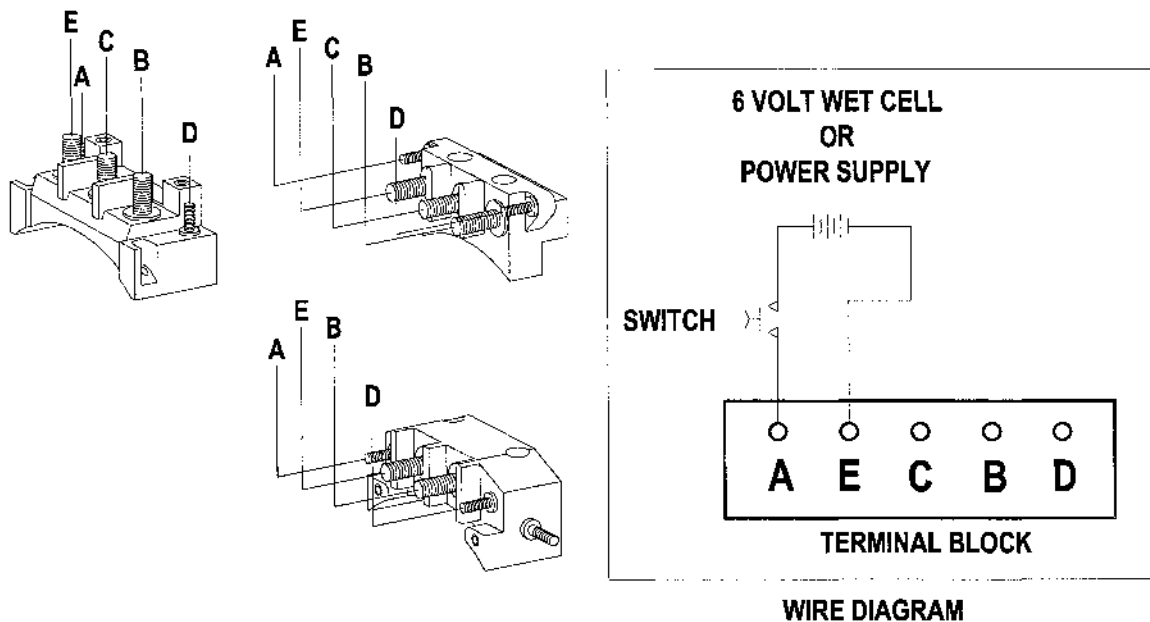


Figure 6003 - Polarizing the Field

12. Insulating Tape Replacement on Brush Access Cover

A. Removal

Remove all signs of the insulating tape and clean the inside surface of the brush access cover. See Figure 6004.

B. Procedure

- (1) If the plating on the inner surface of the brush access cover is damaged, touch-up the surface coating using the instructions in Goodrich Corporation SPD 1002.
- (2) Cut the leading edge of the fiberglass insulating tape square with the sides of the roll.
- (3) Starting at one of the two sides of the access cover band, line up the sides of the tape roll with the sides of the access cover band.

NOTE: The installed tape must be within 0.06 in. (1.5 mm) of the band edges and ends. The tape can overlap the band edge by 0.02 in. (0.5 mm) on one side only, but must not overlap the band.

- (4) Apply the fiberglass tape directly from the roll to the inner surface of the access cover.
- (5) Cut off unwanted tape at the band ends.

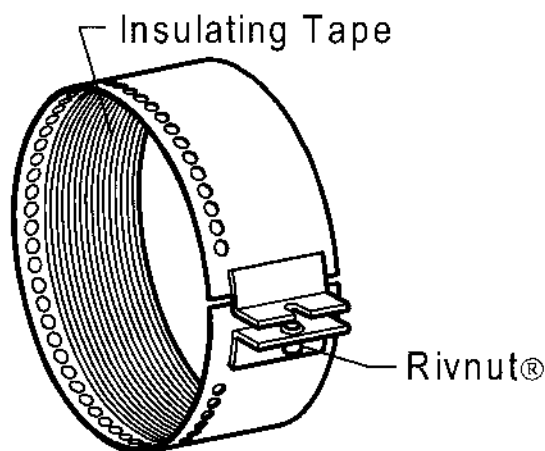


Figure 6004 - Brush Access Cover Repair

13. **Rivnut® Replacement (See Figure 6004)**

A. Removal

Remove the fastener by drilling through the fastener head with a drill bit approximately 20% larger than the diameter of the threads. Punch out the remaining shank.

B. Install a new rivnut®.

- (1) Using a PlusNut Fastener Header Tool (Figure 9005), loosen 1/2" nut by turning COUNTERCLOCKWISE until stud is fully extended.
- (2) Thread the tool completely into the fastener until it stops.
- (3) Tighten 1/2" nut until hand-tight against washer and tool body.

CAUTION: TO AVOID IRREPARABLE THREAD DAMAGE, DO NOT OVERTIGHTEN NUT.

- (4) Using an 0.75" open-ended wrench on tool body and an 0.75" socket/ratchet wrench on the nut, tighten nut until firm resistance is felt (several turns). Rivet should be physically reformed (see Figure 6005). If rivet is loose, tighten nut until rivet is secure.
- (5) Break the nut loose from the washer and body of the tool by turning nut counterclockwise, and remove tool from rivet by unscrewing it from the rivet.
- (6) TORQUE test rivet to 60 INCH-POUNDS (6.8 N-m).
- (7) Touch up the chemfilm and/or paint, as required.

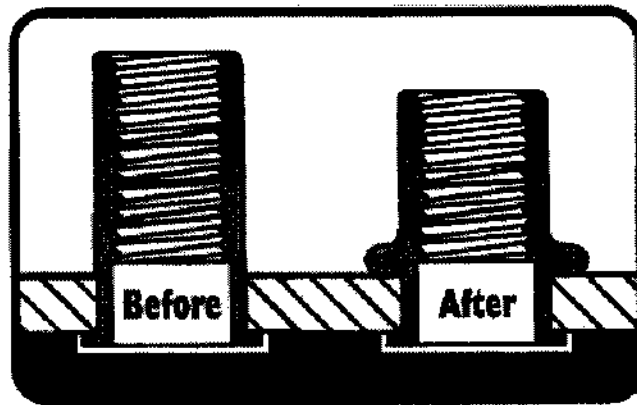


Figure 6005 - Attaching Rivnut® Fastener

C. Repair of Brush Holder

NOTE: In this section numbers in parentheses () refer to the numbers in Figure 10002.

- (1) Disassembly of complete brush holder (15) - see Figure 6006.
 - (a) Using a #40 drill bit, remove 2 rivets (70) from terminal board (60).
 - (b) Remove and discard brush lead terminal board and insulating sheet (65).
 - (c) Repeat steps (a) and (b) for each brush holder.

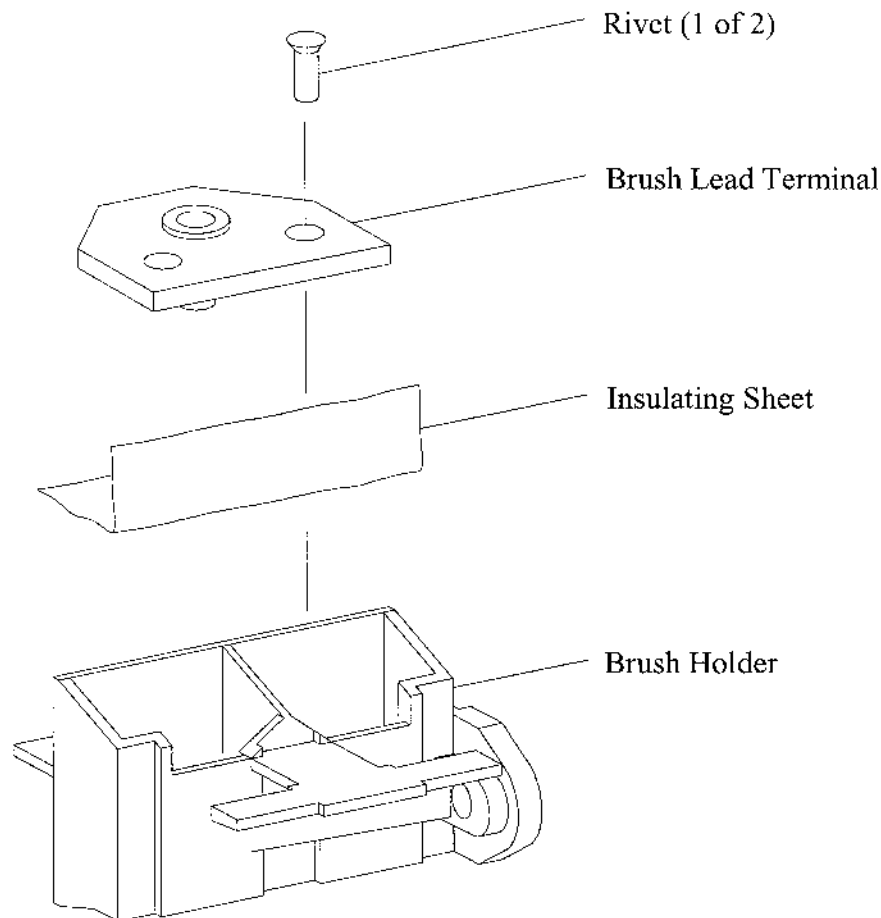


Figure 6006 - Complete Brush Holder

- (2) Assembly of Complete brush holder (15) - see Figure 6007.
 - (a) Secure rivet alignment fixture to press table.
 - (b) Secure rivet press tool to arbor press.
 - (c) Place insulating sheet (65) and brush lead terminal board (60) onto the brush holder and insert 2 rivets (70) through the holes.

NOTE: When parts are correctly aligned, the section of insulating sheet which extends outside the brush lead terminal board will fold up against the brush holder.
 - (d) Set the brush holder upside down onto rivet alignment fixture and align the rivets with the 2 cones on the fixture.
 - (e) Press rivets. The diameter of the rivet head shall be a minimum of 1.5 times the diameter of the rivet.

- (f) Repeat steps (2)(a) through (2)(e) for the remaining brush holder assemblies.

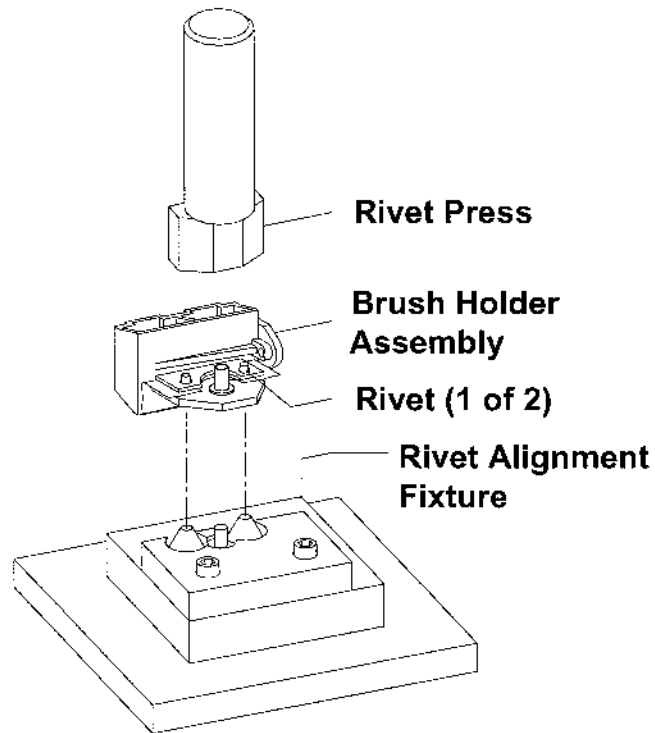


Figure 6007 - Brush Holder Rivet Alignment



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

ASSEMBLY

1. Introduction

This Section gives assembly instructions for 23079 Series DC Starter-Generator. Assemble the unit in a clean work area away from machining or other metal removing operations. Clean all parts IAW instructions in the CLEANING section before final assembly.

2. Required Tools for assembly

In addition to normal shop tools, tools in Table 7001 are needed for starter-generator assembly.

Tool Description	Reference
Arbor Press	Not Illustrated
Armature Support, Anti-Drive End	Figure 9003
Armature Support, Drive End	Figure 9002
Anti-Drive End Hub Support	Figure 9015
Compressed Air	Commercially Available
Drive End Hub Support	Figure 9014
Inner Race Bearing Drivers	Figure 9006
Outer Race Bearing Drivers	Figure 9007
Leather or Plastic Mallet	Not Illustrated
Retaining Ring Pliers	Not Illustrated
Spline Wrench	Figure 9016
Horizontal Stator and Housing Assembly Support Fixture	Figure 9012
Vertical Stator and Housing Assembly Support Fixture	Figure 9013
Wire Hook Tool	Not Illustrated
High Potential Tester	Not Illustrated
Dampener Hub Driver	Figure 9004

Table 7001 - Assembly tools

3. Materials Required for Assembly

Table 7002 lists the materials required to assemble the starter-generator.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

WARNING: BEFORE USING ANY OF THE FOLLOWING MATERIALS, BE AWARE OF ALL HANDLING, STORAGE, AND DISPOSAL PRECAUTIONS RECOMMENDED BY THE MANUFACTURER OR SUPPLIER. FAILURE TO COMPLY WITH THE RECOMMENDATIONS MAY RESULT IN SERIOUS INJURY, PHYSICAL DISORDER, OR DEATH.

Item	Description/Specification	Source (CAGE Code)
Acrylic coating	MIL-I-46058/HumiSeal 1B31, Type AR	Chase Corporation Woodside, NY
Electrical Tape	Fiberglass, MIL-I-15126, Type GFT	Commercially available
Epoxy bonding cement	Two part resin/Goodrich Corporation P/N 05-651222	Goodrich Corporation ESA Englewood, NJ (Vendor Code V19527)
Grease	Lubricating, MIL-PRF-81322	Commercially available
Insulation sleeving	Heat shrinkable, MIL-I-23053/5m size as req'd	Commercially available
Isopropyl / Alcohol	TT-I-735, Grade A Flash Point:53°F (12°C) Flammable Refer to the Material Safety Data (MSD) Sheet for the material for additional safety information	Commercially available
Lubricating and Assembly Paste	Altemp QNB50 FLAMMABLE. Read the WARNING before using this material. Refer to the applicable MSDS for additional safety information	Kluber Lubrication 54 Wentworth Avenue Londonderry, NH 03053
Paper	Protective/MIL-B-121, grade A, Type II, class 1	National Paper, Walton Hills, OH United Packaging, Cleveland, OH
Sandpaper	180 Grit, non-aluminum oxide	Commercially available
Sealant	Silicone Rubber, RTV732, Black	Dow Corning Corp. 47799 T.Halyard Drive, Plymouth, MI 48170

Table 7002 - Assembly Materials



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

Item	Description/Specification	Source (CAGE Code)
Sealing compound (Anti-seize)	Goodrich Corporation P/N 05-648318 (Gasola)	Federal Process Co. Beechwood, Ohio (Vendor Code V97785)
Thread locking adhesive	MIL-S-22473/Loctite grade D	Loctite Corporation Aurora, IL (Vendor Code VO5972)

Table 7002 - Assembly Materials (Continued)

4. **Assembly Procedure for the stator and housing assembly**

Unless otherwise specified, numbers in parentheses () refer to the item numbers of Figure 10004 of the ILLUSTRATED PARTS LIST. Where applicable, refer to the torque specifications provided in the FITS AND CLEARANCES section.

A. **Attach the terminal block (30) to the stator housing (5).**

NOTE: Before acceptance testing, attach an unserviceable terminal block to the stator housing. Acceptance testing can damage the internal capacitors of a new or serviceable terminal block.

- (1) Set the stator housing onto a horizontal stator support
- (2) On models 23079-002, 23079-002-1, 23079-003, and 23079-004:

CAUTION: DO NOT BEND THE LARGE STATOR LEADS MORE THAN NECESSARY TO INSTALL THE TERMINAL BLOCK.

- (a) Carefully bend back the large stator leads enough to insert terminal studs B, C, and E up into the large stator leads (See Figure 7001).
- (b) Set the terminal block onto the stator housing and slide it into position under the large stator leads.
- (c) Carefully lower the large stator leads onto the terminal block.
- (3) On all other models, set the terminal block onto the stator housing and slide it into position into the large stator leads.
- (4) Put the small stator leads on terminal studs A and D.
- (5) Put a spring lockwasher (40) and a flat washer (45) on each of the two screws (35).
- (6) Put the two screws in the through holes of the terminal block and torque to 20 to 25 in-lbs (2.3 to 2.8 Nm).

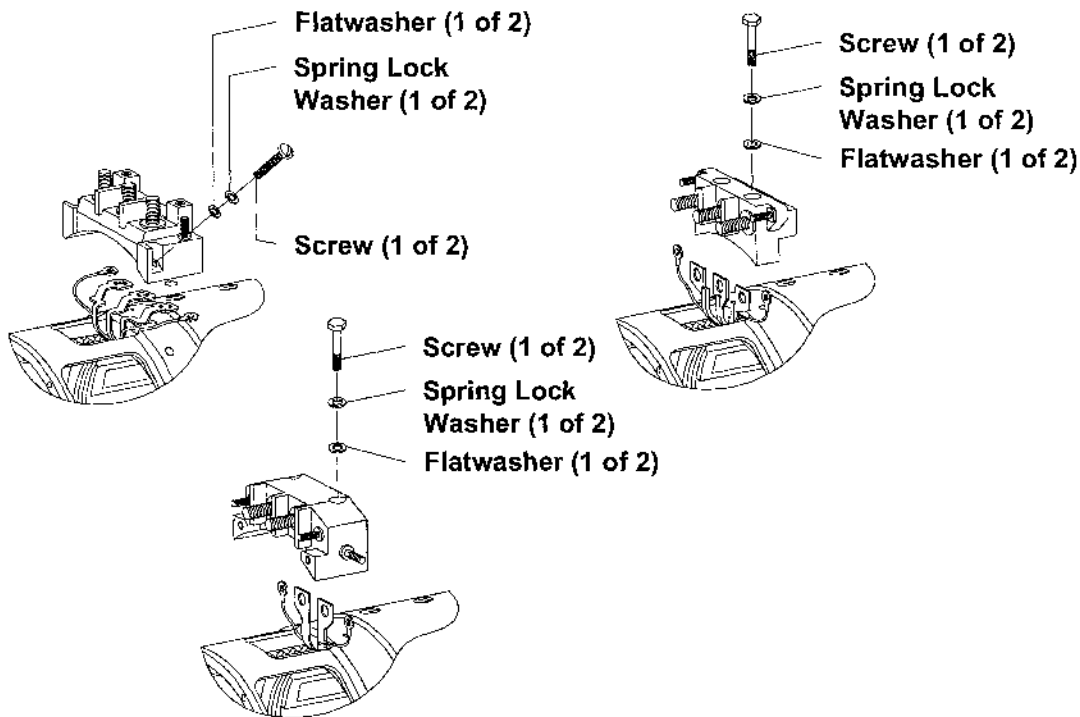


Figure 7001 - Terminal Block Attachment

5. Assembly of the Drive End End Bell Assembly

Unless otherwise specified, numbers in parentheses () refer to item numbers in Figure 10003 of the ILLUSTRATED PARTS LIST. Where applicable, refer to torque specifications provided in the FITS AND CLEARANCES section.

A. Attach the screen (15) to the drive end end bell (5).

- (1) Put a flat washer (25) on each drive screw (20).
- (2) Attach the screen to the drive end end bell assembly with the drive screws.

6. Assembly of the Anti-Drive End End Bell Assembly

Unless otherwise specified, numbers in parentheses () refer to item numbers given in Figure 10002 of the ILLUSTRATED PARTS LIST.

A. Install the brush springs (75) on the brush holder assemblies (15).

CAUTION: DO NOT WIND THE BRUSH SPRINGS TOO TIGHTLY OR THIS CAN RESULT IN REDUCED BRUSH SPRING PRESSURE.

Wind the brush springs about 3/4 turn and put two onto each brush holder assembly. See Figure 7002.

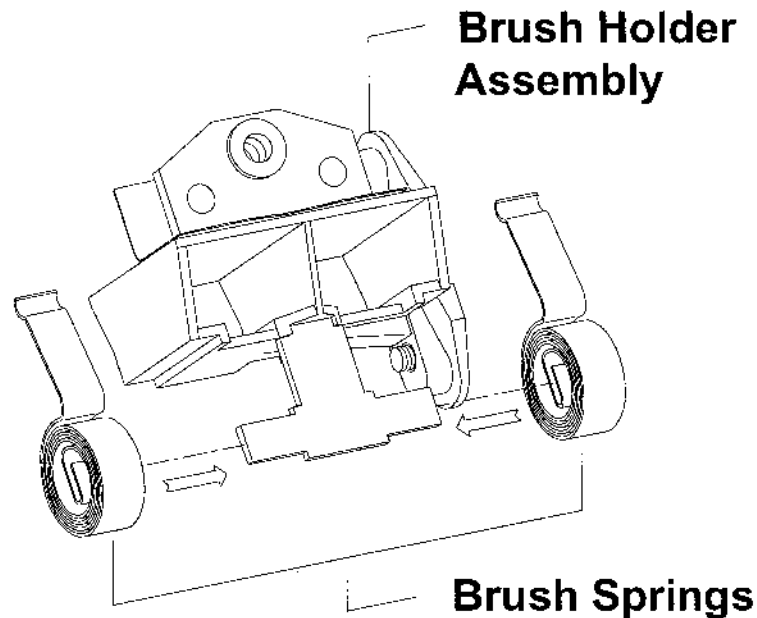


Figure 7002 - Brush Spring Installation

- B. Attach each brush holder assembly (15) to the anti-drive end end bell (5). See Figure 7003 and refer to SPD 1004.**

CAUTION: CORRECT INSTALLATION OF THE INSULATING SLEEVES IS CRITICAL. USE GREAT CARE WHEN INSERTING THE INSULATING SLEEVES INTO THE ANTI-DRIVE END END BELL. IF A SLEEVE IS PINCHED BETWEEN THE ANTI-DRIVE END END BELL AND A BRUSH HOLDER ASSEMBLY, CARBON DUST WILL COLLECT IN THE AREA WHERE THE INSULATING SLEEVE IS DAMAGED, CAUSING A PATH TO GROUND AND EQUIPMENT FAILURE. THE DAMAGE IS NOT DETECTABLE DURING THE DIELECTRIC TEST AFTER ASSEMBLY.

- (1) Insert two insulating sleeves (50) into two adjacent through holes in the anti-drive end end bell. Make sure the insulating sleeves extend out of the through holes on both sides of the anti-drive end end bell.
- (2) Put an insulating board (55) onto the ends of the exposed insulating sleeves on the inboard side of the anti-drive end end bell.
- (3) Put an insulating washer (40) onto the ends of each exposed insulating sleeve on the outboard side of the anti-drive end end bell. Make sure the insulating sleeves do not slide and that the insulating board does not fall off.
- (4) Put flat washers (45) on two attaching bolts (35).



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

WARNING: THREAD LOCKING ADHESIVE IS FLAMMABLE. DO NOT USE NEAR OPEN FLAMES, NEAR WELDING AREAS, OR ON HOT SURFACES.

- (5) Apply thread-locking adhesive grade D to the threads of the two attaching bolts.
- (6) Insert the attaching bolts into the two insulating sleeves on the outboard side of the anti-drive end end bell. Make sure that the insulating sleeves do not slide and that the insulating board does not fall off.
- (7) Put the brush holder assembly against the two bolts on the inboard side of the anti-drive end end bell.
- (8) Position and align brush holders (25) in the brush holder alignment fixture per SPD 1004. Tighten bolts as required.
- (9) Attach the brush holder assembly to the anti-drive end end bell with the two attaching bolts. Torque the bolts to 25 to 30 in-lbs. (2.8 to 3.4 Nm).
- (10) Repeat steps 1 through 9 to attach the remaining brush holder assemblies.

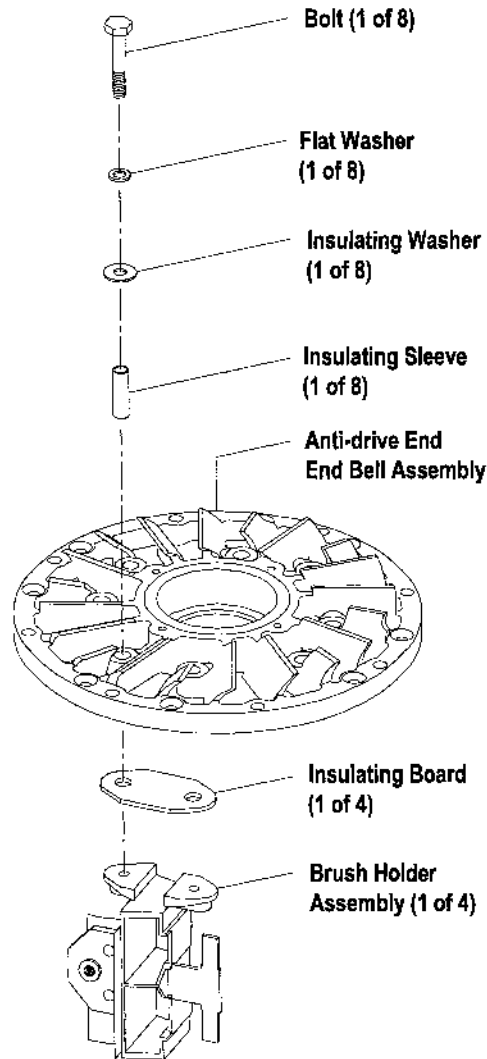


Figure 7003 - Brush Holder Attachment to the Anti-drive End End Bell

C. Perform a dielectric test IAW Paragraph 8.M.(5) of the CHECK section.

7. Final Assembly of the DC starter-generator

Unless otherwise specified, numbers in parentheses () refer to item numbers given in Figure 10001 of the ILLUSTRATED PARTS LIST.

A. Install a baffle disc (225) and a ball bearing (220) onto the drive end of the armature (205) shaft. (See Figure 7004)

NOTE: On spring loaded units using a P/N 23065-1663 drive end end bell assembly, the anti-drive end bearing must be installed with the Goodrich Corporation part mark facing the inboard side toward the armature. On non-spring loaded units using a P/N 23065-1660 or P/N 23065-1662 drive end end bell assembly, specific bearing orientation is not required.

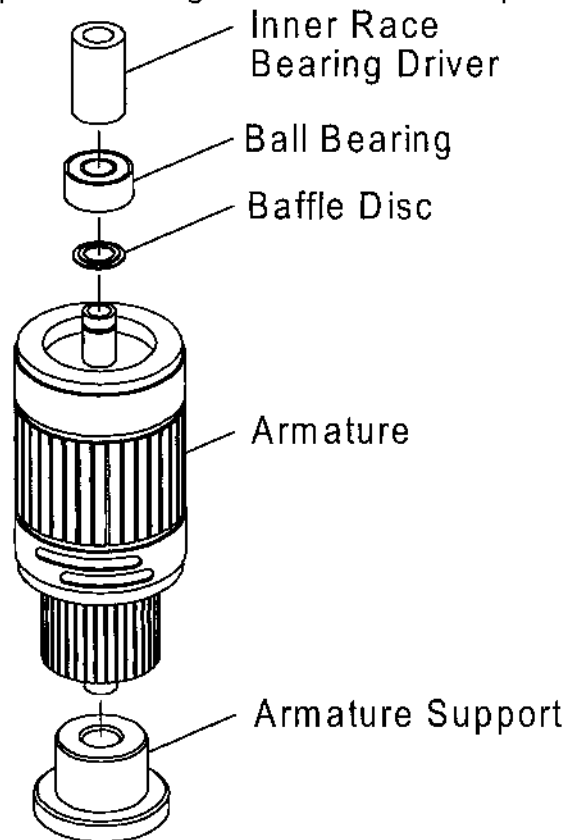


Figure 7004 - Baffle Disc and Ball Bearing Installation on Drive End of Armature Shaft

CAUTION: DO NOT TOUCH THE POLISHED SURFACE OF THE COMMUTATOR WITH BARE HANDS. SKIN ACIDS AND OILS CAN CONTAMINATE CONDUCTING SURFACES CAUSING CORROSION AND/OR POOR ELECTRICAL CONTACT.

- (1) Put an armature support on the arbor press table.
- (2) Put the armature, commutator end down on the armature support.
- (3) Put a baffle disc on the drive end of the armature shaft with the concave side facing toward the armature.
- (4) Set a ball bearing onto the armature shaft.

- (5) Set an inner race bearing driver on the ball bearing.
- (6) Press the ball bearing down onto the armature shaft
- (7) Make sure the ball bearing is fully seated against the baffle disc.

B. Install a spacer (230) or a baffle disc (225) and a retaining ring (155) onto the drive end of the armature shaft. (See Figure 7005).

- (1) On spring loaded units using P/N 23065-1663 drive end end bell assembly, put a spacer onto the drive end of the armature shaft.
- (2) On non-spring loaded units using a P/N 23065-1660 or P/N 23065-1662 drive end end bell assembly, put a second baffle disc on the drive end of the armature shaft with the concave side facing away from the armature shaft.
- (3) Put a retaining ring in the groove on the drive end of the armature shaft using snap ring pliers. Make sure it is fully engaged in the groove of the armature shaft.

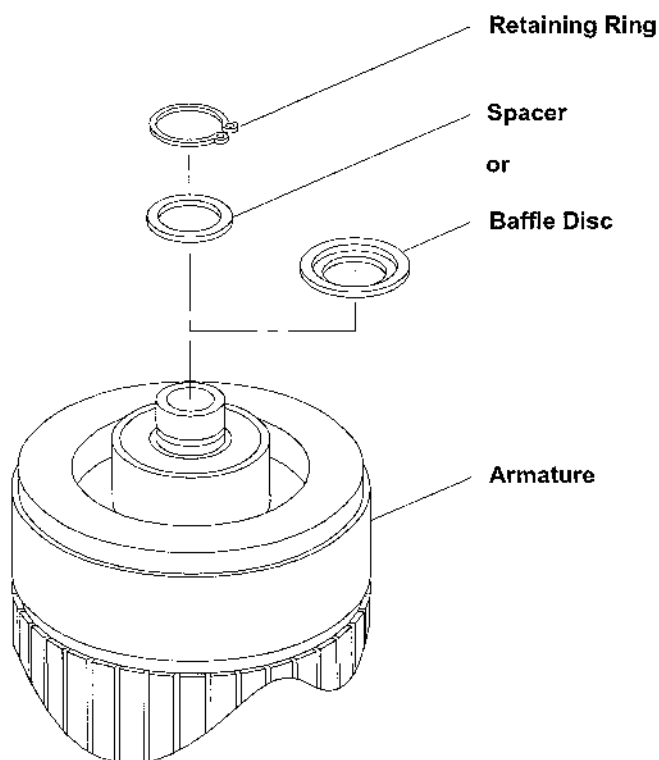


Figure 7005 - Spacer/Baffle Disc and Retaining Ring Installation on Armature Shaft



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

C. Press a ball bearing (220) into the anti-drive end end bell assembly (160). (See Figure 7006).

NOTE: On spring loaded units using a P/N 23065-1663 drive end end bell assembly, the anti-drive end bearing must be installed with the Goodrich Corporation part mark facing the inboard side toward the armature. On non-spring loaded units using a P/N 23065-1660 or P/N 23065-1662 drive end end bell assembly, specific bearing orientation is not required.

- (1) Set an anti-drive end hub support on the arbor press table.
- (2) Set the anti-drive end end bell assembly, brush holder side down, on the anti-drive end hub support.
- (3) Set a ball bearing onto the bearing liner of the anti-drive end end bell assembly.
- (4) Apply lubricating and assembly paste to the inside liner of the anti-drive end end bell assembly.
- (5) Set an outer race bearing drive onto the ball bearing.
- (6) Press the ball bearing into the bearing liner of the anti-drive end end bell assembly.
- (7) Make sure that the ball bearing is fully seated against the inner lip of the bearing liner.

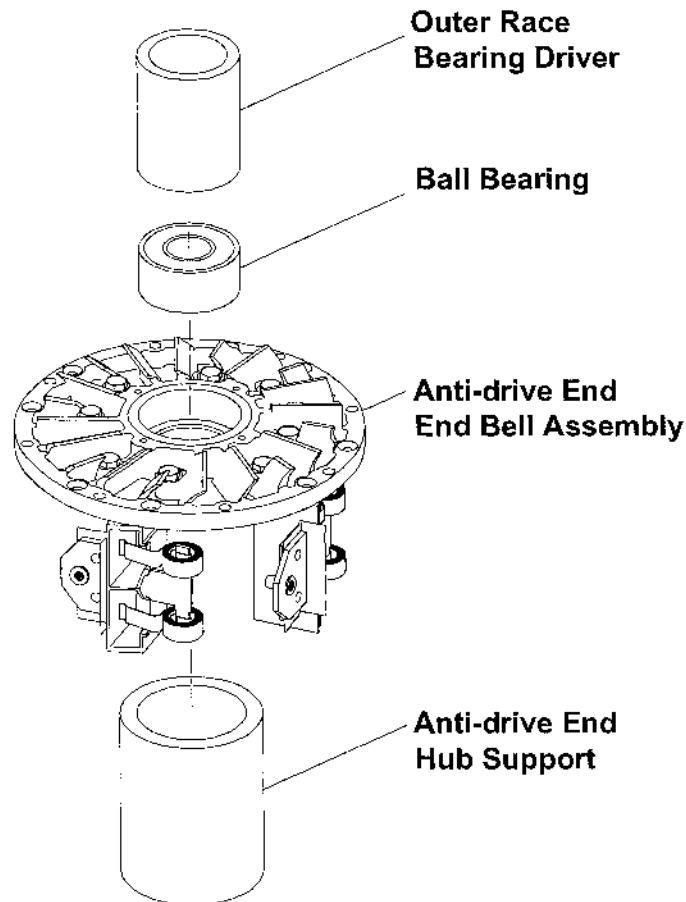


Figure 7006 - Ball Bearing Installation into Anti-drive End End Bell Assembly

- D. Attach the bearing retainer (210) to the anti-drive end end bell assembly (160). (See Figure 7007).**

WARNING: SEALING COMPOUND IS FLAMMABLE. DO NOT USE NEAR OPEN FLAMES, NEAR WELDING AREAS, OR ON HOT SURFACES.

- (1) Apply sealing compound to the threads of the four attaching screws (215).
- (2) Attach the bearing retainer to the anti-drive end end bell assembly using the attaching screws.

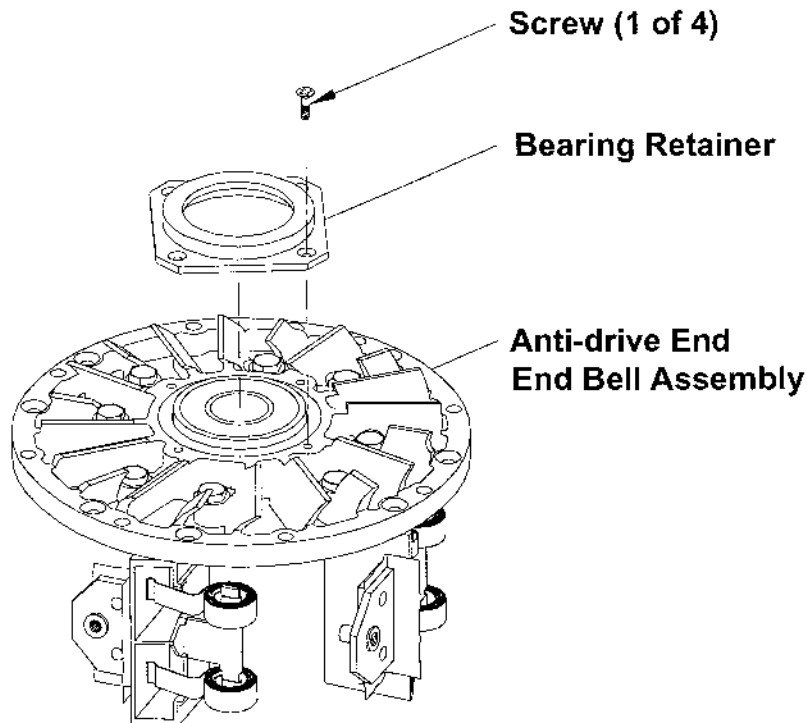


Figure 7007 - Attaching the Bearing Retainer

- E. Attach the anti-drive end end bell assembly (160) to the armature (205). (See Figure 7008).**

CAUTION: DO NOT TOUCH THE POLISHED SURFACE OF THE COMMUTATOR WITH BARE HANDS. SKIN ACIDS AND OILS CAN CONTAMINATE CONDUCTING SURFACES CAUSING CORROSION AND/OR POOR ELECTRICAL CONTACT.

- (1) Put the armature support on the arbor press table.
- (2) Set the armature, drive end down, on the armature support.
- (3) Set the anti-drive end end bell assembly, brush holder side down, onto the armature shaft.
- (4) Set an inner race bearing driver onto the ball bearing installed in the anti-drive end end bell assembly.
- (5) Press the ball bearing in the anti-drive end end bell assembly onto the armature shaft.
- (6) Make sure that the ball bearing is fully seated against the shoulder of the armature shaft.

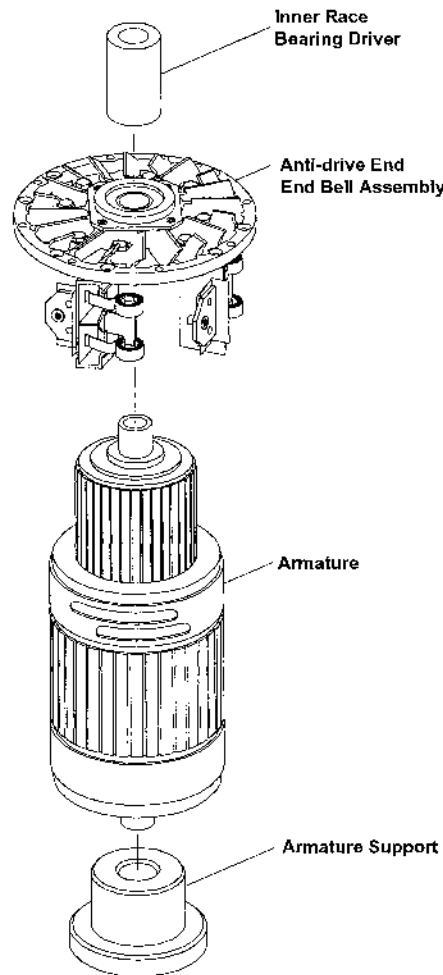


Figure 7008 - Attaching Anti-drive End End Bell Assembly and Armature

F. Attach the drive end end bell assembly (185) to the stator and housing assembly (235). (See Figure 7009).

- (1) Set the stator and housing assembly on a vertical stator support.

WARNING: SEALING COMPOUND IS FLAMMABLE. DO NOT USE NEAR OPEN FLAMES, NEAR WELDING AREAS, OR ON HOT SURFACES.

- (2) Apply sealing compound to the threads of the three attaching screws (190).
- (3) Put the three screws through the holes in the drive end end bell assembly.
- (4) Attach the drive end end bell assembly to the stator and housing assembly, aligning the attaching screws to the threaded holes in the stator and housing assembly.

- (5) Make sure the mounting surfaces of the drive end end bell assembly and the stator and housing assembly are fully seated against each other.

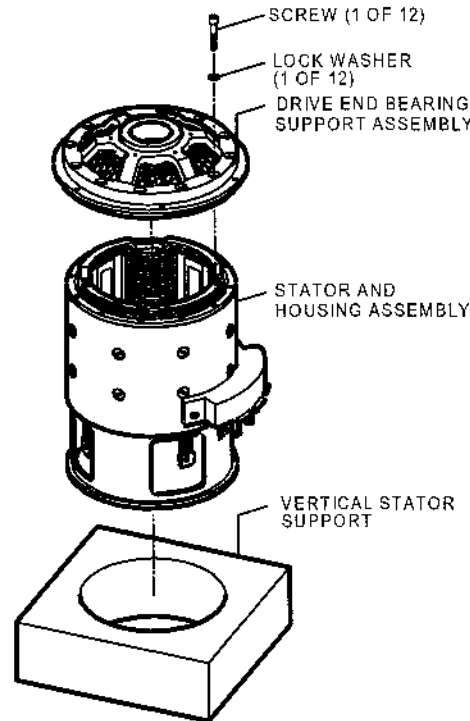


Figure 7009 - Attaching Drive End End Bell to Stator and Housing Assembly

G. Determine the shim requirements for the spring wave washer (195).

NOTE: This paragraph is applicable only for spring loaded units using a P/N 23065-1663 drive end end bell assembly.

- (1) Measure the distance from the mounting surface of the anti-drive end end bell assembly (10002) to the outer race of the bearing (220) on the drive end of the armature (205) shaft. Record as dimension "A". (See Figure 7010).
- (2) Measure the distance from the anti-drive end mounting surface of the stator and housing assembly (10004) to the bottom of the bearing liner in the drive end end bell assembly (10003). Record as Dimension "B". (See Figure 7010).
- (3) Subtract dimension "A" from dimension "B" and record the difference.

NOTE: The calculated difference between dimensions "A" and "B" is the gap for the spring wave washer. The desired gap of the compressed spring wave washer is 0.041 ± 0.005 in. (1.04 ± 0.13 mm).

- (4) Make up a shim pack with enough shims (200) to reduce the gap to within the specified tolerance range.

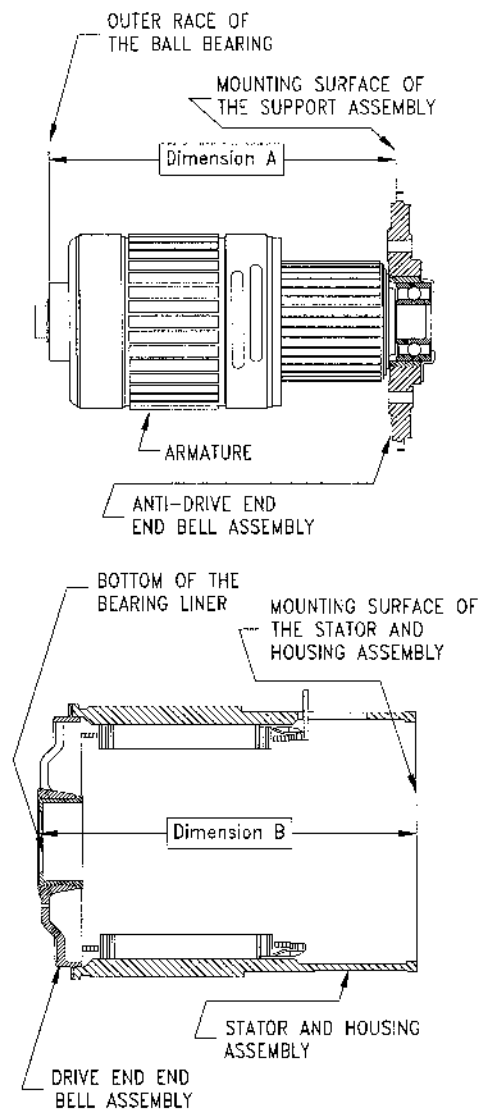


Figure 7010 - Determining Shim Requirements

H. Install the shim (200) pack and the spring wave washer (195) into the drive end end bell assembly.

NOTE: This paragraph is applicable only for spring-loaded units using a P/N 23065-1663 drive end end bell assembly.

- (1) Put the shim pack into the bottom of the bearing liner of the drive end end bell assembly.
- (2) Put the spring wave washer into the bearing liner on top of the shim pack.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

- I. **Attach the anti-drive end end bell assembly (160), with the attached armature (205), to the stator and housing assembly (235). (See Figure 7011).**
- (1) Set a hub support onto the arbor press table.
 - (2) Apply lubricating and assembly paste to the inside diameter of the bearing liner of the drive end end bell assembly.
 - (3) Carefully set the stator and housing assembly, with the attached drive end end bell assembly (185), on the hub support.
 - (4) Insert the armature, with the attached anti-drive end end bell assembly, into the stator and housing assembly.
 - (5) Make sure the drive end bearing (220) on the armature shaft is correctly centered over the bearing liner of the drive end end bell assembly.
 - (6) Make sure the brush stator leads are aligned with each brush holder assembly. If necessary, use a probe to move the brush stator leads to their correct position.
 - (7) Set an inner race bearing driver onto the ball bearing installed in the anti-drive end end bell assembly.
 - (8) Press the ball bearing on the drive end of the armature shaft into the bearing liner of the drive end end bell assembly.

WARNING: SEALING COMPOUND IS FLAMMABLE. DO NOT USE NEAR OPEN FLAMES, NEAR WELDING AREAS, OR ON HOT SURFACES.

- (9) Apply sealing compound to the threads of the eight attaching screws (165).
- (10) Attach the anti-drive end end bell assembly to the stator and housing assembly with the eight attaching screws.
- (11) Make sure that the mounting surfaces of the anti-drive end end bell assembly and the stator and housing assembly are fully seated against each other.

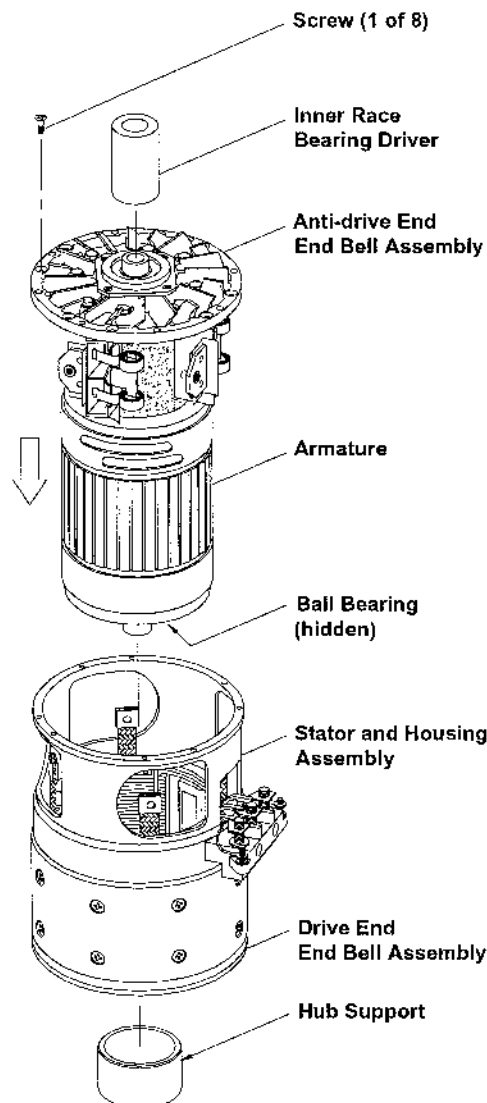


Figure 7011 - Attaching Anti-drive End End Bell with Attached Armature to Stator and Housing Assembly

J. Install the dampener backplate (150) onto the armature (205) shaft.

- (1) Set the stator and housing assembly on a horizontal stator support.
- (2) Twist the dampener backplate onto the drive end of the armature shaft until it is fully seated.

K. Install the dampener hub (135) and dampener plate (140) onto the drive shaft (125).

CAUTION: DO NOT FORCE THE DAMPENER HUB ONTO THE DRIVE SHAFT MATING TAPER OR THE HUB CAN FRACTURE.

- (1) If the dampener hub was removed from the drive shaft, twist the dampener hub by hand onto the drive shaft taper and make sure it is fully seated.
- (2) Insert the drive shaft through the dampener plate while aligning the plate and hub splines.
- (3) Lightly tap the drive end of the drive shaft using a leather or plastic mallet to seat the dampener plate onto the hub.

L. Insert the drive shaft (125) into the armature (205) shaft. (See Figure 7012)

CAUTION: DO NOT USE FORCE TO ENGAGE THE DRIVE SHAFT AND ARMATURE SHAFT MATING SPLINES. FAILURE TO ENGAGE THE DRIVE SHAFT SPLINES CORRECTLY CAN DAMAGE THE DRIVE AND ARMATURE SHAFTS.

- (1) Put the friction ring (145) in the recess in the dampener backplate (150).
- (2) Insert the drive shaft into the drive end of the armature shaft.
- (3) Push the drive shaft through the armature shaft until the dampener plate (140) is fully engaged against the friction ring.
- (4) Turn the drive shaft in the direction of rotation to make sure that the armature shaft and drive shaft splines are correctly engaged.

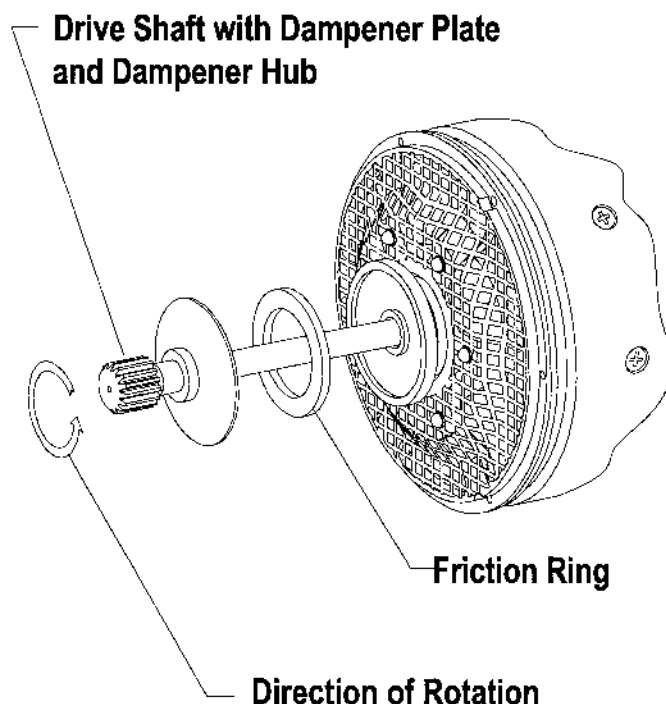


Figure 7012 - Installing Drive Shaft into Armature Shaft

M. Attach the fan (110) to the drive shaft (125). (See Figure 7013).

NOTE: To keep the drive shaft from turning, use a spline wrench on the drive spline when tightening the self-locking nut onto the drive shaft.

- (1) Put the fan onto the drive shaft.
- (2) Attach the fan to the drive shaft with a flat washer (120) and a self-locking nut (115). Torque to 100-120 in-lbs. (11.3 to 13.6 Nm).

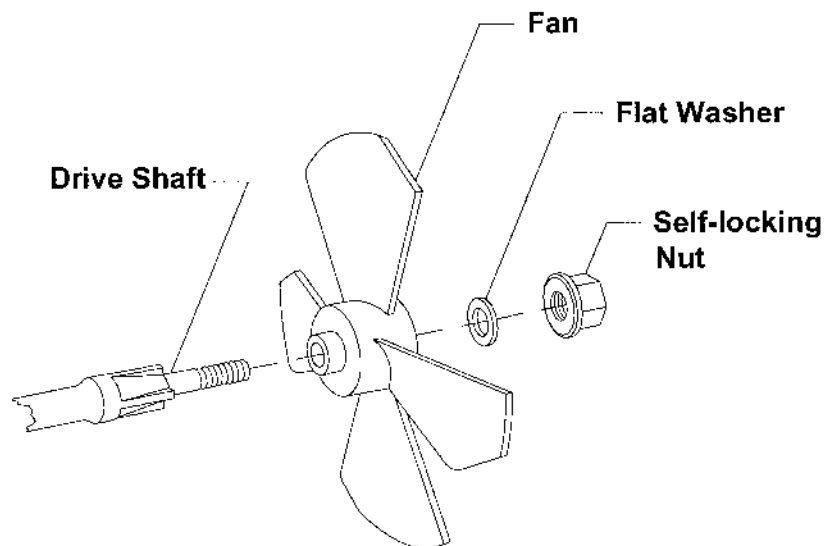


Figure 7013 - Installing the Fan

N. Attach the Thermal switch (170) to the stator and housing assembly (235). (See Figure 7014).

NOTE: For models 23079-006 and 23079-007 only.

- (1) Apply a coating of silicon rubber sealant to the threads of the thermal switch (170) and to the lockwasher (180).
- (2) Put the lockwasher onto the threads of the thermal switch.
- (3) Insert the thermal switch with lockwasher into the hole in the stator and housing assembly. Torque to 4 in-lbs (0.45 Nm) Max.

NOTE: The thermal switch will not be attached to the terminal block until after acceptance testing.

- (4) Isolate the thermal switch lead with electrical insulating tape or equivalent. Secure the lead to the stator and housing assembly.

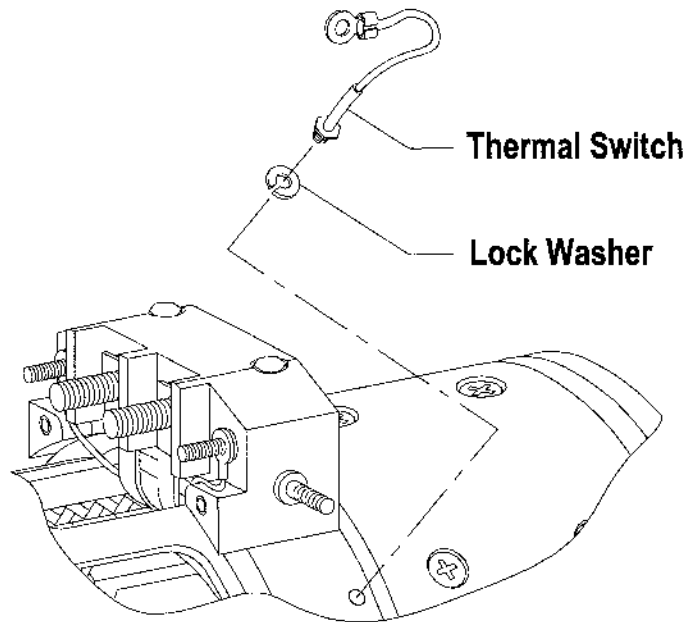


Figure 7014 - Attaching the Thermal Switch to the Stator and Housing Assembly

O. Insert the brushes (100) into the starter-generator. (See Figure 7015).

CAUTION: RAISE AND LOWER THE BRUSH SPRINGS SLOWLY. DO NOT LIFT THE BRUSH SPRINGS MORE THAN NEEDED TO INSERT THE BRUSH SETS INTO THE BRUSH HOLDER ASSEMBLY.

- (1) Set the starter-generator, anti-drive end up, onto a vertical stator support.
- (2) At each brush holder location, lift one brush spring at a time with a wire hook tool and insert the brush set into a brush holder assembly.
- (3) Slowly lower the springs on top of the brushes. Make sure the brush leads are not caught under the brush springs.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

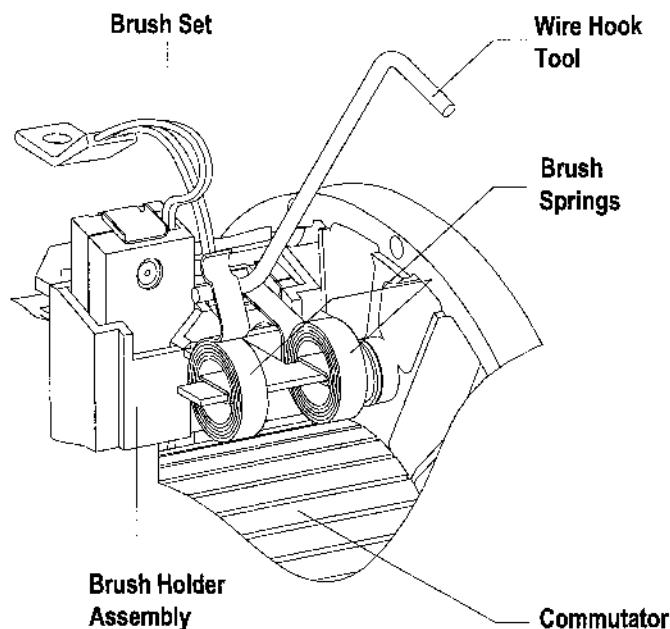


Figure 7015 - Installing Brush Sets

CAUTION: FAILURE TO FULLY SEAT THE BRUSHES CAN DECREASE BRUSH LIFE, LOWER STARTER- GENERATOR PERFORMANCE, OR CAUSE STARTER-GENERATOR FAILURE.

P. Coarse seat the new brushes (100).

NOTE: Coarse seat the Brushes IAW Goodrich Corporation SPD 1006.

NOTE: The sandpaper should first touch the leading edge (short side) of the brush when the armature (205) is rotated in its normal direction.

Q. Attach the stator leads and brush leads to each brush holder assembly. (See Figure 7016).

- (1) Set the four stator lead terminals on top of the four brush lead terminal boards.
- (2) Set the terminal of the brush leads on top of the four stator lead terminals.
- (3) Attach each stator and brush lead to the brush lead terminal boards with a screw (105).
- (4) Bend the brush leads as shown so that they do not cause interference when the brush access cover (85) is installed.

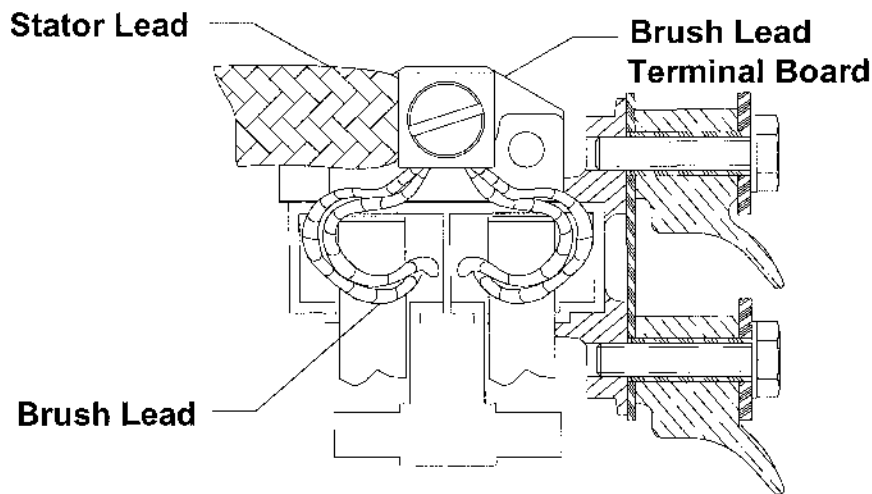


Figure 7016 - Attaching the Stator and Brush Leads

R. Attach the commutator viewing adapter to the starter-generator.

- (1) Put the commutation viewing adapter over the openings in the stator and housing assembly (235).

CAUTION: DO NOT CENTER THE BRACKETS DIRECTLY OVER AN OPENING IN THE STATOR AND HOUSING ASSEMBLY.

- (2) Center the brackets of the commutation viewing adapter between two brush access openings in the stator and housing assembly
- (3) Tighten the screw into the rivet nut that is attached to the commutation viewing adapter.

S. Attach the air inlet adapter (70) with the air inlet (55), if present, to the starter-generator. (See Figure 7017).

NOTE: For models 23079-006 and 23079-007 only. The air inlet adapter can be installed separately if the air inlet is not already attached.

- (1) Put a flat washer (80) onto each of the eight attaching screws (75) of the air inlet adapter.

WARNING: SEALING COMPOUND IS FLAMMABLE. DO NOT USE NEAR OPEN FLAMES, NEAR WELDING AREAS, OR ON HOT SURFACES.

- (2) Apply sealing compound to the screws.
- (3) Install the air inlet adapter, and air inlet if present, with the attaching screws and flat washers.

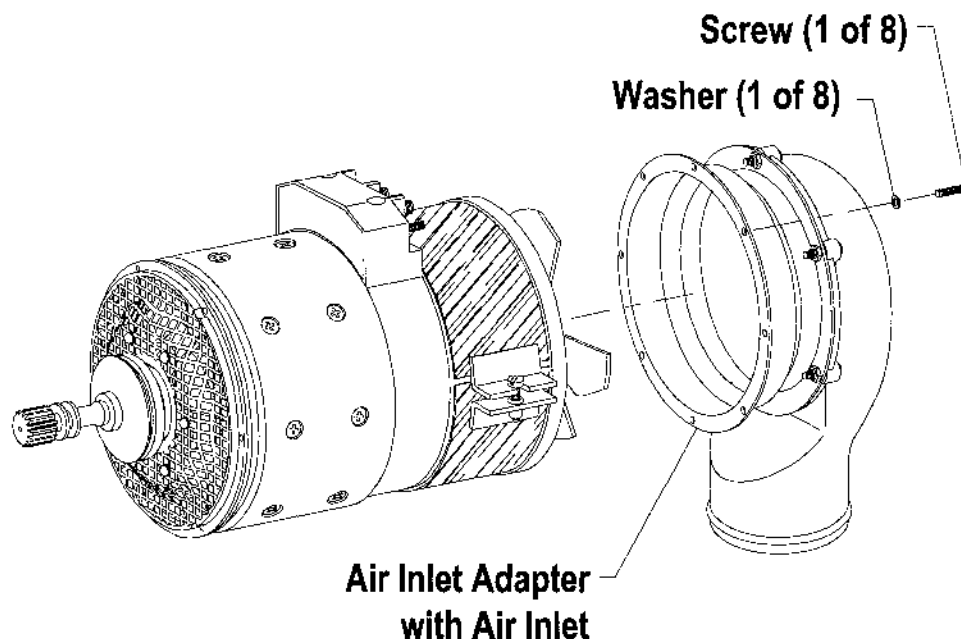


Figure 7017 - Attaching the Air Inlet Adapter with Air Inlet Attached

T. Attach the air inlet (55) to the starter-generator. (See Figure 7018)

NOTE: For models 23079-006 and 23079-007 only. The air inlet is usually attached to the air inlet adapter

- (1) Put a flat washer (65) onto each attaching screw (60) of the air inlet.

WARNING: SEALING COMPOUND IS FLAMMABLE. DO NOT USE NEAR OPEN FLAMES, NEAR WELDING AREAS, OR ON HOT SURFACES.

- (2) Apply sealing compound to the screws.
- (3) Install the air inlet with the attaching screws and flat washers.

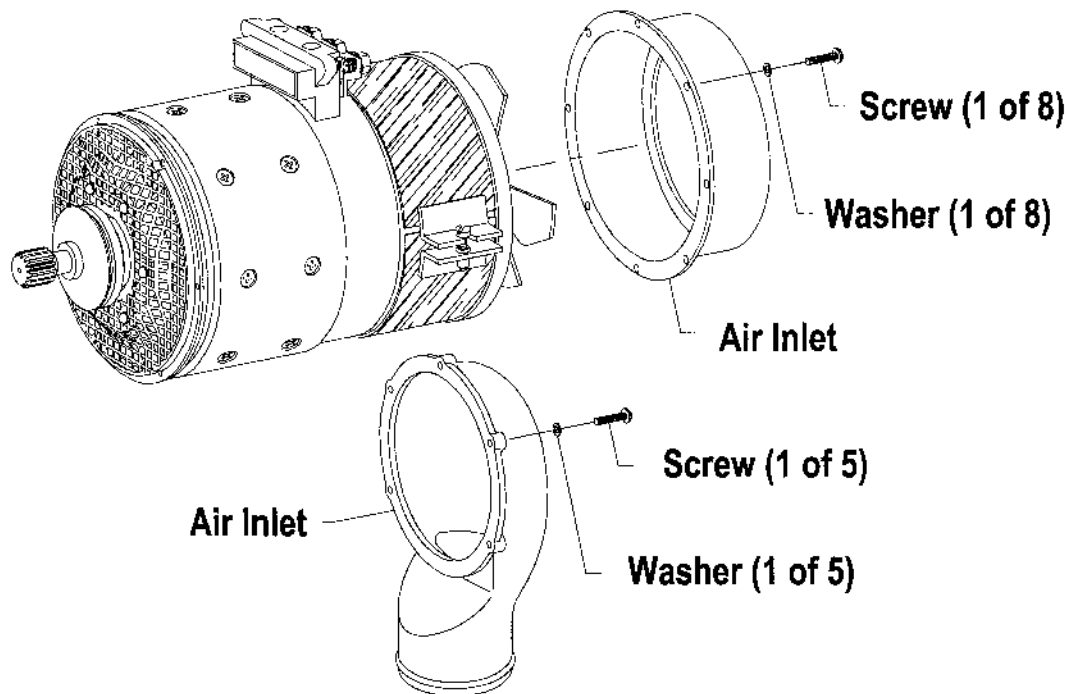


Figure 7018 - Attaching the Air Inlet

U. Perform brush (100) run-in.

CAUTION: DO NOT ALLOW THE STARTER-GENERATOR TO HANG UNSUPPORTED DURING INSTALLATION ONTO AND REMOVAL FROM THE DRIVE STAND. TOO MUCH LOAD ON THE DRIVE SHAFT SHEAR SECTION CAN DAMAGE THE UNIT.

Run-in the brushes IAW Goodrich Corporation SPD 1006.

V. Attach the nameplate (10), modification status label (30), caution label (35), patent label (40), and information plate (20), if applicable, to the stator and housing assembly (235).

NOTE: See the ILLUSTRATED PARTS LIST for the applicable model and modification status when re-identifying a starter-generator.

CAUTION: DO NOT STAMP INFORMATION DIRECTLY ONTO ANY PART OF THE STATOR AND HOUSING ASSEMBLY.

- (1) If starter-generator is modified and the original nameplate is not damaged:
 - (a) Put a blank modification status label on a wooden work surface.
 - (b) Stamp the letter "X" under the applicable letter on the modification status label.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

- (c) Apply the modification status label to the stator and housing assembly next to the nameplate and orientated to read in the same direction as the nameplate.
- (2) If the starter-generator was modified and the nameplate was damaged:
 - (a) If the starter-generator has a modification status label, note the existing modification status, then remove and discard it.
 - (b) Remove and retain the old nameplate.
 - (c) Put a new replacement nameplate on a wooden work surface.
 - (d) Transfer all the information from the old nameplate by stamping the information with a 1/8" letter punch set and a light hammer on the replacement nameplate.
 - (e) Discard the old nameplate.
 - (f) Stamp the letter "X" under the applicable letters on the replacement nameplate.
 - (g) Attach the replacement nameplate to the stator and housing assembly using four drive screws.
 - (h) Apply acrylic coating to the replacement nameplate.
 - (i) Allow the coating to air dry.
- (3) If the starter-generator was not modified but the nameplate was damaged:
 - (a) Put a new replacement nameplate on a wooden work surface.
 - (b) Transfer all the information from the old nameplate by stamping the information with a 1/8" letter punch set and a light hammer on the replacement nameplate.
 - (c) Discard the old nameplate.
 - (d) Attach the replacement nameplate to the stator and housing assembly using four drive screws.
 - (e) Apply acrylic coating to the replacement nameplate.
 - (f) Allow the coating to air dry.
- (4) If the caution label is damaged or unreadable:
 - (a) Remove the label from the stator and housing assembly.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

WARNING: ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. DO NOT USE NEAR OPEN FLAMES, WELDING AREAS, OR ON HOT SURFACES. INHALATION OF VAPORS CAN CAUSE DROWSINESS, DIZZINESS, AND HEADACHE. CONTACT WITH SKIN CAN CAUSE IRRITATION. USE IN A WELL-VENTILATED AREA.

- (b) Clean the surface of the stator and housing assembly using a cloth moistened with isopropyl alcohol.
 - (c) Apply a new label to the stator and housing assembly next to the terminal block.
- (5) If the patent label is damaged or unreadable.
- (a) Remove the patent label from the stator and housing assembly.

WARNING: ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. DO NOT USE NEAR OPEN FLAMES, WELDING AREAS, OR ON HOT SURFACES. INHALATION OF VAPORS CAN CAUSE DROWSINESS, DIZZINESS, AND HEADACHE. CONTACT WITH SKIN CAN CAUSE IRRITATION. USE IN A WELL-VENTILATED AREA.

- (b) Clean the surface of the stator and housing assembly using a cloth moistened with isopropyl alcohol.
- (c) Apply a new patent label to the stator and housing assembly next to the nameplate and orientated to read in the same direction as the nameplate.

NOTE: For models 23079-006 and 23079-007 only.

- (6) If the information plate was damaged:
- (a) Remove and retain the old information plate.
 - (b) Put a new replacement information plate on a wooden work surface.
 - (c) Transfer all the information from the old information plate by stamping the information with a 1/8" letter punch set and a light hammer on the replacement information plate.
 - (d) Discard the old information plate.
 - (e) Attach the replacement information plate to the stator and housing assembly using four drive screws (-25).
 - (f) Apply acrylic coating to the replacement information plate.
 - (g) Allow the coating to air dry.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

8. Mandatory Acceptance Testing

All starter-generators that have been overhauled, repaired, or modified must be acceptance tested IAW Paragraphs 5.A through 5.L of the TESTING AND FAULT ISOLATION section before they are put into storage or returned to service.

9. Generator storage/shipment preparation

Numbers in parentheses () refer to item numbers given in Figure 10001 through Figure 10005 of the ILLUSTRATED PARTS LIST.

A. Attach a serviceable terminal block (10004-30) to the starter-generator.

NOTE: A dummy terminal block is used during acceptance testing.

- (1) Remove the dummy terminal block according to the instructions in Paragraph 6 of the DISASSEMBLY section.
- (2) Attach a terminal block according to instructions in Paragraph 4 of this section.

B. Attach the brush access cover (10001-85) to the starter-generator.

- (1) Put the brush access cover over the openings in the stator and housing assembly (10001-235).

CAUTION: ORIENT AIR BLEED HOLES IN THE BRUSH BAND TOWARD THE TERMINAL BLOCK.

CAUTION: DO NOT CENTER THE BRACKETS DIRECTLY OVER AN OPENING IN THE STATOR AND HOUSING ASSEMBLY. POSITION THE SPLIT LINE OVER A HOUSING RIB.

- (2) Center the brush access cover between the four brush access openings in the stator and housing assembly.
- (3) Tighten the screw (10001-95) into the rivet nut (10001-90) that is attached to the brush access cover. Torque screw to 20-30 in-lbs (2.3 to 3.4 Nm).

C. Put the attaching hardware onto the terminal block (10004-30).

- (1) Put a flat washer (10004-25) and a self-locking nut (10004-20) onto terminal studs A and D. Tighten the self-locking nuts sufficiently to retain them during shipment.
- (2) Put a flat washer (10004-15) and a self-locking nut (10004-10) onto terminal studs B, E, and, if present, C. Tighten the self-locking nuts sufficiently to retain them during shipment.

D. Attach the lead wire of the thermal switch (10001-170) to the terminal block (10004-30).

NOTE: For models 23079-006 and 23079-007 only.

- (1) Remove any electrical insulation, if present, from the thermal switch lead.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

- (2) Attach the terminal switch to the terminal block with the self-locking nut (-175).

E. Attach the terminal block cover (10001-45) to the starter-generator if present.

NOTE: For models 23079-002, 23079-002-1, 23079-003, 23079-004, 23079-006, and 23079-007.

- (1) Put the terminal block cover onto the terminal block.
- (2) Secure the terminal block cover with two attaching screws (10001-50).

F. If present, attach the QAD mounting kit (10001-5) to the starter-generator.

NOTE: When a starter-generator has been removed from service, the QAD mounting kit usually stays on the aircraft.

- (1) Engage the guide pins of the mounting adapter (10005-25) to the pin holes in the drive end end bell assembly (10001-185).
- (2) Be sure the guide pins are located correctly and that the pilot diameters of the mounting adapter and the drive end end bell assembly are fully engaged.
- (3) While holding the mounting adapter firmly in place, put the V-band clamp (10005-5) over the flanges of the mounting adapter and the drive end end bell assembly.
- (4) Tighten the self-locking nut (10005-10) onto the T-bolt (10005-15). Torque to 70 in-lbs. (7.9 Nm) to lock the QAD mounting kit to the starter-generator.

G. Prepare the starter-generator for storage/shipment

- (1) Apply lubricating grease to the drive spline of the drive shaft (10001-125) using MIL-PRF-81322.
- (2) Wrap the drive spline with MIL-B-121, grade A, type II, class 1 protective paper. Secure with waterproof tape.
- (3) Put a section of cardboard tubing over the drive shaft to prevent the drive shaft from puncturing the shipping container.
- (4) Wrap the starter-generator in chemically neutral protective paper or polyethylene (plastic) wrap and seal with waterproof tape to prevent corrosion and airborne contamination from getting inside the starter-generator.
- (5) Put the starter-generator in its original container if possible.

NOTE: An alternate shipping container can be constructed of double walled corrugated cardboard or equivalent. The inside dimensions of the shipping container shall have a minimum of 6 inches (150 mm) clearance from the starter-generator. The starter-generator must be protected from the sides of the shipping container and from movement inside the shipping container. The packing material shall be soft, shock absorbing material such as foam rubber, Styrofoam, or bubble wrap.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

- (6) For models (23079-006 and 23079-007), put a new O-ring (10001-130) into a suitable protective container and label the container with instructions to the user to lubricate and install the O-ring onto the drive end of the drive shaft before installation of the starter-generator into the aircraft.
- (7) Include all appropriate inspection, test, repair, and installation documents in the shipping container with the starter-generator.
- (8) Close the shipping container using heavy duty staples. Apply waterproof tape to the seams of the shipping container. As an added precaution against container separation while in transit, the use of steel or plastic bands wrapped tightly around the shipping container is recommended.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

FITS AND CLEARANCES

1. Acceptance Limits

Table 8001 lists the acceptance limits applicable to starter-generator components and subassemblies. Paragraph designations in the CHECK column refer to applicable inspection procedures found in the CHECK section for cross reference.

CHECK	Nomenclature	Inspection	Acceptance Limits
8.B.	Mounting Adapter	Guide Pin Height	0.055 - 0.075 in. (1.40 - 1.91 mm)
		Pilot Flange Dia.	4.120/4.122" (104.65/104.70mm)
		Pilot Bore Dia.	5.870/5.872" (149.1/149.15 mm)
8.I.	Drive Shaft	Drive Spline Diameter	0.921 in. (23.39 mm) minimum over a pin diameter of 0.096 in. (2.438 mm)
8.J.	Friction Ring	Thickness	0.060 in. (1.524 mm) minimum
8.K	Dampener Plate	Thickness	0.038 in. (0.97 mm) minimum
		Spline Diameter	0.655 in. (16.637 mm) maximum between two 0.090 (2.286 mm) pins
8.M.	Anti-drive End End Bell Assy	Bearing Bore Diameter	1.8501 - 1.8504 in. (46.993 - 47.000 mm) See Figure 8001
	Brush Spring	Pressure	3.1 - 4.0 lb. (13.8 - 17.8 N)
8.N.	Drive End End Bell Assembly	Bearing Bore Diameter	1.8501 - 1.8504 in. (46.993 - 47.000 mm) See Figure 8001
8.O.	Armature	Bearing Journal Diameter	0.7872 - 0.7875 in. (19.995 - 20.003 mm) See Figure 8002
		Commutator Diameter	2.50 in. (63.50 mm) Minimum
		Commutator Undercut	Depth: 0.045 in. (1.143 mm) Minimum Width: 0.035-0.045 in. (0.889-1.143 mm)
		Commutator Surface Finish	64 - 100 micro inch (1.6 - 2.5 μ m)
		Commutator Concentricity	Bar to bar: 0.0002 in. (5.1 μ m) T.I.R.: 0.0008 in. (20.3 μ m)
		Balance	5 grain in. (8.2 gr-mm) maximum at each end
8.T	Terminal Block	Capacitance Values	Micro farads See Figure 8003

Table 8001 - Acceptance Limits



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

NOTE: During overhaul inspection, a bearing liner is acceptable if up to 25% of the circumference is 0.0001 in. (0.0025 mm) above the maximum diameter limit. See Figure 8001. This criteria does not apply to new or repaired bearing liners.

NOTE: An armature journal diameter is acceptable if up to 25% of the circumference is 0.0001 in. (0.0025 mm) less than the minimum diameter limit. See Figure 8002. This criteria does not apply to new or repaired armature journals.

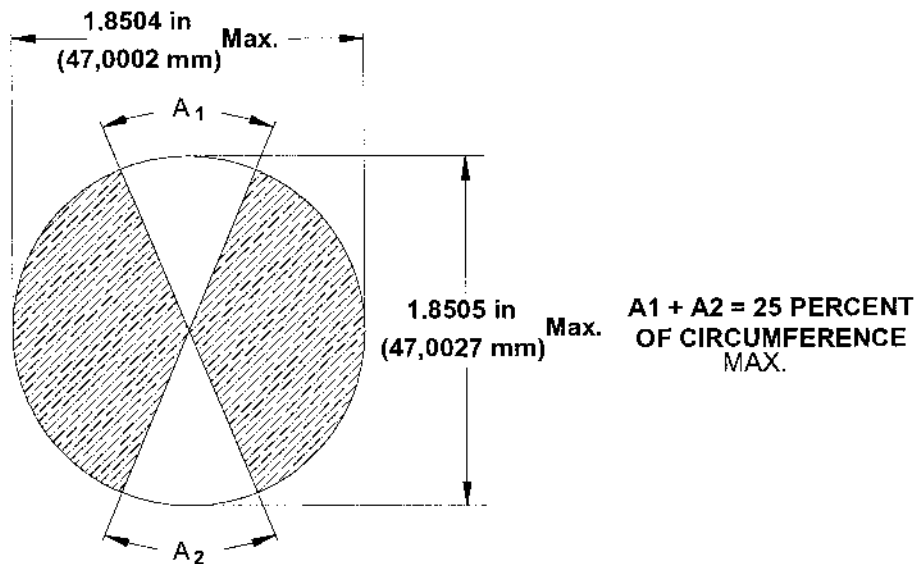


Figure 8001 - Measuring Liner Diameter



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

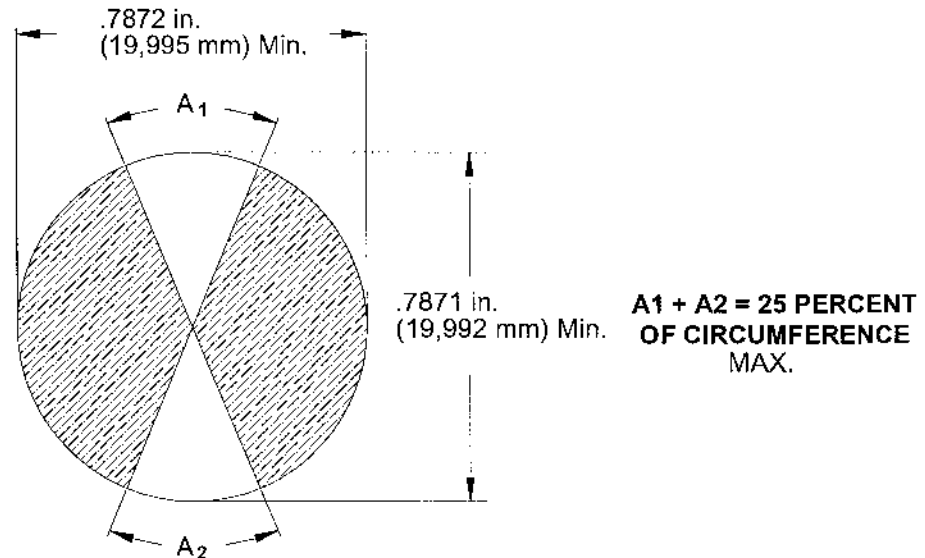


Figure 8002 - Measures the Journal Diameter

NOTE: Terminal block capacitance values are measured at a frequency of 120 Hz $\pm$ 10 Hz and at an ambient temperature between 50° to 104°F (10° to 40°C). All capacitance values shown are in micro farads.

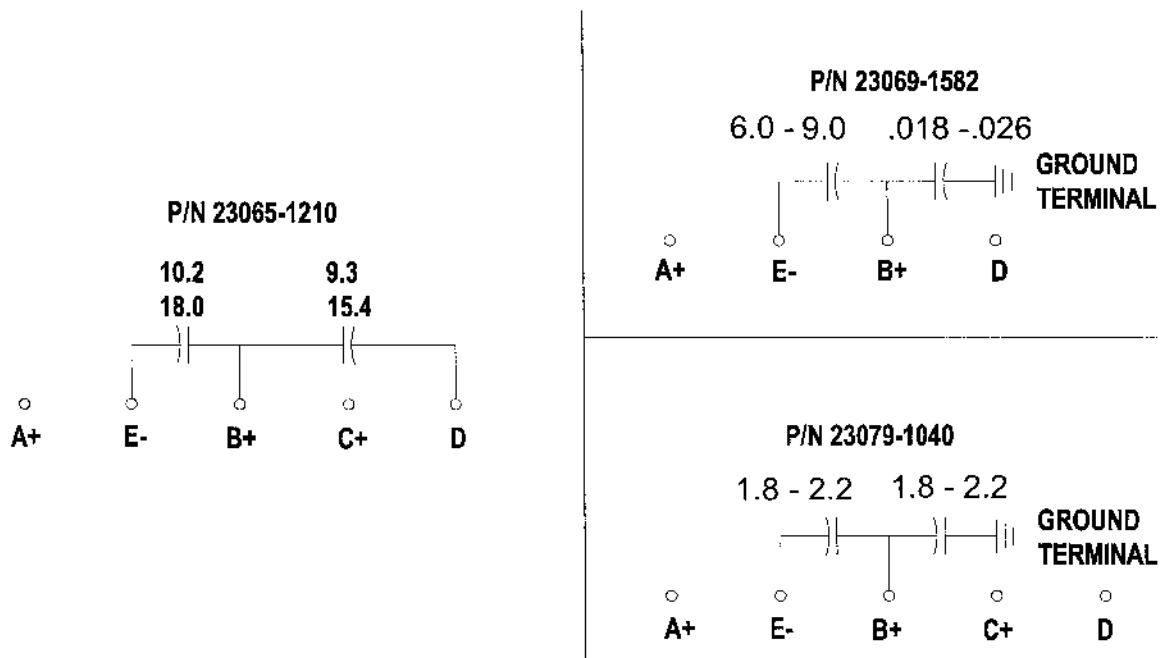


Figure 8003 - Terminal Block Schematics



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

2. Torque Limits

The following torque limits are used to maintain the DC Starter-Generator.

Fig. Item No.	Nomenclature	Torque (Newton Meters)
10001-90	Rivnut®	60 in.-lbs (6.8 N-m) Min. torque out
10001-95	Screw	20 to 30 in.-lbs (2.3 to 3.4 N-m)
10001-115	Nut, Self-Locking	100 to 120 in.-lbs (11.3 to 13.5 N-m)
10001-170	Thermostatic Switch	4 in.-lbs (0.45 N-m)
10002-35	Bolt, Machine	25 to 30 in.-lbs (2.8 to 3.4 N-m)
10005-5	Coupling, V-Retainer	70 in.-lbs (7.9 N-m)

Table 8002 - Torque Limits



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

SPECIAL TOOLS, FIXTURES, AND EQUIPMENT

1. Introduction

Table 9001 lists the approved special tools, fixtures, and equipment necessary for generator service.

Nomenclature	Figure No.
Adapter, commutation, Viewing	9001
Adapter, Armature Shaft (drive end)	9002
Adapter, Armature Shaft (anti-drive end)	9003
Driver, Dampener Hub	9004
Driver, PlusNut™ Fastener Header	9005
Driver, Bearing, Inner-race	9006
Driver, Bearing, Outer-race	9007
Fixture, Commutator Turning	9008
Fixtures, Rivet Alignment and Rivet Press	9009
Support, Armature	9010
Support, Anti-drive end end bell Assembly	9011
Support, Stator, Horizontal	9012
Support, Stator, Vertical	9013
Support, Hub (drive end)	9014
Support, Hub (anti-drive end)	9015
Wrench, Spline	9016

Table 9001 - Special Tools, Fixtures, and Equipment

2. Tool Descriptions

The drawings in this section provide material and fabrication instructions for the tools listed in Table 9001. Where dimensions and fabrication instructions are provided, the tool or fixture is approved for local manufacture.

A. Commutation Viewing Adapters. (See Figure 9001).

This adapter is constructed using a used brush access cover. Two designs are shown. Design one uses four pieces of acrylic, 2.00 in. (50.8 mm) high by 1.75 in. (44.5 mm) wide by 0.63 in. (1.60 mm) thick. Design two uses one piece of acrylic, 19.00 in. (482.6 mm) long by 2.30 in. (58.4 mm) by 0.63 in. (1.60 mm) thick. Construction of these designs is as follows:

(1) Design One

- (a) Cut four holes in a brush access cover. Each hole should be positioned above one of the four brush holder assemblies to allow for viewing the brushes as they make contact with the commutator during test.
- (b) Cut four pieces of clear acrylic to be larger than the brush access holes in the stator and housing assembly.
- (c) Apply adhesive on the inside surface of the viewing adapter to the frame of the cut out viewing windows.
- (d) Attach the four pieces of acrylic to the viewing windows on the inner surface of the viewing adapter using four pop rivets.

(2) Design Two

- (a) Remove the brackets from a used brush access cover.
- (b) Attach the brackets to a piece of clear acrylic.

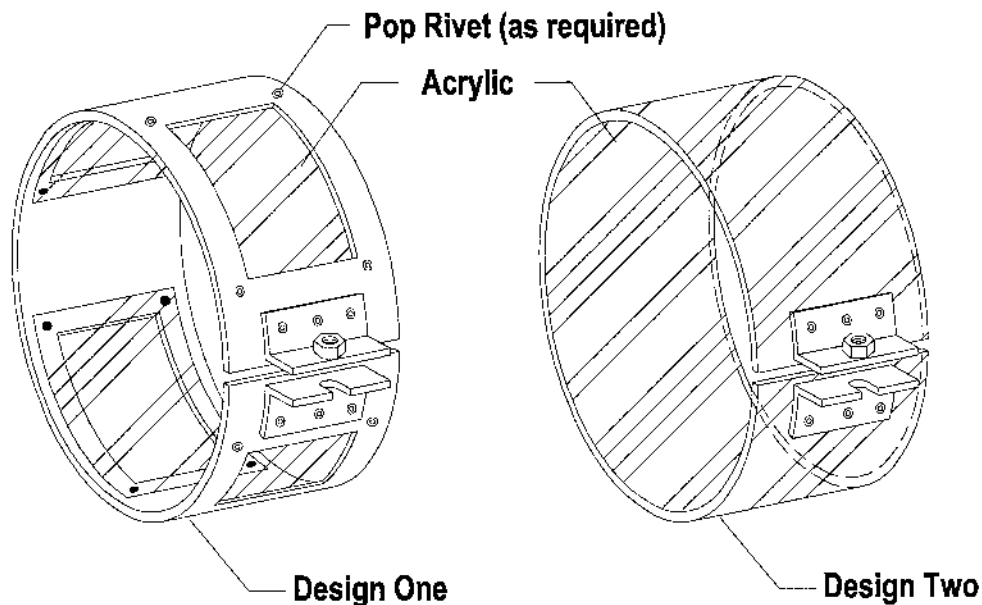


Figure 9001 - Commutation Viewing Adapter

B. Drive End Armature Shaft Adapter. (See Figure 9002).

Unless otherwise specified, dimensions are in inches. Tolerances on decimals are: .XX±0.01, XXX±0.005. Angles are ±5°0'. Machined ends to be parallel within 0.001 inch and square with sidewalls to ±0.05 degrees. Material is CD-260 Brass or equivalent (Stock size: 1.00" diameter).

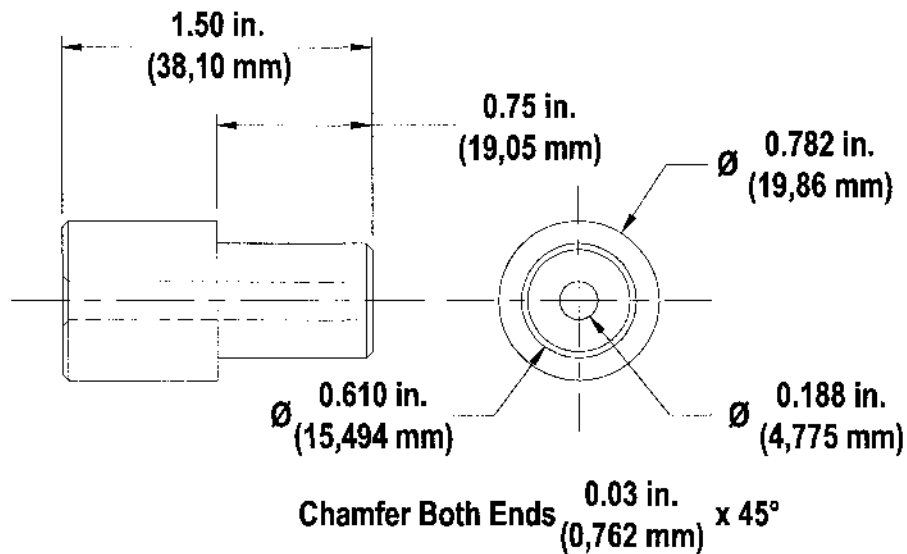


Figure 9002 - Drive End Armature Shaft Adapter



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

C. Anti-Drive End Armature Shaft Adapter. (See Figure 9003).

Unless otherwise specified, dimensions are in inches. Tolerances on decimals are: .XX $\pm$ 0.01, XXX $\pm$ 0.005. Angles are $\pm$ 5°0'. Machined ends to be parallel within 0.001 inch and square with sidewalls to $\pm$ 0.05 degrees. Material is CD-260 Brass or equivalent (Stock size: 1.00" diameter).

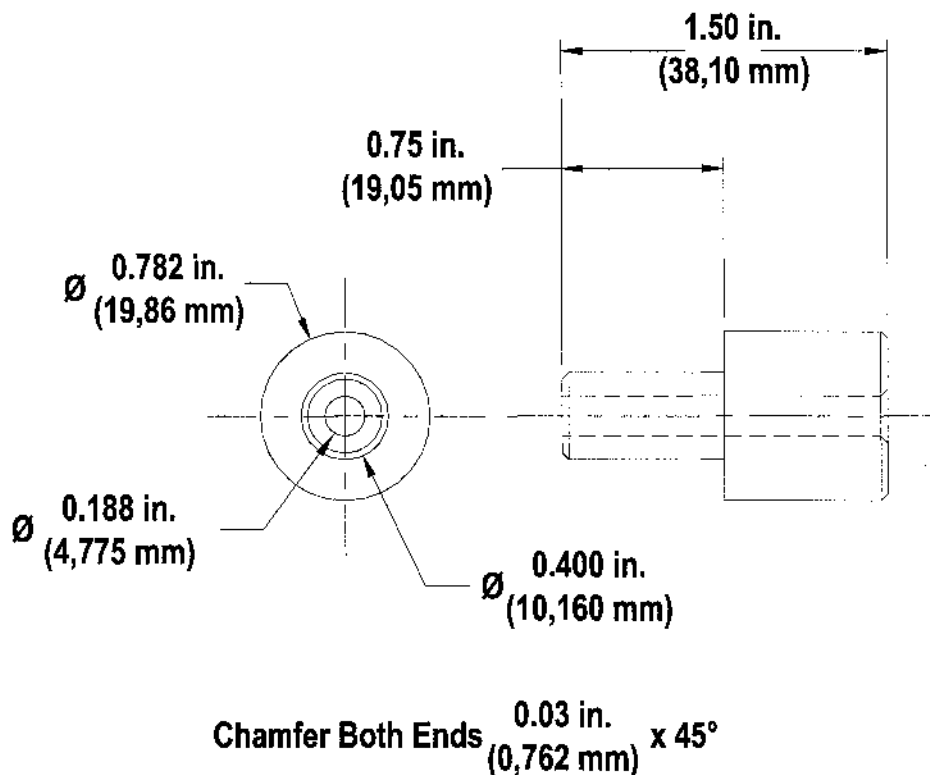


Figure 9003 - Anti-Drive End Armature Shaft Adapter



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

D. Dampener Hub Driver. (See Figure 9004).

Unless otherwise specified, dimensions are in inches. Tolerances on decimals are: .XX $\pm$ 0.01, XXX $\pm$ 0.005. Angles are $\pm$ 5°0'. Material is 01 tool or equivalent hardened to Rc 55-60. Stock size is 3.00 diameter.

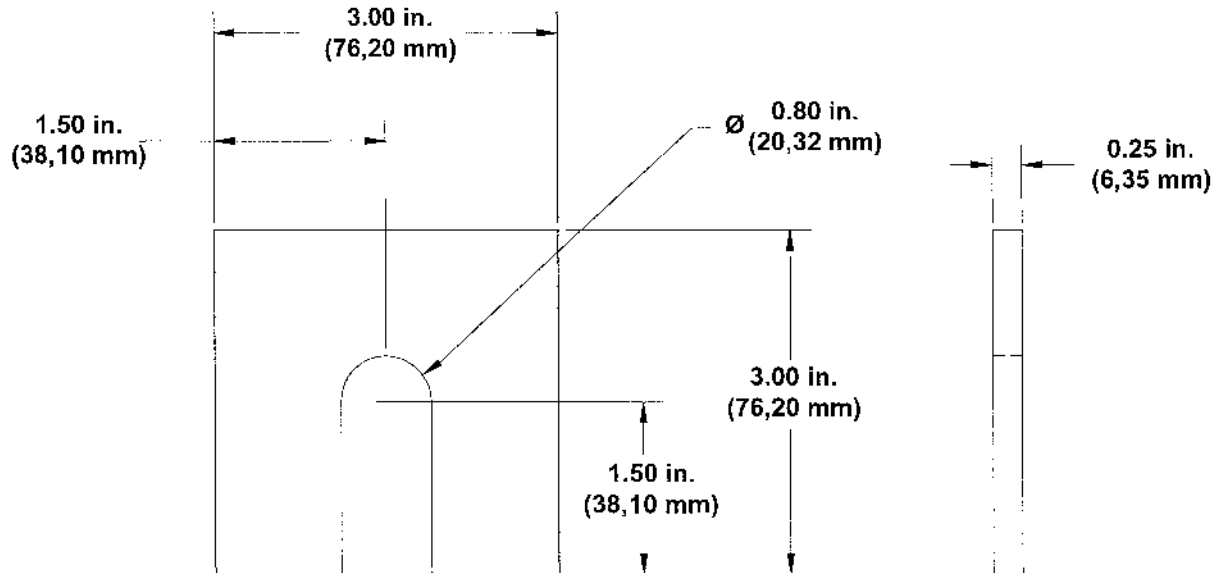


Figure 9004 - Dampener Hub Driver

E. PlusNut™ Fastener Header (See Figure 9005).

NOTE: For rivnut® replacement - Reference REPAIR. Unless otherwise specified, dimensions are in inches. Tolerances on decimals are: .XX $\pm$ 0.01, XXX $\pm$ 0.005. Angles are $\pm$ 5°0'. Machined ends to be parallel within 0.001 inch and square with sidewalls to $\pm$ 0.05 degrees. Material is 01 tool steel or equivalent hardened to Rc 35-40. Use 2.00" diameter stock.

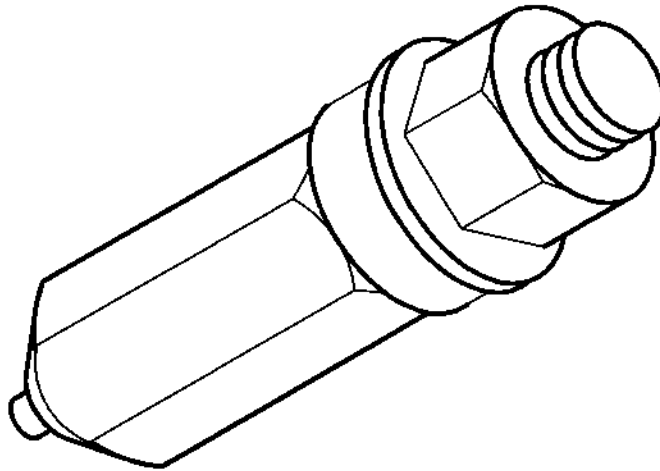


Figure 9005 - PlusNut™ Fastener Header

F. Inner Race Bearing Driver. (See Figure 9006).

Unless otherwise specified, dimensions are in inches. Tolerances on decimals are: .XX $\pm$ 0.01, XXX $\pm$ 0.005. Angles are $\pm$ 5°0'. Machined ends to be parallel within 0.001 inch and square with sidewalls to $\pm$ 0.05 degrees. Material is 01 tool steel or equivalent hardened to Rc 55-60. Stock size: 2.00" diameter.

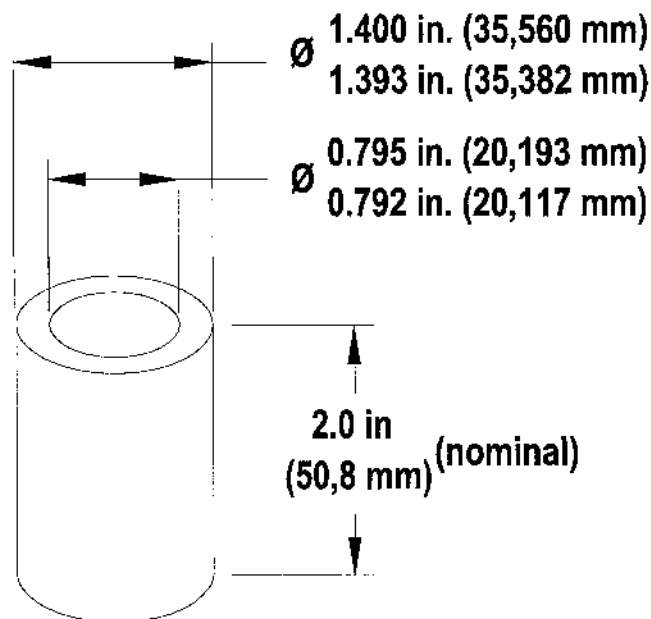


Figure 9006 - Inner Race Bearing Driver



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

G. Outer Race Bearing Driver. (See Figure 9007).

Unless otherwise specified, dimensions are in inches. Tolerances on decimals are: $.XX \pm 0.01$, $XXX \pm 0.005$. Angles are $\pm 5^\circ 0'$. Machined ends to be parallel within 0.001 inch and square with sidewalls to ± 0.05 degrees. Material is 01 tool steel or equivalent hardened to Rc 55-60. Stock size: 2.00" diameter.

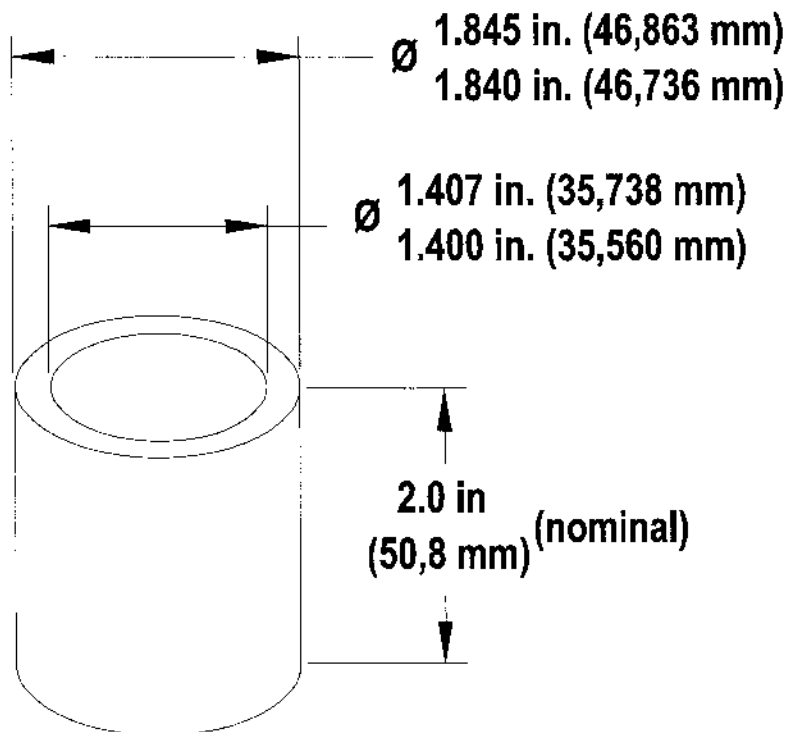


Figure 9007 - Outer Race Bearing Driver

H. Commutator Turning Fixture. (See Figure 9008).

Unless otherwise specified, dimensions are in inches. Tolerances on decimals are: .XX $\pm$ 0.01, XXX $\pm$ 0.005. Angles are $\pm$ 5°0'. Materials are 0.250" 01 tool steel or equivalent hardened to Rc 35-40 and one ball bearing, IPL Item 10001-220.

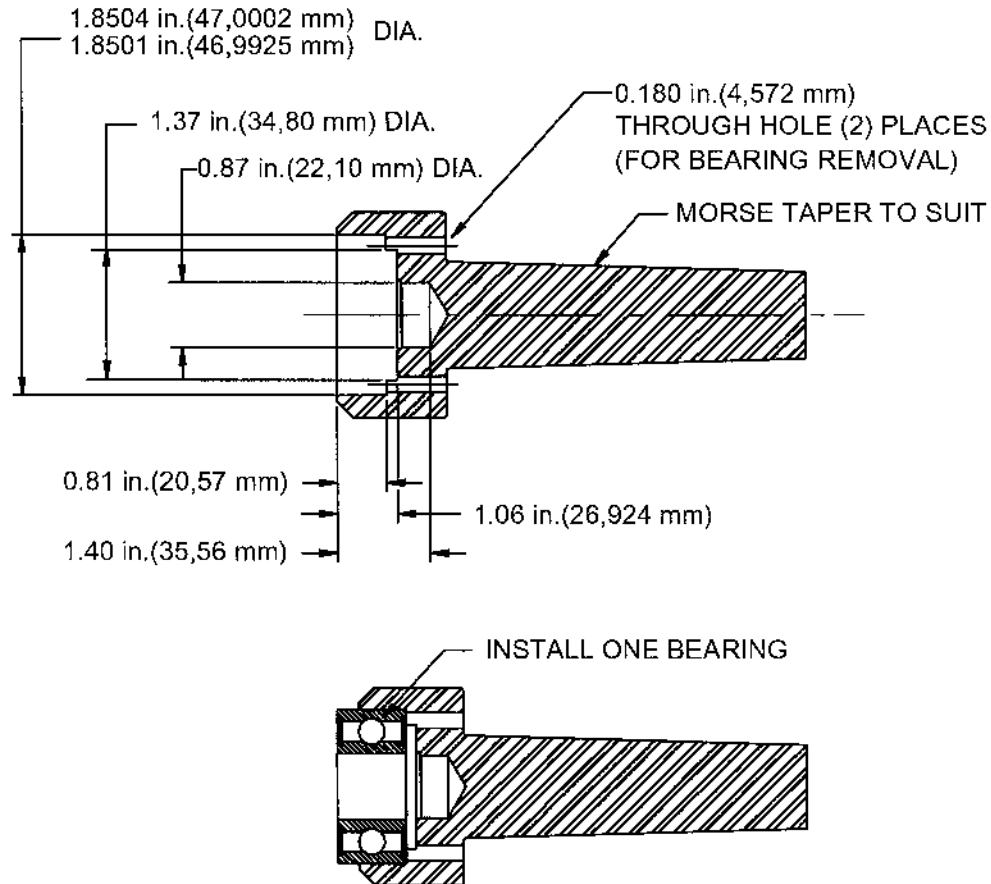
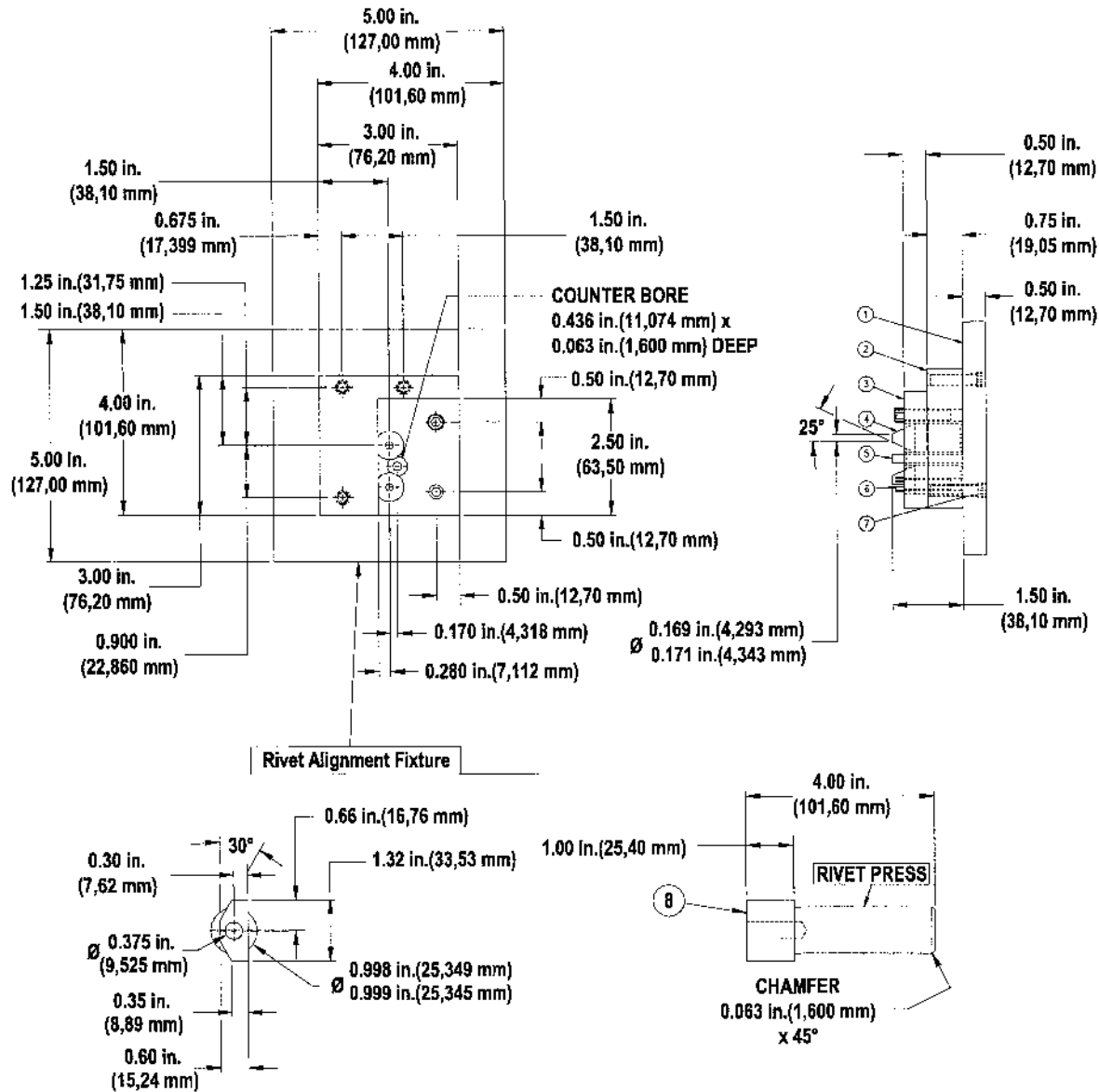


Figure 9008 - Commutator Turning Fixture

I. Rivet Alignment and Rivet Press Fixtures. (See Figure 9009).

Unless otherwise specified, dimensions are in inches. Tolerances on decimals are: XX±0.01, XXX±0.005. Angles are ±5°0'. Material is as indicated on the chart found on the figure.



ITEM	QTY.	DIMENSIONS	DESCRIPTION
8	1	1 1/2 DIA. X 4 1/8 LONG	A2 TOOL STEEL, HARDEN Rc 55-60
7	3	.250-20	SCREW, CAP SOCKET HEAD, COUNTER SUNK
6	2	.190-24	SCREW, CAP SOCKET HEAD
5	2	5/8 DIA. X 1 3/4 LONG	01 TOOL STEEL, DRILL ROD
4	1	3/16 DIA. X 1 3/4 LONG	01 TOOL STEEL, DRILL ROD
3	1	1/2 X 2 1/2 X 2 1/2	1010 COLD ROLL STEEL
2	1	3/4 X 3 X 3	1010 COLD ROLL STEEL
1	1	1/2 X 5 X 5	1010 COLD ROLL STEEL

Figure 9009 - Rivet Alignment and Rivet Press Fixtures

J. Armature Support. (See Figure 9010).

Unless otherwise specified, dimensions are in inches. Tolerances on decimals are: .XX $\pm$ 0.01, XXX $\pm$ 0.005. Angles are $\pm$ 5°0'. Machined ends to be parallel within 0.001 inch and square with sidewalls to $\pm$ 0.05 degrees. Material is 3.50" diameter 01 tool steel or equivalent hardened to Rc 35-40.

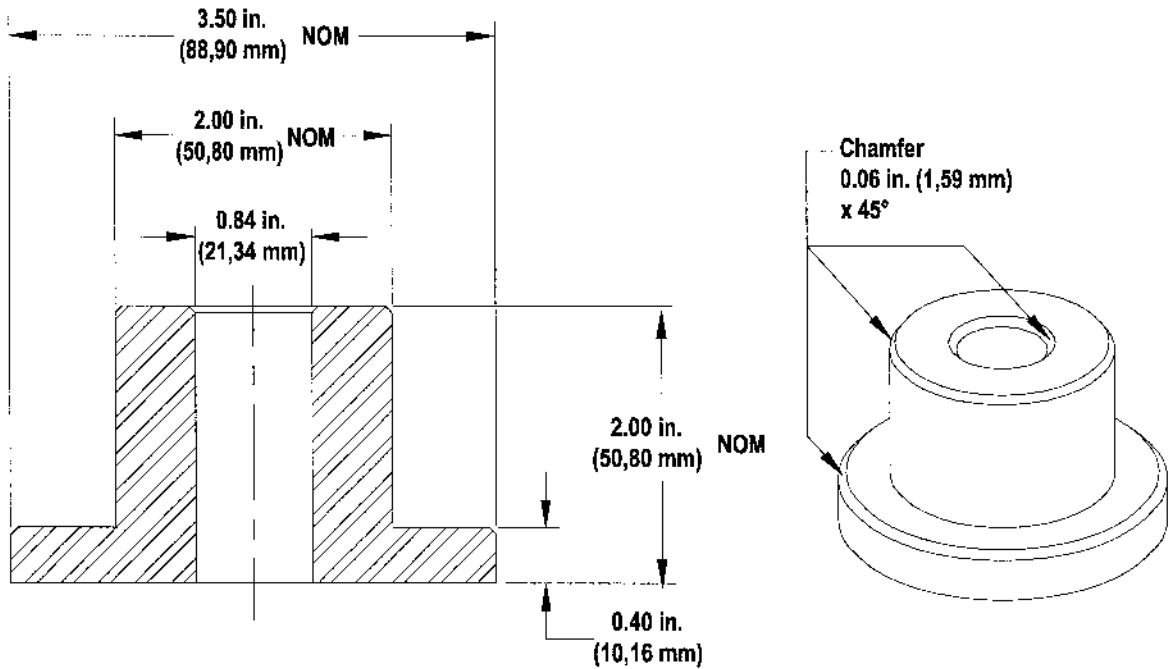


Figure 9010 - Armature Support

K. Anti-drive End End Bell Assembly Support. (See Figure 9011).

Unless otherwise specified, dimensions are in inches. Tolerances on decimals are: .XX $\pm$ 0.01, XXX $\pm$ 0.005. Angles are $\pm$ 5°0'. Material is 6.00" diameter 01 tool steel or equivalent.

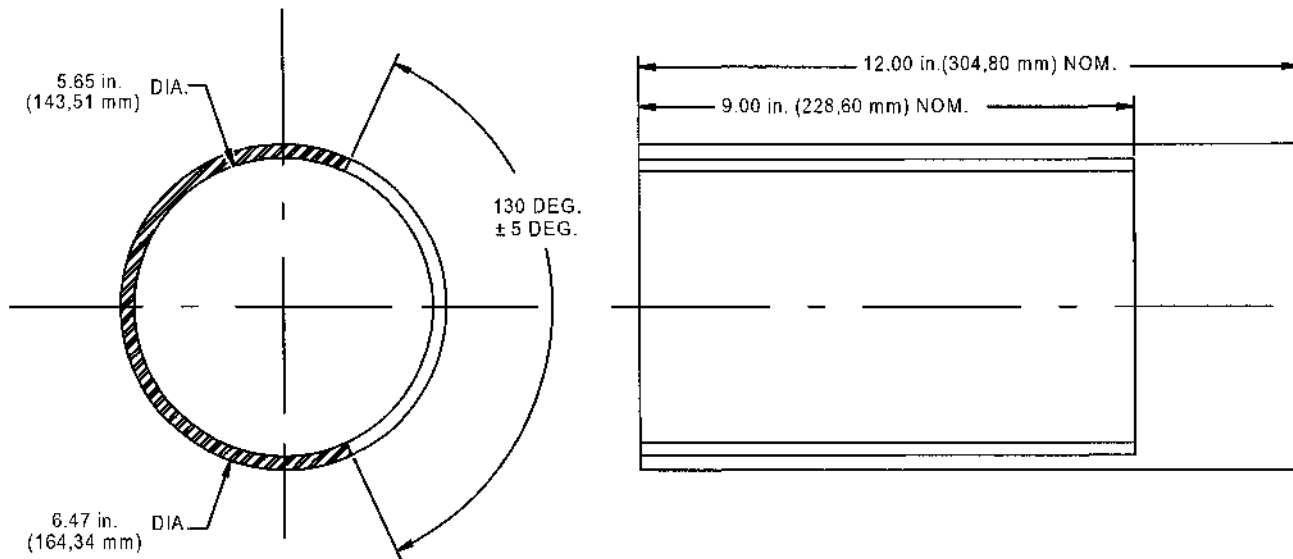


Figure 9011 - Anti-drive End End Bell Assembly Support

L. Horizontal Stator Support. (See Figure 9012).

Unless otherwise specified, dimensions are in inches. Tolerances on decimals are: .XX $\pm$ 0.01, XXX $\pm$ 0.005. Angles are $\pm$ 5°0'. Material is any suitable hardwood. (Stock size 7.5" X 6" X 4".)

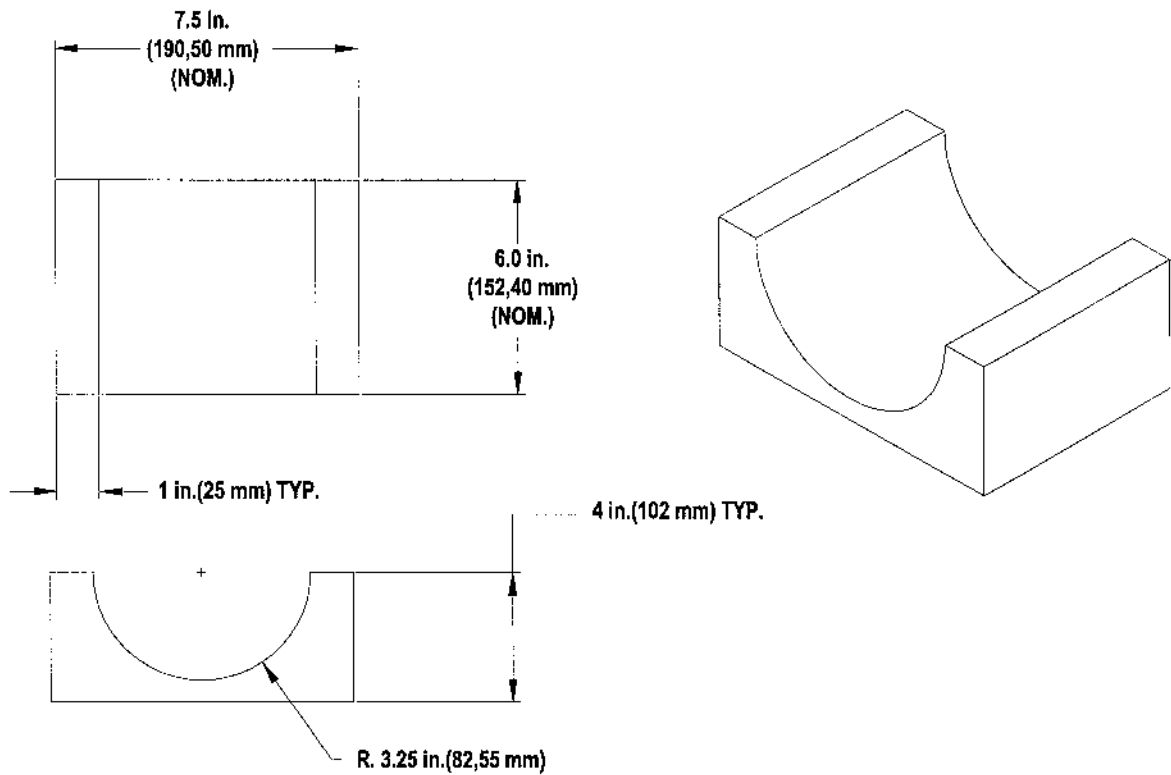


Figure 9012 - Horizontal Stator Support Fixture

M. Vertical Stator Support. (See Figure 9013).

Unless otherwise specified, dimensions are in inches. Tolerances on decimals are: .XX $\pm$ 0.01, XXX $\pm$ 0.005. Angles are $\pm$ 5°0'. Material is any suitable hardwood. (Stock size 9" X 9" X 3".)

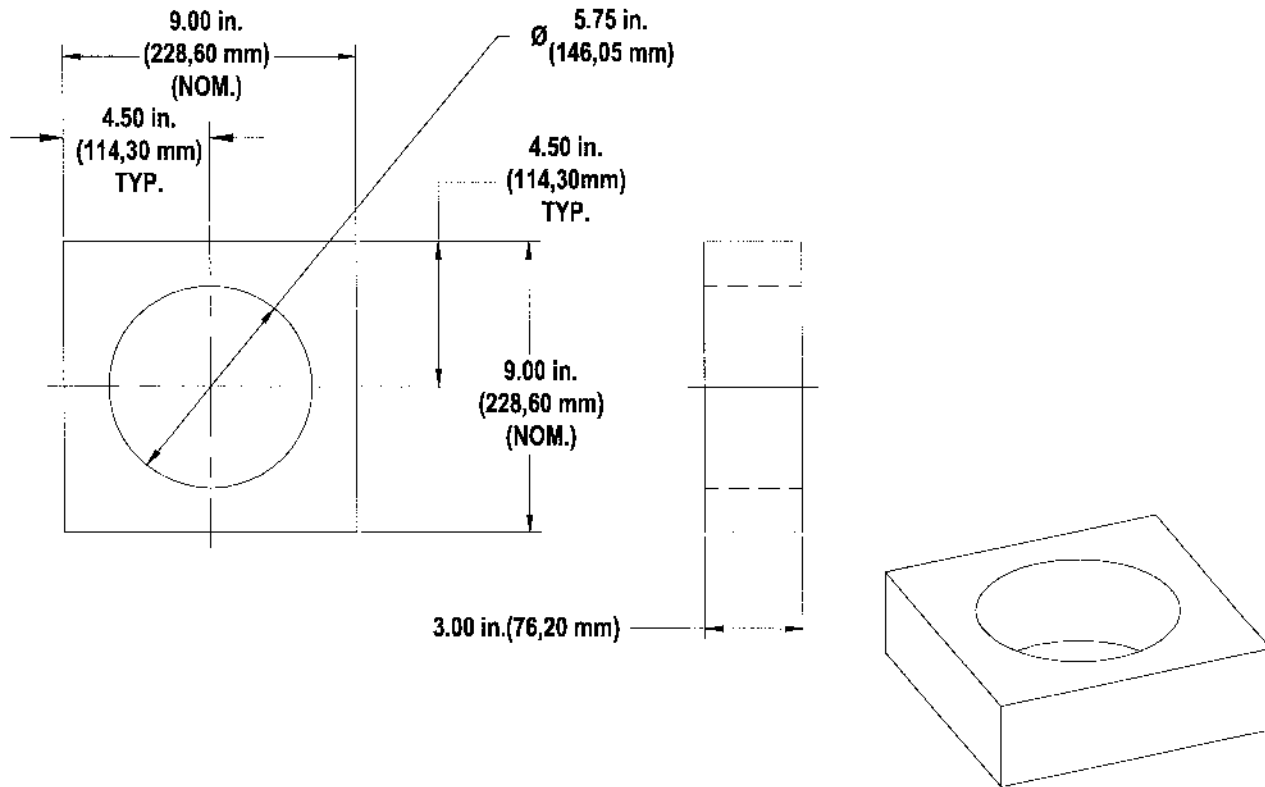


Figure 9013 - Vertical Stator Support Fixture

N. Drive End Hub Support Fixture. (See Figure 9014).

Unless otherwise specified, dimensions are in inches. Tolerances on decimals are: $XX \pm 0.01$, $XXX \pm 0.005$. Angles are $\pm 5^\circ$. Material is 2.50" diameter 01 tool steel or equivalent, hardened to Rc 35-40.

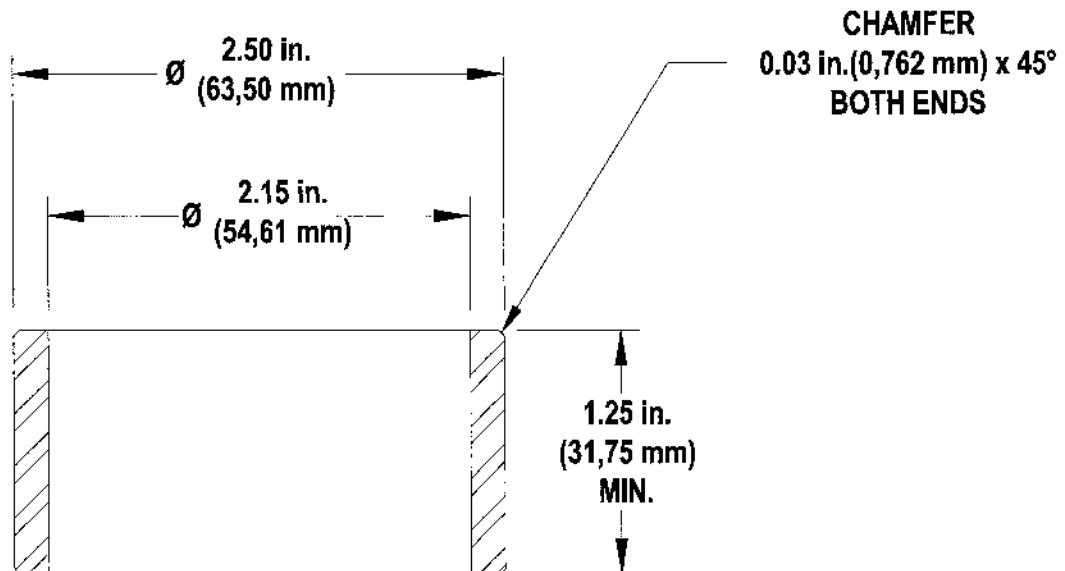


Figure 9014 - Drive End Hub Support



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

O. Anti-drive End Hub Support. (See Figure 9015).

Unless otherwise specified, dimensions are in inches. Tolerances on decimals are: .XX $\pm$ 0.01, XXX $\pm$ 0.005. Angles are $\pm$ 5°0'. Material is 2.50" diameter 01 tool steel or equivalent, hardened to Rc 35-40.

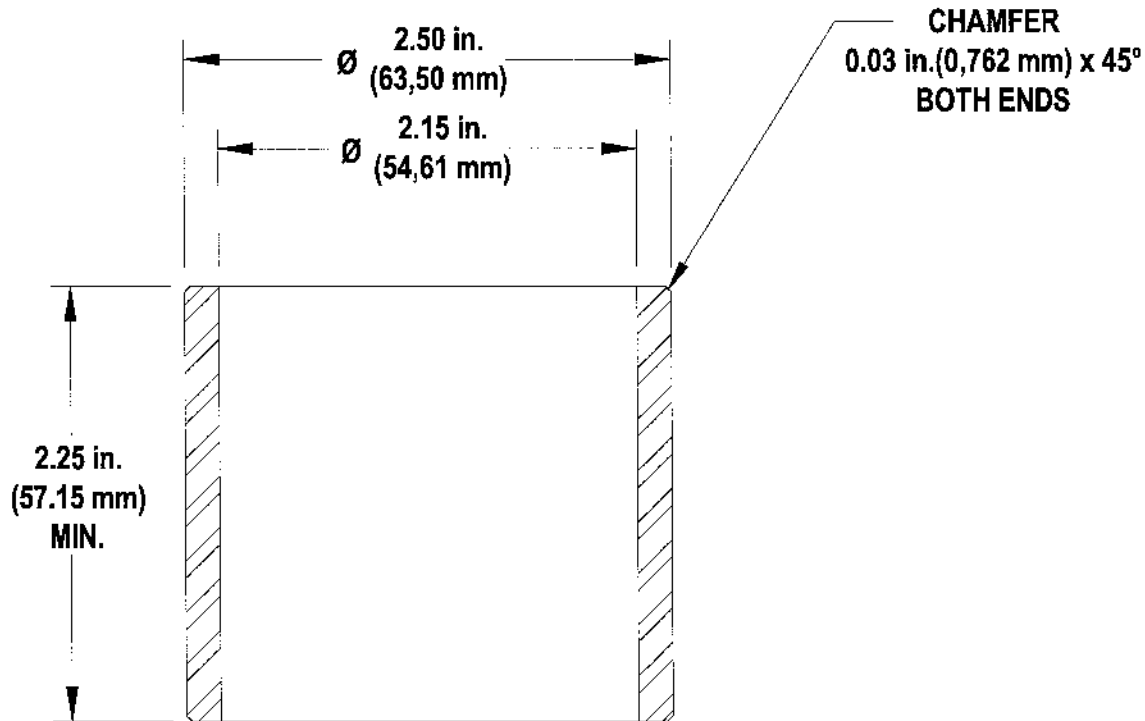
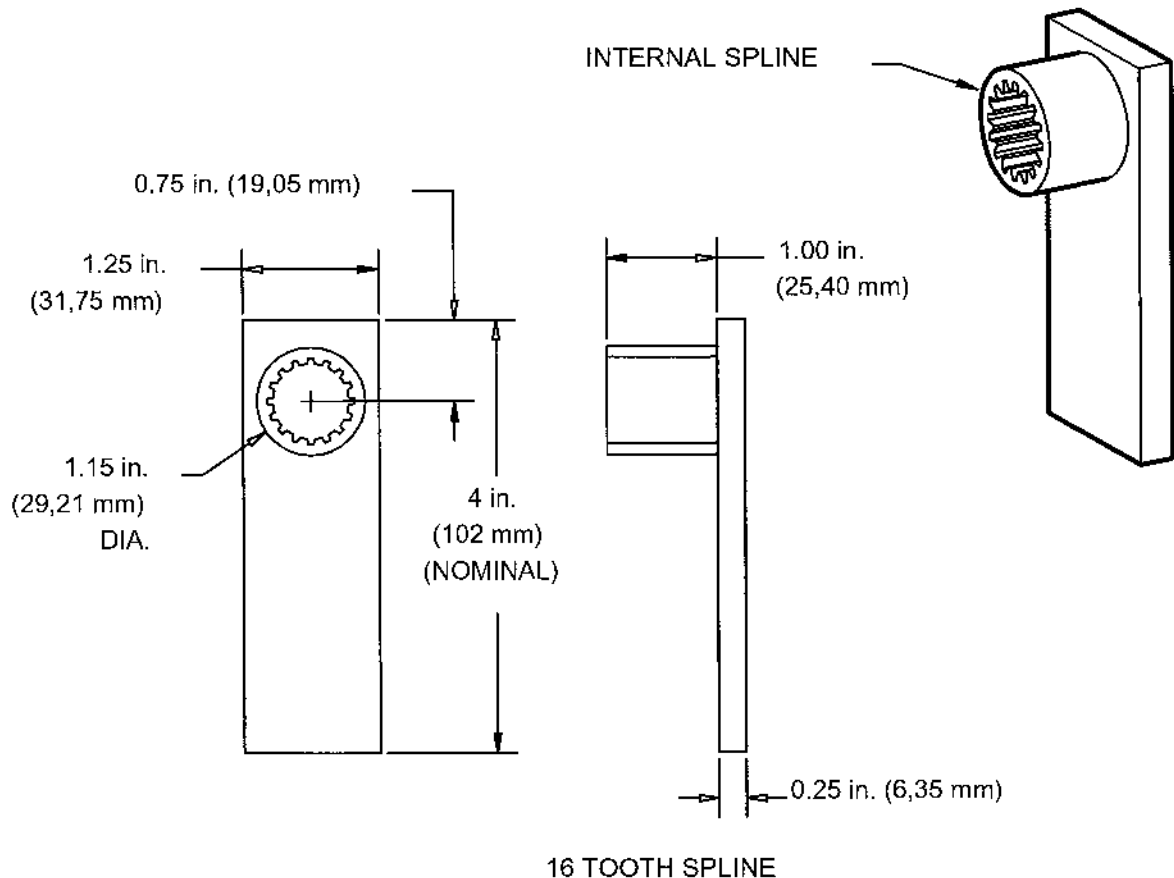


Figure 9015 - Anti-drive End Hub Support

P. Spline Wrench. (See Figure 9016).

Unless otherwise specified, dimensions are in inches. Tolerances on decimals are: XX±0.01, XXX±0.005. Angles are ±2°0'. Machined ends to be parallel within 0.001 inch and square with sidewalls to ±0.05 degrees.

Material: 01 Tool Steel hardened to Rc 55-60



Internal Involute Spline Data		
Fillet Root Side Fit	inch	mm
Number of Teeth	16	
Diametral Pitch	20 / 40	
Pressure Angle	30 DEG.	
Pitch Diameter	0.8000	20,320
Minor Diameter (min.)	0.7270	18,590
Major Diameter (min.)	0.8640	21,946
Chord Space (min.)	0.0728	1,849
Pin Diameter	0.0720	1,829
Between three .072 Pins (min.)	0.7096	18,024

Figure 9016 - Spline Wrench



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

ILLUSTRATED PARTS LIST

1. Introduction

This section contains lists of assemblies and detail parts for the 23079 Series of Starter-Generators. Each list is arranged in disassembly sequence, except that attaching parts are listed immediately after the parts they attach.

All replacement parts are manufactured or source-controlled by Goodrich Corporation, Aurora with the exception of those parts assigned a military standard (MIL, MS, etc.), a National Aerospace Standard (NAS), Army Navy (AN) standard, or an industry standard (ANSI, ASME, ISO, etc.) part number.

WARNING: ANY USE OF PARTS, MATERIALS OR PROCESSES NOT AUTHORIZED BY GOODRICH CORPORATION, AURORA FOR MAINTENANCE OR REPAIR OF THESE UNITS MAY AFFECT CONTINUED AIRWORTHINESS OR INVALIDATE CERTIFICATION.

To order authorized parts contact your Goodrich Corporation Equipment Service Center.

2. Arrangement of Parts List

A. Figure and Item Number Column

The first number indicates the figure where the part is illustrated. Each number (without a dash) corresponds to a part illustrated in the figure. Item numbers preceded by a dash identify units and/or assemblies which have not been illustrated.

B. Part Number Column

The part numbers listed in this column are the only authorized parts for replacement and repair of the starter-generator.

C. Nomenclature Column

The proper name of each part is provided in this column. The description of each item is indented by columns to show the relationship to the next higher assembly. The number of indentures depicts the relationship of the item to the associated next higher assembly as follows:

0 1 2 3 4 5
Assembly
Detail Parts for Assembly
Attaching Parts for Assembly
Sub-Assembly
Detail Parts for Sub-Assembly
Attaching Parts for Sub-Assembly

The interchangeability relationship between parts is identified in the NOMENCLATURE column of the parts list. A list of the terms used to show interchangeability and their definition is as follows:



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

- **Replaced by:** The part number listed is discontinued and should be replaced by the new part number. The original part may be used until the current stock is depleted and then it must be replaced by the new part listed.
- **Replaces:** The part number listed should be used in place of the discontinued part after the original part stock is depleted.
- **Alternate:** Either part may be used.
- **Superseded by:** The part number listed is obsolete and must be replaced by the new part number during the next repair or service of the generator. The superseded part is not to be installed or reinstalled.
- **Supersedes:** The part number listed must replace the obsolete part number at the next repair or service of the generator.
- **Order separately:** The part is not furnished as part of the generator as defined in this parts list.

D. Effect Code

Each generator model is assigned a unique effectivity code (A through Z, I and O are not used). Subsequent items within the parts list pertaining to Figure 10001 will refer to the effectivity codes that correspond to a particular model. In subsequent figures, the effectivity code is redefined for each major subassembly and remains independent of all other main assembly and subassemblies defined in other figures within this section. In cases where a part is used on all assemblies, the effect code is left blank.

E. Units Per Assembly

This column indicates the quantity of parts used at the location shown. For bulk items, the abbreviation AR is used to indicate the part quantity is used on an "as required" basis. The abbreviation RF indicates that the item is listed for "reference" only.

3. Vendor's Code

In the case of an item supplied by a vendor and not listed under the prime manufacturer's part number, a vendor code is prefixed by the capital letter V appearing in the nomenclature column. This code number designates the original manufacturer, and is in accordance with Cataloging Handbooks H4-1, H4-2, and H4-3. The Customer Support division in your region is to be contacted for parts dispatch.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

The following is an index of the manufacturers names, addresses, and codes as used in the listing.

Vendor Code	Name and Address
V05972	Loctite Corp. 705 N. Mountain Rd. Newington, CT 06111
V0LAX1	National Paper and Packaging Co. 1240 E. 55th Street Cleveland, OH 44103
V0SR97	Chase Corporation Humi-Seal Division 26-60 Brooklyn-Queens Expressway P.O. Box 446 Woodside, NY 11377
V0ZVN9	Omni Fastener, Inc. 9000 Sweet Valley Drive Valley View, OH 44125
V19527	Goodrich Corporation Equipment Services Americas 30 Van Nostrand Avenue Englewood, NJ 07631-4396 USA
V38360	Markem Corporation 150 Congress Street Keene, NH 03431-4307
V6W965	United Packaging Co. 16800 S. Waterloo Rd. Cleveland, OH 44110-3807
V71643	Saint-Gobain Performance Plastics, Inc. 407 East Street New Haven, CT 06509
V72688	John C. Dolph Co. West New Rd. P.O. Box 267 Monmouth Junction, NJ 08852-9513
V94058	The Brulin Corp. P.O. Box 270 Indianapolis, IN 46206
V97785	Federal Process Co. 3737 Park East Drive Beachwood, OH 44122-4307

NOTE: Spec 200/2000, automated ordering users, refer to the Exclusive Distributor Section in the front of this manual when ordering parts identified with V19527, Goodrich Corporation.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

MODEL NOTES:

- (1) Models 23079-000, 23079-000-1, 23079-000-2, 23079-002, 23079-002-1, 23079-003, 23079-004, and 23079-007 are not supplied with a QAD kit. The P/N 23065-504 QAD kit is required for installation and may be ordered separately.
- (2) Model 23079-005 is not supplied with a QAD kit. QAD kit P/Ns 23065-504, 23065-512, and 23065-513 are application specific. The QAD kit is required for installation and may be ordered separately.
- (3) Model 23079-006 is supplied with a P/N 23065-504 QAD kit. The QAD kit is required for installation and may be called out in aircraft documentation for this model.
- (4) Models 23079-009 and 23079-010 are not supplied with a QAD kit. QAD kit P/N 23065-513 is required for installation and may be ordered separately.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

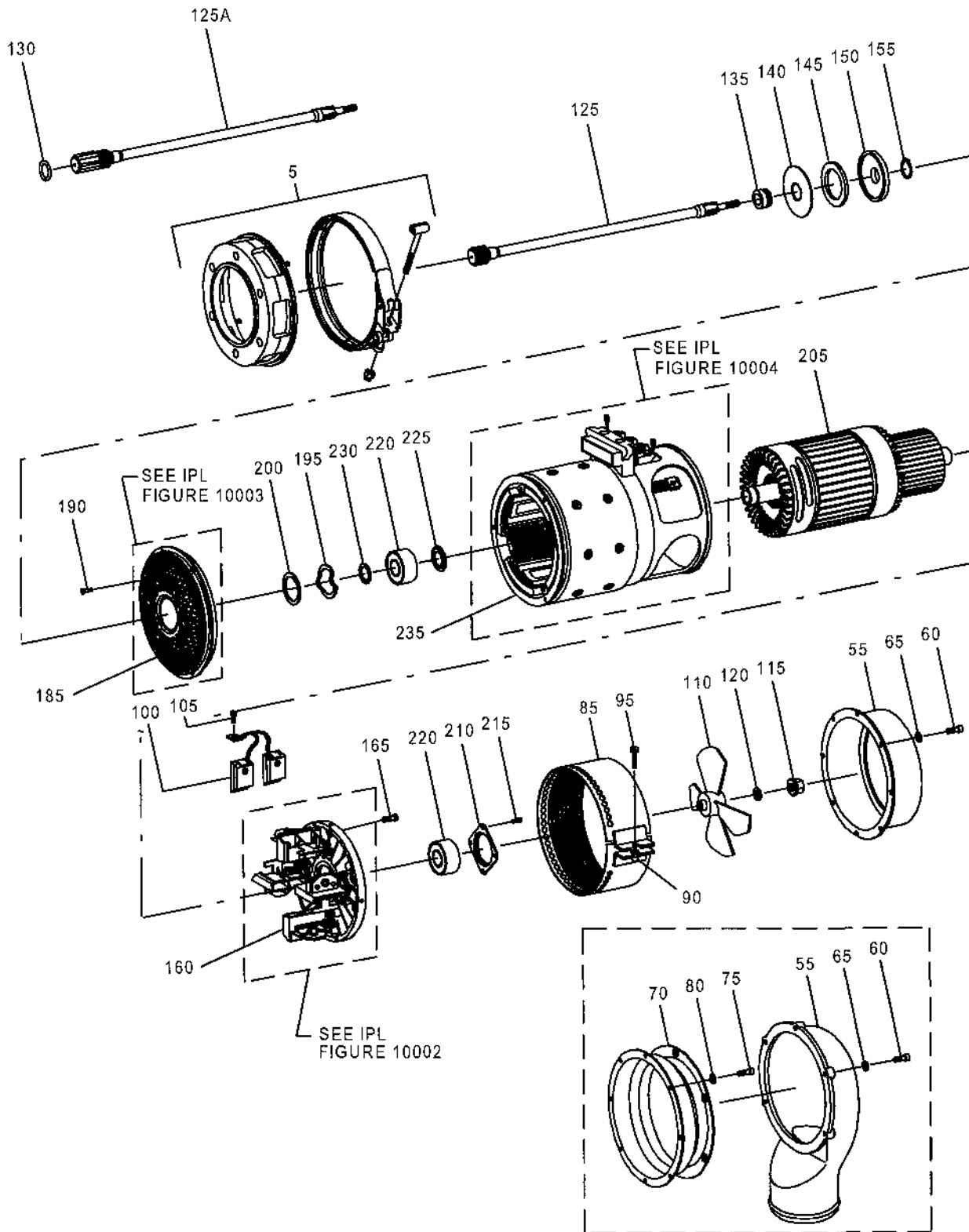


Figure 10001 - DC Starter-Generator Assembly, 23079 Series

ILLUSTRATED PARTS LIST

GOODRICH

Component Maintenance Manual with Illustrated Parts List DC Starter-Generator, 23079 Series

FIGURE AND ITEM	PART NUMBER	NOMENCLATURE 1 2 3 4 5	EFFECT CODE	QTY PER ASSY
10001-	23079-000	DC STARTER-GENERATOR. See Note (1)	A	REF
	23079-000-1	DC STARTER-GENERATOR. See Note (1)	B	REF
	23079-000-2	DC STARTER-GENERATOR. See Note (1)	C	REF
	23079-002	DC STARTER-GENERATOR. See Note (1)	D	REF
	23079-002-1	DC STARTER-GENERATOR. See Note (1)	E	REF
	23079-003	DC STARTER-GENERATOR. See Note (1)	F	REF
	23079-004	DC STARTER-GENERATOR. See Note (1)	G	REF
	23079-005	DC STARTER-GENERATOR. See Note (2)	H	REF
	23079-006	DC STARTER-GENERATOR. See Note (3)	J	REF
	23079-007	DC STARTER-GENERATOR. See Note (1)	K	REF
	23079-009	DC STARTER-GENERATOR. See Note (4)	L	REF
	23079-010	DC STARTER-GENERATOR. See Note (4)	M	REF
5	23065-504	• KIT, Q.A.D., Mounting. See IPL Figure 10005 (Superseded by 23065-513)	ABCDEF GHJK M	1 REF
	23065-512	• KIT, Q.A.D., Mounting. See IPL Figure 10005	H	1
	23065-513	• KIT, Q.A.D., Mounting. See IPL Figure 10005 (Supersedes 23065-504) SB 23079-010-24-01	HL M	1 1
-10	06-209285	• NAMEPLATE, Replacement. (ATTACHING PARTS)		1
-15	MS21318-14	• • SCREW, Drive (Superseded by MS21318-13)	ABCDEF GHJKL	REF
	MS21318-13	• • SCREW, Drive (Supersedes MS21318-14)	ABCDEF GHJKL	4

ILLUSTRATED PARTS LIST



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

FIGURE AND ITEM	PART NUMBER	NOMENCLATURE					EFFECT CODE	QTY PER ASSY
		1	2	3	4	5		
-15 (Cont.)		• • SCREW, Drive					M	4
		----- * -----						
-20	06-0017-01	• INFORMATION PLATE, Replacement . . (ATTACHING PARTS)					JK	1
-25	MS21318-14	• • SCREW, Drive (Replaced by MS21318-13)					JK	REF
	MS21318-13	• • SCREW, Drive (Replaces MS21318-14)					JK	2
		----- * -----						
-30	06-201100	• LABEL, Modification Status						AR
-35	06-201020	• LABEL, Caution						1
-40	06-200001	• LABEL, Patent					BCEGHJK LM	1
-41	06-123301	• LABEL, FAA-PMA					BDGHL	1
-45	23079-1060	• COVER, Terminal Block					DEFG	1
	23069-1241	• COVER, Terminal Block (ATTACHING PARTS)					JK	1
-50	NAS1189-06P-12L	• • SCREW, Machine, Flat Head					DEFG	2
	NAS1189-06P-14L	• • SCREW, Machine, Flat Head (Alternate 05-350108)					JK	2
	05-350108	• • SCREW, Machine, Flat Head (Alternate NAS1189-06P-14L)					JK	2
		----- * -----						
55	23065-1470	• INLET, Air					ABCDEHL M	1
	23065-1340	• INLET, Air					FG	1
	G29-383	• INLET, Air (ATTACHING PARTS)					JK	1
60	MS35265-43	• • SCREW, Machine Drilled, Fillister Hd.					ABCDEF GHLM	8
	MS35266-65	• • SCREW, Machine Drilled, Fillister Hd.					JK	5
65	MS35333-38	• • WASHER, Lock					ABCDEF GHLM	8
	05-700105	• • WASHER, Lock (Alternate to MS35333-38)					LM	8
	AN960-10L	• • WASHER, Flat					JK	5
		----- * -----						

ILLUSTRATED PARTS LIST



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

FIGURE AND ITEM	PART NUMBER	NOMENCLATURE 1 2 3 4 5	EFFECT CODE	QTY PER ASSY
70	23065-1370	• ADAPTER, Air Inlet (ATTACHING PARTS)	JK	1
75	MS35265-43	• • SCREW, Machine Drilled, Fillister Hd.	JK	8
80	MS35333-38	• • WASHER, Lock. (Alternate 05-700105)	JK	8
	05-700105	• • WASHER, Lock. (Alternate MS35333-38)	JK	8
85	23065-1055	• COVER, Brush Access.	ABCDEF G	1
	23048-1123	• COVER, Brush Access.	HJKLM	1
90	05-321002	• • RIVNUT® (Replaced by NAS1329H3K80L)		REF
	NAS1329H3K80L	• • RIVNUT® (Replaces 05-321002) (Replaced by NAS1329H3K130L)		REF
	NAS1329H3K130L	• • RIVNUT® (Replaces NAS1329H3K80L) (ATTACHING PARTS)		1
95	AN502-10-14	• • SCREW, Machine Drilled, Fillister Hd.	ABCDEF G	1
	AN502-10-18	• • SCREW, Machine Drilled, Fillister Hd. ----- *	HJKLM	1
100	30300-1284	• BRUSH, Electrical Contact.	ABDEFG HJKM	4
	30300-1415	• BRUSH, Electrical Contact. (Replaced by 23079-1281)	L	REF
	23079-1281	• BRUSH, Electrical Contact. (Replaces 30300-1415) SB 23079-009-24-07 MOD G	L	4
		• (ATTACHING PARTS)	C	4
105	05-340201	• • SCREW, Machine, Binding Head . . .	ADF	4
	05-340209	• • SCREW, Machine, Binding Head . . . ----- *	BCEGHJK LM	4
110	23065-1743	• FAN	ABCDEHL M	1
	23065-1740	• FAN (ATTACHING PARTS)	FGJK	1
115	02-4107-01	• • NUT, Self-Locking. (Replaced by MS21042-4)	ABCDEF GHJKL	REF

ILLUSTRATED PARTS LIST



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

FIGURE AND ITEM	PART NUMBER	NOMENCLATURE					EFFECT CODE	QTY PER ASSY
		1	2	3	4	5		
115 (Cont.)	MS21042-4	•	•	NUT, Self-Locking			ABCDEF GHJKL	1
				(Replaces 02-4107-01)				
				SIL 23703				
		•	•				M	1
120	AN960-416L	•	•	WASHER, Flat			ABCDEF GHJKL	REF
				(Replaced by AN960-416)				
	AN960-416	•	•	WASHER, Flat			ABCDEF GHJKL	1
				(Replaced AN960-416L)				
		•	•				M	1
				----- *				
125	23065-1520	•		SHAFT, Drive			ABCDEF GHL M	1
125A	02-6102-03	•		SHAFT, Drive			JK	1
130	M83248/1-210	•		O-RING			JK	1
135	23032-1900	•		HUB, Dampener				1
140	30059-1480	•		PLATE, Dampener				1
145	02-5600-06	•		RING, Friction				REF
				WARNING; MAY CONTAIN ASBESTOS				
				(Superseded by 02-5600-14)				
	02-5600-14	•		RING, Friction			ABCDEF GHJKL	1
				(Supersedes 02-5600-06)				
				SIL 02-5600-01				
		•					M	1
150	30059-1471	•		BACKPLATE, Dampener				1
155	MS16628-1078	•		RING, Retaining				1
160	23046-306	•		END BELL ASSEMBLY, Anti-Drive End .			AD	1
				Alternate 160A/D/E (See Figure 10002)				
				Replaced by 160J				
		•		END BELL ASSEMBLY, Anti-Drive End .			F	1
				Alternate 160D				
				Replaced by 160J				
160A	23046-373	•		END BELL ASSEMBLY, Anti-Drive End .			AD	1
				Alternate 160, 160D/E				
				Replaced by 160J				
160B	23075-331	•		END BELL ASSEMBLY, Anti-Drive End .			BEGHJKL M	1
				Alternate 160C/F/G				
				Replaced by 160H				
160C	23075-335	•		END BELL ASSEMBLY, Anti-Drive End .			BEGHJKL M	1
				Alternate 160B/F/G				
				Replaced by 160H				
160D	23065-360	•		END BELL ASSEMBLY, Anti-Drive End .			AD	1
				Alternate 160, 160A/E				
				Replaced by 160J				

ILLUSTRATED PARTS LIST



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

FIGURE AND ITEM	PART NUMBER	NOMENCLATURE					EFFECT CODE	QTY PER ASSY
		1	2	3	4	5		
		• END BELL ASSEMBLY, Anti-Drive End. Alternate 160 Replaced by 160J					F	1
160E	23065-361	• END BELL ASSEMBLY, Anti-Drive End. Alternate 160, 160A/D Replaced by 160J					AD	1
160F	23065-362	• END BELL ASSEMBLY, Anti-Drive End. Alternate 160B/C/G Replaced by 160H					BEGHJKL M	1
160G	23065-363	• END BELL ASSEMBLY, Anti-Drive End. Alternate 160B/C/F Replaced by 160H					BEGHJKL M	1
160H	23065-380	• END BELL ASSEMBLY, Anti-Drive End. Replaces 160B/C/F/G SB 23079-0XX-24-03 MOD D					BEGHJK	1
		• END BELL ASSEMBLY, Anti-Drive End. Replaces 160B/C/F/G SB 23079-009-24-05 MOD E					L	1
		• END BELL ASSEMBLY, Anti-Drive End. Replaces 160G SB 23079-0XX-24-03 MOD B					M	1
		•					C	1
160J	23065-381	• END BELL ASSEMBLY, Anti-Drive End. Replaces 160, 160A/D/E SB 23079-0XX-24-03 MOD D					AD	1
		• END BELL ASSEMBLY, Anti-Drive End. Replaces 160, 160D SB 23079-0XX-24-03 MOD D (ATTACHING PARTS)					F	1
165	NAS1189-08P12L	• SCREW, Machine, Flat Head. ----- * -----						8
-170	23065-1620	• SWITCH ASSEMBLY, Thermal (ATTACHING PARTS)					JK	1
-175	MS21043-08	• • NUT, Self Locking					JK	1
-180	MS35333-74	• • WASHER, Lock. ----- * -----					JK	1
185	23065-1660	• END BELL ASSEMBLY, Drive End Replaced by 23065-1662					ABHLM	1
		• END BELL ASSEMBLY, Drive End Replaced by 23065-1663					DEFGJK	1
185A	23065-1662	• END BELL ASSEMBLY, Drive End Replaces 23065-1660 MOD C Replaced by 23065-1663					ABHM	1

ILLUSTRATED PARTS LIST



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

FIGURE AND ITEM	PART NUMBER	NOMENCLATURE					EFFECT CODE	QTY PER ASSY
		1	2	3	4	5		
185B	23065-1663	• END BELL ASSEMBLY, Drive End Replaces 23065-1660 MOD D Replaced by 23065-1663					L	1
		• END BELL ASSEMBLY, Drive End Replaces 23065-1662 SB 23079-0XX-24-04 MOD G					ABM	1
		• END BELL ASSEMBLY, Drive End Replaces 23065-1660 SB 23079-0XX-24-04 MOD C					DEFGJK	1
		• END BELL ASSEMBLY, Drive End Replaces 23065-1662 SB 23079-0XX-24-04 MOD F					H	1
		• END BELL ASSEMBLY, Drive End Replaces 23065-1662 SB 23079-009-24-06 MOD H					L	1
		• (ATTACHING PARTS)					C	1
190	MS24693-S23	• • SCREW, Machine, Flat, Countersunk Head ----- *						3
195	02-4231-02	• WASHER, Spring Wave Post SB 23079-0XX-24-04 Post SB 23079-009-24-06						1
200	23080-1940	• SHIM, Steel, 0.003 inch (0,08 mm) thick Post SB 23079-0XX-24-04 Post SB 23079-009-24-06						AR
	23080-1941	• SHIM, Steel, 0.020 inch (0,51 mm) thick Post SB 23079-0XX-24-04 Post SB 23079-009-24-06						AR
	23080-1942	• SHIM, Steel, 0.030 inch (0,76 mm) thick Post SB 23079-0XX-24-04 Post SB 23079-009-24-06						AR
205	23079-1020	• ARMATURE (Replaced by 23079-330)					ABDEFG HJKLM	1
	23079-330	• ARMATURE (Replaces 23079-1020) SB 23079-0XX-24-05 MOD E					ABDEFG HJK	1
		MOD F					L	1
		MOD C					M	1
		•					C	1
210	23046-1111	• RETAINER, Bearing (ATTACHING PARTS)						1
215	MS24694-S3	• • SCREW, Machine, Fillister Head						4

ILLUSTRATED PARTS LIST



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

FIGURE AND ITEM	PART NUMBER	NOMENCLATURE					EFFECT CODE	QTY PER ASSY
		1	2	3	4	5		
		----- *						
220	03-6010-08	• BEARING, Ball Superseded by 03-6010-15					ABDEFG HJK	REF
	03-6010-10	• BEARING, Ball Superseded by 03-6010-14 Supersedes 03-6010-14 SB 23079-010-24-02 MOD D					L M	REF 2
	03-6010-14	• BEARING, Ball Supersedes 03-6010-10 SB 23079-009-24-01 MOD A Superseded by 03-6010-17 Superseded by 03-6010-10					L M	REF REF
	03-6010-15	• BEARING, Ball Supersedes 03-6010-08 SB 23079-00X-24-01 MOD A Superseded by 03-6010-18					ABDEFG HJK	REF
	03-6010-17	• BEARING, Ball Supersedes 03-6010-14 SB 23079-009-24-03 MOD B Superseded by 03-6010-18					L	REF
	03-6010-18	• BEARING, Ball Supersedes 03-6010-15 SB 23079-00X-24-02 MOD B Supersedes 03-6010-17 SB 23079-0XX-24-02 MOD C					ABDEFG HJK L	2 2
		•					C	2
225	G128-178	• BAFFLE DISK, (Inboard) Superseded by 23080-3110						REF
	23080-3110	• BAFFLE DISK, (Inboard) Supersedes G128-178 (225) SIL 23080-3110-01						1
-225A	G128-178	• BAFFLE DISK, (Outboard) Replaced by 23032-1145 (230)						1
230	23032-1145	• SPACER Replaces G128-178 (-225A) SB23079-0XX-24-04 SB23079-009-24-06						1
235	23079-310	• STATOR & HOUSING ASSEMBLY (See Figure 10004)					ABCHLM	1
	23079-320	• STATOR & HOUSING ASSEMBLY					DEFG	1
	23065-325	• STATOR & HOUSING ASSEMBLY					JK	1

ILLUSTRATED PARTS LIST

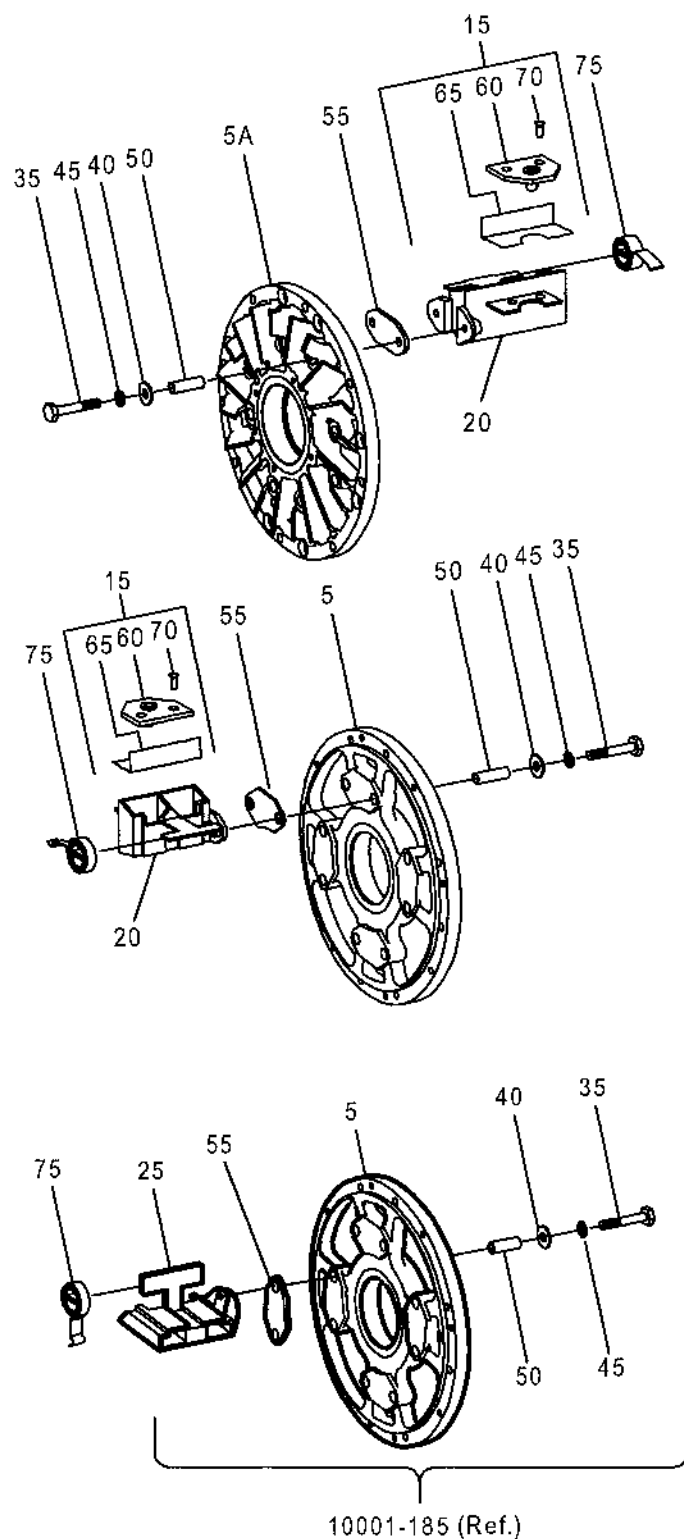


Figure 10002 - Anti-Drive End Endbell Assembly



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

FIGURE AND ITEM	PART NUMBER	NOMENCLATURE					EFFECT CODE	QTY PER ASSY
		1	2	3	4	5		
10002-	23046-306	END BELL ASSEMBLY, Anti-Drive End. . .					A	REF
	23046-373	END BELL ASSEMBLY, Anti-Drive End. . .					B	REF
	23075-331	END BELL ASSEMBLY, Anti-Drive End. . .					C	REF
	23075-335	END BELL ASSEMBLY, Anti-Drive End. . .					D	REF
	23065-360	END BELL ASSEMBLY, Anti-Drive End. . .					E	REF
	23065-361	END BELL ASSEMBLY, Anti-Drive End. . .					F	REF
	23065-362	END BELL ASSEMBLY, Anti-Drive End. . .					G	REF
	23065-363	END BELL ASSEMBLY, Anti-Drive End. . .					H	REF
	23065-380	END BELL ASSEMBLY, Anti-Drive End. . .					J	REF
	23065-381	END BELL ASSEMBLY, Anti-Drive End. . .					K	REF
	5 23065-1100	• END BELL, Anti-Drive End.					ABCD	1
-10	23065-1950	• END BELL, Anti-Drive End.					EFGH	1
	23065-1952	• END BELL, Anti-Drive End.					JK	1
	NASM21209C0815	• • INSERT, Helicoil TM (Replaces MS21209C0815)					ABCD	4
	MS21209C0815	• • INSERT, Helicoil TM (Replaced by NASM21209C0815)					ABCD	4
	NASM21209C0815	• • INSERT, Helicoil TM (Replaces MS21209C0815)					EFGHJK	12
	MS21209C0815	• • INSERT, Helicoil TM (Replaced by NASM21209C0815)					EFGHJK	12
	15 23075-1203	• BRUSH HOLDER, Complete (Isolated) .					CGJ	4
	23075-1207	• BRUSH HOLDER, Complete (Isolated) .					DH	4
	20 23075-1193	• • BRUSH HOLDER, Machining. (Isolated brush holders)					CGJ	4
	23075-1253	• • BRUSH HOLDER, Machining. (Isolated brush holders)					DH	4
25	23048-1380	• BRUSH HOLDER ASSEMBLY. (Non-isolated brush holders)					BF	4
	23064-1180	• BRUSH HOLDER, Machining (Non-isolated brush holders)					AEK	4
-30	NASM21209-F1-15	• THREAD INSERT, Locking. (Replaces MS21209-F1-15)					AEK	4
	MS21209-F1-15	• THREAD INSERT, Locking. (Replaced by NASM21209-F1-15) (ATTACHING PARTS)					AEK	4

ILLUSTRATED PARTS LIST



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

FIGURE AND ITEM	PART NUMBER	NOMENCLATURE					EFFECT CODE	QTY PER ASSY
		1	2	3	4	5		
10002-								
35	AN3-7A	•	•	BOLT, Machine, Hex Head			ACEGJK	8
	AN3-10A	•	•	BOLT, Machine, Hex Head			BDFH	8
40	05-374094	•	•	WASHER, Insulating			ABCDEF GH	8
	05-374095	•	•	WASHER, Insulating			JK	8
45	AN960-10L	•	•	WASHER, Flat			ABCDEF GH	8
	NAS1149F0332P	•	•	WASHER, Flat			JK	8
50	05-631145	•	•	SLEEVE, Insulating				8
55	23014-1039	•	•	BOARD, Insulating			ABCDEF GH	4
	23085-1471	•	•	BOARD, Insulating			JK	4
60	23075-1400	•		TERMINAL BOARD, Brush Lead			CDGHJ	1
65	23075-1230	•		SHEET, Insulating			CDGHJ	1
				(ATTACHING PARTS)				
70	MS20426AD3-5	•	•	RIVET, Solid, Precision head, Countersunk			CDGHJ	2
75	23033-1200	•		SPRING, Brush			CDGHJ	8
	23033-1120	•		SPRING, Brush (Alternate 23064-1350)			ABEF	8
		•					K	8
	23064-1350	•		SPRING, Brush (Alternate 23033-1120)			ABEF	8
-80	NAS1329H3K130L	•		RIVNUT®			CDGHJ	4



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

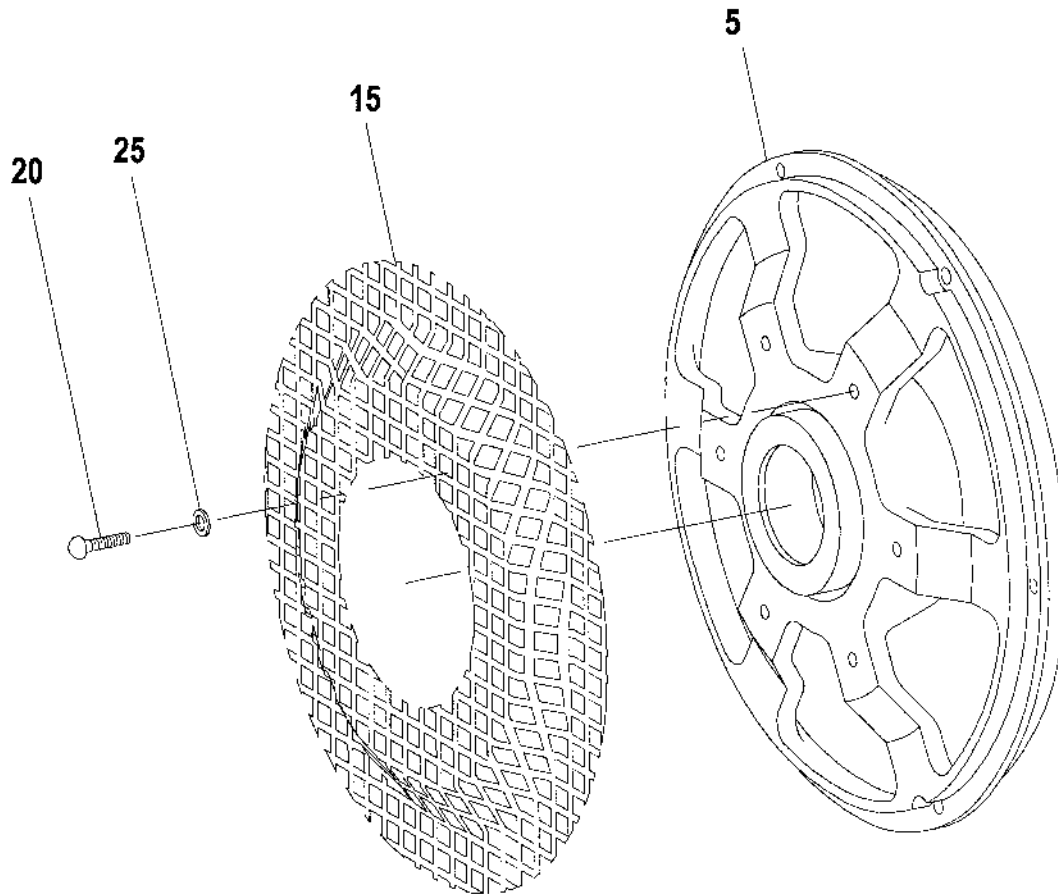
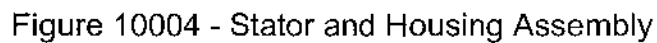


Figure 10003 - Drive End End Bell Assembly



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

FIGURE AND ITEM	PART NUMBER	NOMENCLATURE					EFFECT CODE	QTY PER ASSY
		1	2	3	4	5		
10003-								
	23065-1660	•	END BELL ASSEMBLY, Drive End.				A	REF
	23065-1662	•	END BELL ASSEMBLY, Drive End.				B	REF
	23065-1663	•	END BELL ASSEMBLY, Drive End.				C	REF
5	23065-1640	• •	DRIVE END END BELL				A	1
	23065-1643	• •	DRIVE END END BELL				B	1
	23065-1645	• •	DRIVE END END BELL				C	1
-10		• •	Deleted					
15	23065-1650	• •	SCREEN					1
			(ATTACHING PARTS)					
20	MS21318-15	• •	SCREW, Drive					6
25	AN960-4L	• •	WASHER, Flat					6

Page 10020
Feb 28/03



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

FIGURE AND ITEM	PART NUMBER	NOMENCLATURE					EFFECT CODE	QTY PER ASSY
		1	2	3	4	5		
10004-	23079-310	• STATOR AND HOUSING ASSEMBLY					A	REF
	23079-320	• STATOR AND HOUSING ASSEMBLY					B	REF
	23065-325	• STATOR AND HOUSING ASSEMBLY					C	REF
5	Not Procurable	• • STATOR HOUSING					A	REF
5A	Not Procurable	• • STATOR HOUSING					B	REF
5B	Not Procurable	• • STATOR HOUSING					C	REF
10	MS21042-6	• • NUT, Self Locking					AB	3
	MS21042-6	• • NUT, Self Locking					C	2
15	AN960B616	• • WASHER, Flat (Superseded by AN960C616)					BC	REF
	AN960C616	• • WASHER, Flat (Supersedes AN960B616)					A	3
		(Supersedes AN960B616)					B	3
		(Supersedes AN960B616)					C	2
20	MS21042-3	• • NUT, Self Locking						2
25	AN960B10	• • WASHER, Flat Superseded by AN960C10					B	REF
	AN960C10	• • WASHER, Flat Supersedes AN960B10					AC	2
		• •					B	2
30	23065-1210	• • TERMINAL BLOCK.					A	1
30A	23079-1040	• • TERMINAL BLOCK.					B	1
30B	23069-1582	• • TERMINAL BLOCK. (ATTACHING PARTS)					C	1
35	AN3-13A	• • • SCREW, Machine					A	2
35A	MS35266-64	• • • SCREW, Machine					B	2
35B	AN3-14A	• • • SCREW, Machine					C	2
40	MS35338-43	• • • WASHER, Spring Lock						2
45	AN960-10	• • • WASHER, Flat Superseded by AN960C-10						REF
	AN960C-10	• • • WASHER, Flat Supersedes AN960-10						2



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

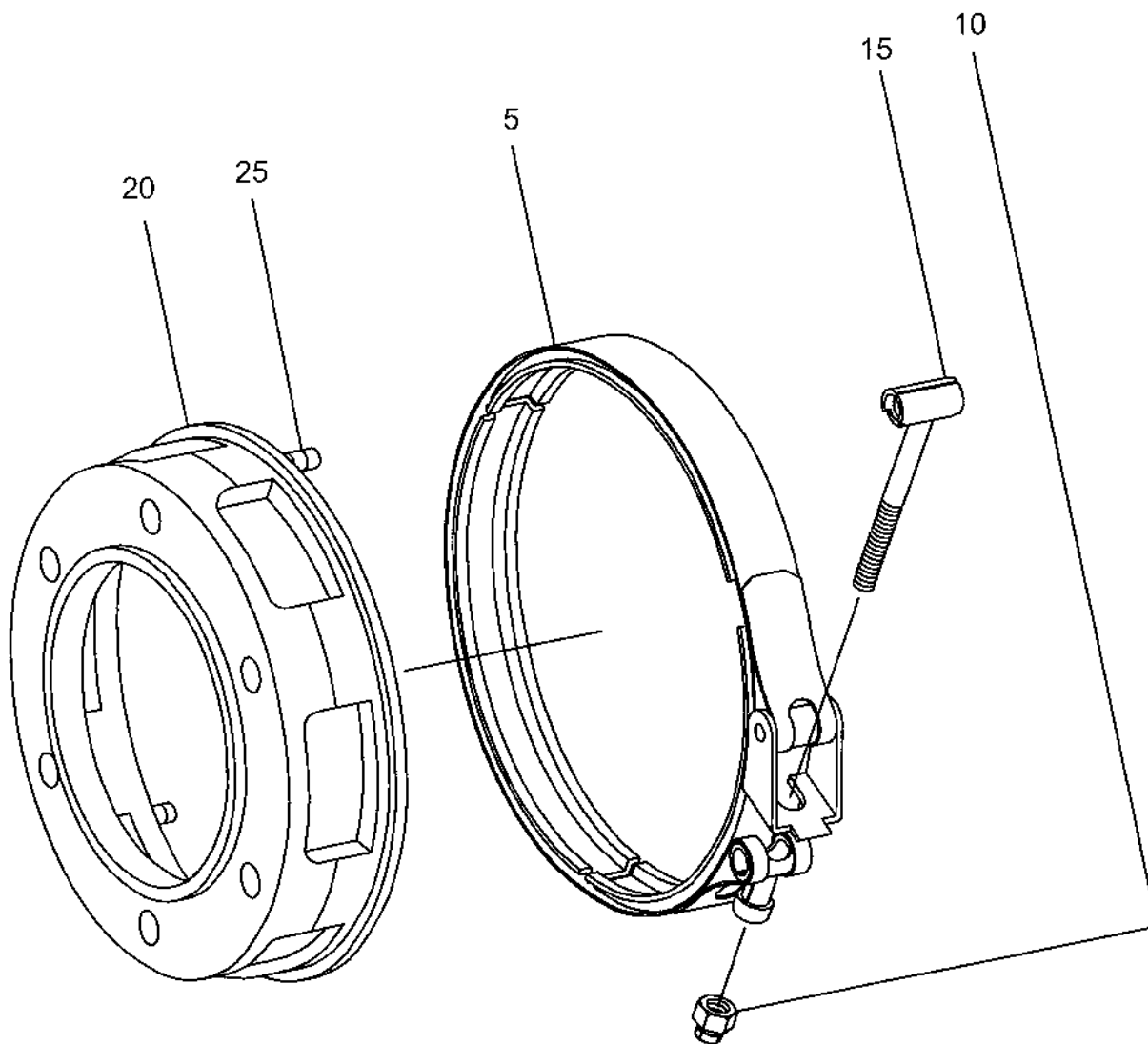


Figure 10005 - Quick Attach/Detach Mounting Kit



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

FIGURE AND ITEM	PART NUMBER	NOMENCLATURE					EFFECT CODE	QTY PER ASSY
		1	2	3	4	5		
10005-								
	23065-504					MOUNTING KIT, QAD	A	REF
	23065-512					MOUNTING KIT, QAD	B	REF
	23065-513					MOUNTING KIT, QAD	C	REF
5	23072-1120					• CLAMP, V-Band (Replaced by 23072-1327 and 23072-1127)	AB	REF
-5A	23072-1327					• CLAMP, V-Band (Replaces 23072-1120)	AB	1
						(Alternate 23072-1127)	C	1
-5B	23072-1127					• CLAMP, V-Band (Replaces 23072-1120)	AB	1
						(Alternate 23072-1327)	C	1
10	MS21045-4					• • NUT, Reduced Hex, Self-Locking Use on 23072-1127		1
	MS21045-L4					• • NUT, Reduced Hex, Self-Locking Use on 23072-1327		1
15	23032-2802					• • T-BOLT Use on 23072-1127		1
	23032-2803					• • T-BOLT Use on 23072-1327		1
20	23065-1430					• ADAPTER, Mounting (Replaced by 23065-1432)	A	REF
-20A	23065-1431					• ADAPTER, Mounting (Replaced by 23065-1433)	B	REF
-20B	23065-1432					• ADAPTER, Mounting (Replaces 23065-1430)	C	1
							A	1
-20C	23065-1433					• ADAPTER, Mounting (Replaces 23065-1431)	B	1
						(ATTACHING PARTS)		
25	02-4412-09					• • PIN, Grooved, Headless		3

ILLUSTRATED PARTS LIST



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

SPECIAL PROCEDURES

1. **Introduction**

This section is unassigned.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

REMOVAL

1. **Introduction**

This section is unassigned.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

INSTALLATION

1. Introduction

This section is unassigned.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

SERVICING

1. Introduction

This section is unassigned.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

STORAGE

1. Storage

Materials recommended for packaging rotating machines containing grease-lubricated bearings for storage or shipment are listed below in Table 15001.

Description	Specification	Quantity
Tag	Commercially available	1 (Domestic) 2 (International)
Grease	MIL-PRF-81322	AR
Chemically neutral protective paper	Commercially available	AR
Polyethylene (plastic) wrap	Commercially available	AR
Packing material - shock absorbing foam rubber, styrofoam, bubble wrap or expanded foam (Goodrich Corporation preferred method) - three (3) inch thick minimum - surrounding machine on all sides	Commercially available	AR
Tape - waterproof, pressure sensitive	Commercially available	AR
Small plastic bag	Commercially available	AR
O-ring (for drive shaft on applicable models)	Check Illustrated Parts List for current p/n	1 ea.
Desiccant	MIL-D-3464 Type I and II	1 ea.
Original shipping container - or equivalent alternate: domestic class, double-wall, corrugated cardboard. Box must be large enough to totally enclose bagged machine and allow for a 3 inch thick layer of cushioning material to surround it on all sides	PPP-B-636 or suitable equivalent	1 ea.
Box - WC5 overseas shipping container, sized to completely enclose domestic class packaging - for overseas shipment only	PPP-B-636 or suitable equivalent	1 ea.

Table 15001 - Packaging Material

A. GENERAL INFORMATION

NOTE: If machine storage time has exceeded 24 months with no use, it is recommended that the ball bearings be replaced. It must then be re-tested according to the TESTING AND FAULT ISOLATION section before being put into service.

- (1) Unit is to have successfully completed tests specified in TESTING AND FAULT ISOLATION section of this manual before preparing unit for shipment or storage.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

- (2) Verify that lockwire and lead seal are in place.
- (3) Record following information on tag(s):
 - Model Number
 - Serial Number
 - Modification (MOD) Status
 - Test Date (PASSED)
 - Packing date
- (4) Select packaging materials as described in Table 15001.

B. DOCUMENTATION

Include all applicable documentation with unit:

- Testing Records
- Repair Reports
- Final Inspection/Check Records
- Packing List and Certificate of Conformance

C. DOMESTIC PACKAGING

Recommended packaging procedure for generators and starter-generators is:

- (1) If machine includes drive shaft with O-ring/oil sealing provision, place O-ring into small plastic bag, include instructions to lubricate and install the O-ring on drive end of shaft prior to installation on aircraft, seal bag, and attach it to the machine.
- (2) Apply lubricating grease to drive spline according to MIL-PRF-81322 and wrap it in protective paper according to MIL-B-121A, Grade A, Type II.
- (3) Put unit and desiccant into a waterproof and vapor-proof heat-seal bag and partially heat-seal it while forcing as much air as possible from bag.
- (4) Insert vacuum tube into bag to draw remaining air out of bag.
- (5) Withdraw vacuum tube quickly and complete heat-sealing of bag.
- (6) Use machine's original shipping container if possible. Put bagged machine into box surrounded with a minimum of three (3) inch thick shock absorbing, cushioning material on all six sides.
- (7) Securely and completely seal all flapped openings of box with tape.
- (8) Tape tag to exterior of box. Make sure all information is visible.

STORAGE



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

D. OVERSEAS SHIPMENT ON SURFACE VESSELS

- (1) Package the machine as given in the DOMESTIC PACKAGING paragraph in this section.
- (2) Put domestically packaged generator or starter-generator into WC5 shipping container.
- (3) Securely and completely seal all flapped-openings of shipping container with tape.
- (4) Tape tag to exterior of shipping container. Make sure all information is visible.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

THIS PAGE INTENTIONALLY LEFT BLANK

STORAGE



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

REWORK

1. Introduction

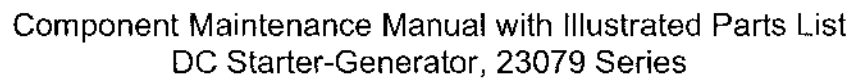
This section is unassigned.

REWORK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23079 Series

THIS PAGE INTENTIONALLY LEFT BLANK



ATTACH A COPY OF EACH AFFECTED PAGE		Date of submittal:
List the technical publication(s) and page(s) included with this form:		
Suggested technical publication(s) addition(s), deletion(s), or correction(s):		
Reason(s) for recommendation(s):		
Company / Organization:	Organization type (check one):	
Address (include country):	☐ Owner/Operator	
	☐ Repair station	
	☐ Other (please specify)	
Name / Title:		
Telephone (include area/country code):	Fax:	
www.	E-Mail:	
Forward completed form along with affected page(s) to:		Goodrich Corporation Technical Publications Supervisor 777 Lena Drive Aurora, Ohio 44202 USA Fax (1) 330 995 1113 E-Mail: aurora.techpubs@goodrich.com